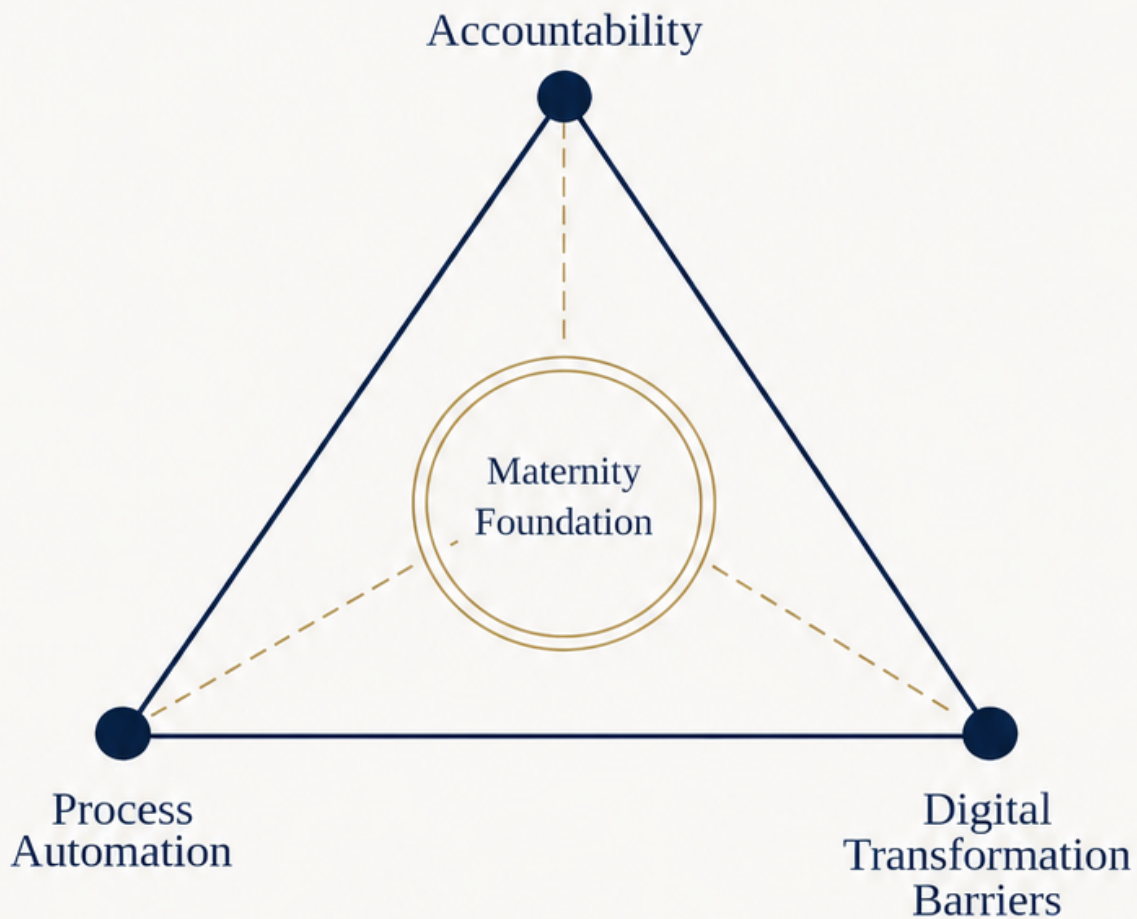


DSR-Study: The Complex Case of Impact Reporting in an NPO



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Abstract

Digital tools and process automation technologies are increasingly promoted as means of improving operational efficiency and accountability. However, the implementation of automation technologies in NPOs remains challenging due to the sector's distinctive organizational structures, accountability demands, and mission-driven value creation. This thesis investigates how an artifact can be designed to support impact reporting in NPOs and what design considerations and principles can be derived from such a design process.

Adopting a Design Science Research (DSR) approach, this study engaged with an NPO seeking to improve a cross-departmental and highly manual impact reporting process. Across three iterative design cycles, the study progressed from an initial ambition of implementing process automation technologies towards the design and evaluation of a proof-of-concept artifact. The final artifact was intended to facilitate cross-departmental discussions regarding the opportunities, limitations, and trade-offs associated with implementing process automation technologies to improve impact reporting. The artifact combined process models, automation concepts, data visualizations, and future-state scenarios to provide stakeholders with a tangible and concrete basis for discussing competing perspectives on impact reporting.

The findings reveal that challenges related to impact reporting are not primarily technological. Specifically, the design process exposed three tensions concerning (1) standardization and flexibility, (2) accountability challenges, as well as (3) process clarity vs organizational flexibility. Based on these findings, three design considerations are proposed to address and acknowledge these tensions: balanced standardization, reflexive impact governance, and process clarification. Furthermore, the study derives three design principles that this study indicates could be useful ways for resolving and mitigating these: multi-stakeholder representation, real-world embedded proof-of-concept artifacts, and visualization of trade-offs.

The study contributes to the literature on digital transformation in NPOs by demonstrating how artifacts can function as communicative and reflective tools for facilitating organizational dialogue, discussion and alignment. More broadly, the findings suggest that establishing a common language may constitute a critical prerequisite for successful digital transformation and process automation initiatives in nonprofit contexts.

AI declaration

ChatGPT – a generative AI-tool – was used in a limited capacity as a supplementary tool during the development of this thesis. Specifically, ChatGPT was used for grammar and language editing, discussing alternative text structures, and assisting in the development of visual elements in the Front Page, Figure 7 and Figure 11. The tool was not used to generate research findings, conduct analyses, or write substantive parts of the thesis. All academic judgments, interpretations, and conclusions were made by the researchers, who assume full responsibility for the final content.

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1. Introduction

Modern technologies such as AI, automation tools, and chatbots promise faster and more efficient ways of working. However, successfully implementing these technologies to support core business processes is no easy task for organizations. Financial constraints, organizational structures, and technical capacity inevitably influence the extent to which an organization can leverage the advantages of digital tools in a meaningful way. For NPOs, the idea of leveraging technological tools to eliminate time-consuming tasks, highlight social impact, and improve decision-making is highly appealing (Kazanskaia, 2025). Unlike for-profit businesses, where competitive advantage is the primary driver, non-profits must justify technology investments in terms of social impact and mission fulfilment. This complex interplay between technology and mission makes the implementation of IT solutions even more challenging. In short, technology can either amplify impact or strain already limited resources (Kazanskaia, 2025).

Adding further complexity, digital transformation processes often alter an organization's value proposition (Vial, 2019). This points to a crucial discussion when NPOs attempt to utilize digital tools to highlight social impact: to what extent should impact reporting follow or define the organization's mission? At first glance, automation tools such as Robotic Process Automation (RPA) appear to be the perfect solution to the growing demand for operational efficiency and impact reporting (Godefroid, 2021; Haddad et al., 2016). However, implementing such automation tools in a meaningful way is by no means simple, especially in NPOs, where practices are often less standardized, value creation follows a different logic, and financial constraints are significant. Furthermore, Radu et al., (2024) argues that there exist 32 specific barriers for NPOs in relation to digital transformation. Specifically, the concept of *communication* is highlighted as a significant barrier but also a possible catalyst for successful initiatives as it enables “a *common language* facilitating bilateral understanding between IT and the rest of the organisation” which is critical for increasing the organizations capabilities and use of digital technologies (Radu et al., 2024, p. 7).

Through the methodological Design Science Research (DSR) framework (Hevner et. Al, 2004; Peffers et al., 2007; Vom Brocke et al., 2020), this thesis explores (1) how to design an artifact to automate a cross-departmental and manual impact-reporting process in an NPO, and (2) how to conceptualize the empirical findings that emerged during the DSR-activities through design

considerations and principles. That is, the ambition is to explore the challenges and implications that emerge when implementing process automation initiatives in NPOs. With this as the outset, the *research question* for this assignment is as follows:

“How can an artifact be designed to support impact reporting in NPOs, and what design considerations & principles can be derived from this?”

1.1. The DSR-Approach – Introduction to our DSR-Journey

Through the adoption of the DSR approach, this study initially set out to implement a somewhat “simple” digital automation tool in an NPO aimed at automating a time-consuming and manual task related to impact reporting. However, by engaging with the practicalities of designing and evaluating the artifact intended to automate their impact reporting process (Gregor & Hevner, 2013), we faced both technical and socio-cultural challenges, which led us to question not only the feasibility of our initial ambition, but also whether designing an implemental solution would be meaningful for the NPO and its mission.

Instead, we embarked on a journey aimed at designing a “proof-of-concept” artifact that was as realistic and aligned with the existing IT infrastructure as possible. This outset served two main purposes: 1) illustrate the concrete, tangible challenges and potentials of implementing a process automation technology to improve impact reporting, and 2) stimulate a cross-departmental discussion – anchored in their actual IT-infrastructure – to illuminate internal cross-department discrepancy regarding potential trade-offs between operational efficiency and supporting the social mission. This specific approach was centred around establishing *common language* regarding their impact reporting process, as “communication is concerned with the methods, strategies and tools that an organisation applies to improve understanding and synergy across different levels.” (Radu et al., 2024, p. 7). As such, our designed artifact can be seen as exactly this – a tool for facilitating communication and a common language.

Through iterative engagement with both the empirical findings and the relevant literature, we then tried to synthesize our empirical findings into prescriptive design considerations and design principles aimed at guiding practitioners on how to address significant barriers when implementing process automation tools to improve impact reporting in NPOs. Rather than viewing the artifact as

the primary contribution, the study thus treats it as a vehicle for exploring and illuminating the organizational, technological, and accountability-related challenges encountered throughout the design process. The following section outlines the theoretical perspectives through which these challenges are conceptualized and interpreted.

1.2. Research Area – Conceptualizing the Empirical Findings

Though one could argue that the isolated ambition of automating a manual and time-consuming would normally be categorized as a *digitalization* process, this narrower focus would not enable us to discuss and interpret the wider challenges and barriers we faced during the evaluations of our artifact. Instead, by conceptualizing the empirical findings through the term of *digital transformation process* (Vial, 2019) – and not merely *digitization* or *digitalization* – it was possible for us to explore the broader strategic, operational and value-creation challenges related to integrating a seemingly “simple” process automation tool in the organization. This concept served as the theoretical foundation of the assignment and was in constant dialogue with specific concepts of accountability in NPOs as well as theories on process automation technologies in NPOs. Taken together, these perspectives provided a foundation for conceptualizing how efforts to redesign impact reporting processes are intertwined with broader organizational questions concerning value creation, accountability, communication, and digital transformation.

While scholars and practitioners have analysed digital transformation processes extensively in recent years, digital transformation processes in the context of NGOs and NPOs remain under-explored (Cipriano & Za 2024; Vogelsang et al., 2021). Further, digital transformation processes in NPOs rarely succeed (Zaitsev & Mankinen, 2022). As Cipriano & Za argues, there exists distinct characteristics in NPOs that must be acknowledged and taken into careful consideration when thinking of implementing new IT-tools. The uniqueness of NPOs– among other things – lies in their commitment to social missions and social impact rather than competitive advantages and for-profit logics as well as relying financially on donations and fundraising. Thus, they are, by nature, committed to building civil society, improving healthcare and social well-being and economic development (Nahrkhalaji et al., 2019) while also being financially dependent on external resources. Balancing these two elements is by no means an easy task.

The constant emergence of digital atomization-tools promises faster and better ways for NPOs to engage with stakeholders, funders and beneficiaries (Rana & Cordery, 2024; Kaba, 2021). However, the growing possibilities of the alluring affordances of new IT-tools might just take away focus from the actual social impact which is the *raison d'être* of NPOs. A key argument is that though digital technologies may seemingly promise to improve transparency and impact reporting, they can also lead to a reinforced funder-centric accountability rather than beneficiary-centred social impact and transform social missions into *calculative performance objects* (Rana & Cordery, 2024) – which can end up changing the initial value-creation structure of NPOs and ultimately minimize the social impact of the NPOs. To accommodate the distinct “peculiarities” (Rana & Cordery, 2024) of NPOs we focus on the concept of developing a *common language* on the potentials and challenges related to process automation in the context of NPOs and impact reporting. As such, the overall focus on (lack of) common language allows us to explore the distinct “peculiarities” of NPOs in beneficial ways. Inspired by the existing research, we aim to build upon it by presenting a *prescriptive design principles* aimed at guiding practitioners on how to develop a *common language* and illuminate specific NPO-tensions relevant hereto.

As such, this study is situated within the complex intersection between (1) the potentials and challenges of process automation in NPOs, (2) balancing social impact with impact reporting and (3) conceptualizing these challenges through the scope of digital transformation processes (Vial, 2019) and (4) the general call for more practical guidance regarding how to approach digital transformation processes as an NPO (Cipriano & Za, 2024). In an ideal word, the alignment of the implementation of process automation with the distinct characteristics of NPOs could optimize both impact reporting, social impact and operational effectiveness.

As our case-study is centred around an organization that is both an NGO (non-governmental organization) and NPO (non-profit organization) we had some difficulties situating our specific academic focus. However, as both NGOs and NPOs belong to the broader non-profit sector (NPS) (Laurett & Ferreira, 2018) we decided to let this sector be the foundation for the assignment and thus both include articles that focus on NGOs and NPOs.

1.3. Thesis Structure

We will now briefly present the structure of the thesis. Through the adoption of the DSR-approach we decided to let the structure presented by Gregor & Hevner (2013) inspire the structure of the assignment. As such, this (1) *Introduction*-section will be followed by (2) a *Literature Review* with specific emphasis on the concepts of *digital transformation in NPOs*, *RPA in NPOs*, and *accountability*. Then (3) we will describe relevant concepts of the DSR-approach and how these informed our analysis in *Methodology and Methods*. This section will also include descriptions of the different methods we used to support the DSR-approach. After this we will describe our approach for collecting data in the section (4) *Data Collection*. This section will include an overview of all the empirical data we collected, and the organizing and coding of the interviews. Fifthly, we will present the (5) *Case Organization*. We will then dive into the (6) *Analysis* which consists of three iterations conducted in accordance with the DSR framework. The central empirical findings from the three DSR-iterations will be conceptualized in (7) *Design Knowledge – Deriving Design Considerations & Principles* through critical dialogue with relevant concepts from the literature. In this section we will further present our prescriptive design considerations and principles aimed at supporting the development of a *common language* to potentially enable meaningful automation of impact reporting processes in NPOs. The overall methodological and theoretical implications will be discussed in the (8) *Discussion*. Lastly, we will relate our findings to the research question in the (9) *Conclusion*.

2. Literature Review

As the DSR approach is an iterative and cyclic methodology, centred around creating and evaluating artifacts to solve real-world problems (Gregor & Hevner, 2013), we tried to conduct the literature review in a seemingly iterative way. After exploring different strategical approaches to conducting literature reviews, we found that the article by Webster & Watson (2002) best complimented our iterative methodology by being concept-oriented and thus not being too rigid or static. Webster & Watson (2002) argue that a literature review should *synthesize* the literature and not just describe it while also highlighting that we must strive to “identify critical knowledge gaps and thus motivate researchers to close this breach.” (Webster & Watson, 2002, p. 19). Thus, a literature review points to a discrepancy between what we know and what we need to know by presenting the “state of the field”. Through constant iterative dialogue between the empirical findings and relevant literature conceptualizing these findings, we decided to organize the literature review into 3 overall *concepts*:

- Digital Transformation Processes & barriers in NPOs
- Process Automation and RPA in the Context of NPOs
- The Complexity of Accountability in NPOs

These concepts thus both informed us throughout the DSR-activities specifically centred around designing and evaluating our artifact. Furthermore, these concepts constituted highly beneficial perspectives for interpreting and contextualizing the empirical findings in the design knowledge-section - centred around deriving more general insights and generating *prescriptive knowledge* through *design principles*.

2.1. Concept 1: Digital Transformation Processes in NPOs

2.1.1. Digital Transformation Processes – a Framework

The aim of this section is to situate the report in relation to the concept of digital transformation processes, as this concept was seen as a beneficial lens for conceptualizing the challenges and barriers that we faced in several of the DSR-activities. The literature on digital transformation is vast and complex (Carroll, 2020), and we therefore used a large amount of time to situate our report within a relevant area of the literature. The article *Understanding digital transformation* (Vial, 2019) proved a reasonable starting point as it reviews 282 works on digital transformation within IS (Information

Systems) research. Additionally, this conceptualization of digital transformation (DT) has gained much scholarly attention making it even more relevant as relevant empirical studies have adopted this framework (Cipriano & Za, 2024; Radu et al., 2024). On a conceptual level, Vial defines digital transformation as “a process wherein organizations respond to changes taking place in their environment by using digital technologies to alter their value creation processes.” (Vial, 2019, p. 119). This can be understood as a process centered around improving an entity (a process, a service, a department), through digital tools which concurrently also changes the characteristics and features of this entity (Vial, 2019). In other words, the concept of digital transformation is understood as a process that *optimizes* an organizational entity (e.g. automating a manual process) while also focusing on how this changes the characteristics of the entity (employee roles, standardization, value proposition) – the *transformation*. In the article, Vial further introduces a framework for conceptualizing digital transformation - consisting of 8 “building blocks”. In summarization, digital transformation is conceptualized as a:

“process where digital technologies create disruptions triggering strategic responses from organizations that seek to alter their value creation paths while managing the structural changes and organizational barriers that affect the positive and negative outcomes of this process.”
(Vial, 2019, p. 118).

Though somewhat broad, this framework of *building blocks* proved beneficial for conceptualizing the different challenges we faced during the different DSR-activities. Specifically, we found it useful as it offers a perspective of the different implications of digital transformation on multiple levels – and that “technology itself is only part of the complex puzzle” which entails strategy, culture, process-understanding, structure etc. (Vial, 2019, p. 118). All these elements must be taken into careful consideration to yield the capability to generate new paths for value creation.

Beyond the conceptual definition, Vial also introduces a figure to illustrate the inductive framework centered around the eight building blocks previously mentioned (Figure 1).

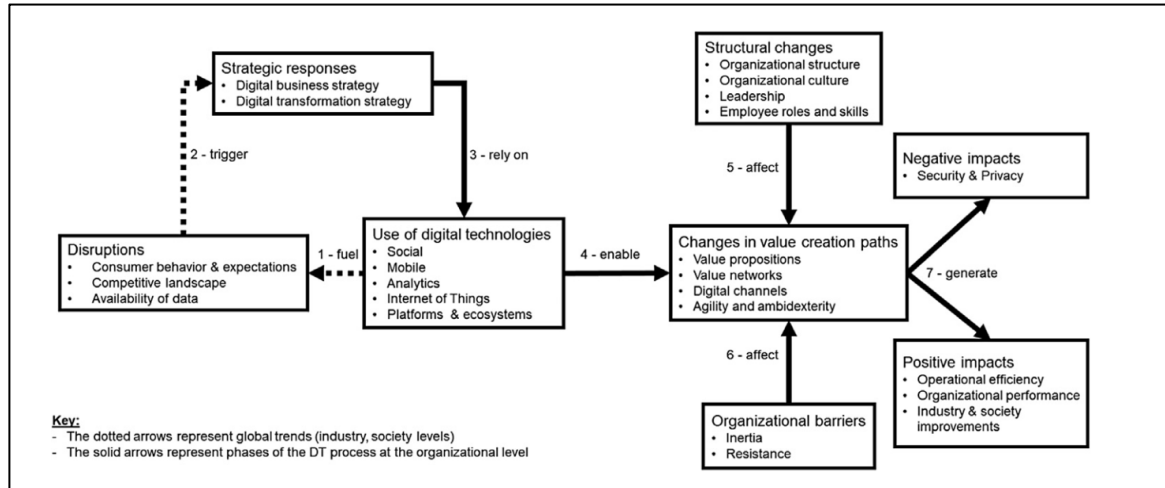


Figure 1: Building Blocks of the DT process (Vial, 2019, p. 122)

This broad conceptualization was beneficial for understanding the challenges and complexities we faced during the different DSR-activities of this study. However, we also found articles from the literature that suggest that this framework might not capture all the complexities relevant for understanding digital transformation processes in the context of NPOs (Vogelsang, et al., 2021; Radu et al., 2024; Cipriano & Za, 2024). In the following sections we will therefore outline the specific characteristics of NPOs regarding (1) *barriers to digital transformation in NPOs* (Vogelsang, et al., 2021; Radu et al., 2024) and (2) *strategic responses* and its relation to *changes in value creation paths* (Cipriano & Za, 2024).

2.1.2. Barriers to Digital Transformation in NPOs

Digital technologies promise countless benefits across sectors - ranging from improvements in business processes and customer services to execution of innovative business models (Vogelsang, et al., 2021; Radu et al., 2024). For NPOs specifically, digital technologies promise enhanced “transparency, accountability, and stakeholder relationships by facilitating more inclusive, high-quality, and impactful provision of services to beneficiaries” (Radu et al., 2024, p. 1). However, barriers often hinder the realization of these promises – “even in leading countries like Sweden” (Radu et al., 2024, p. 1). While prior studies have extensively explored the mechanisms related to digital transformation in the for-profit and public sector, the specific challenges that organizations from the NPO-sector face remain largely under-explored (Radu et al., 2024). This is problematic considering the significant differences between NPOs and organizations in the private and public sectors.

Through a thematic analysis of the – relatively scarce – literature on the concept, Radu et al. (2024) identify 17 barriers significant to the NPO-sector. In addition, through a case-study of a Swedish NPO they add 15 new barriers uniquely relevant for the NPO-context. These 32 barriers are represented in the below figure 2:

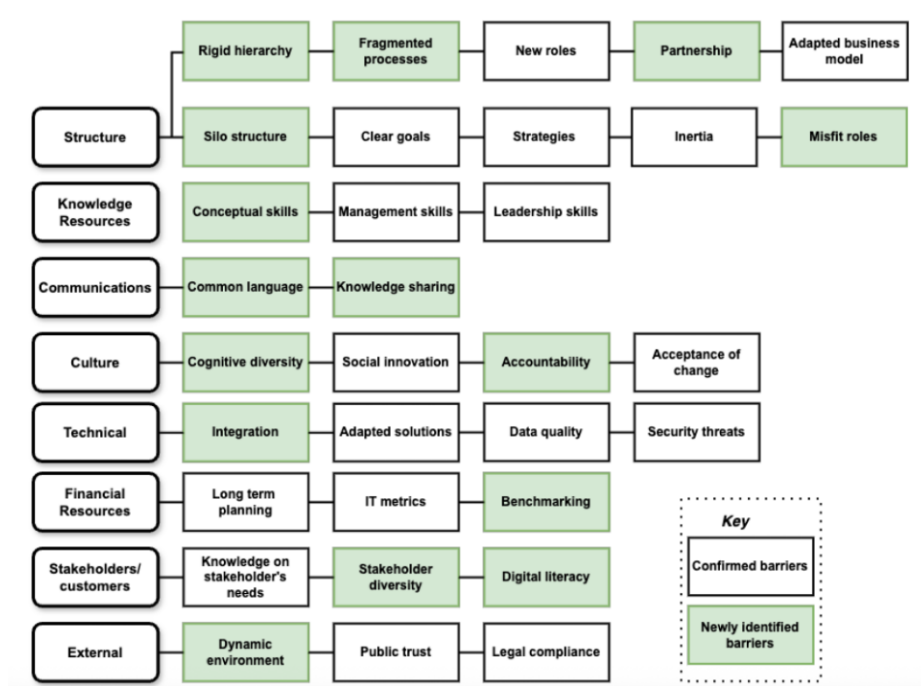


Figure 2: 32 DT Barriers in NPOs (Radu et al., 2024, p. 6)

The 32 barriers are grouped under 8 categories; structure, knowledge resources, communications, culture, technical, financial resources, stakeholders/customers and external. The relationships between these are further conceptualized through figure 3.

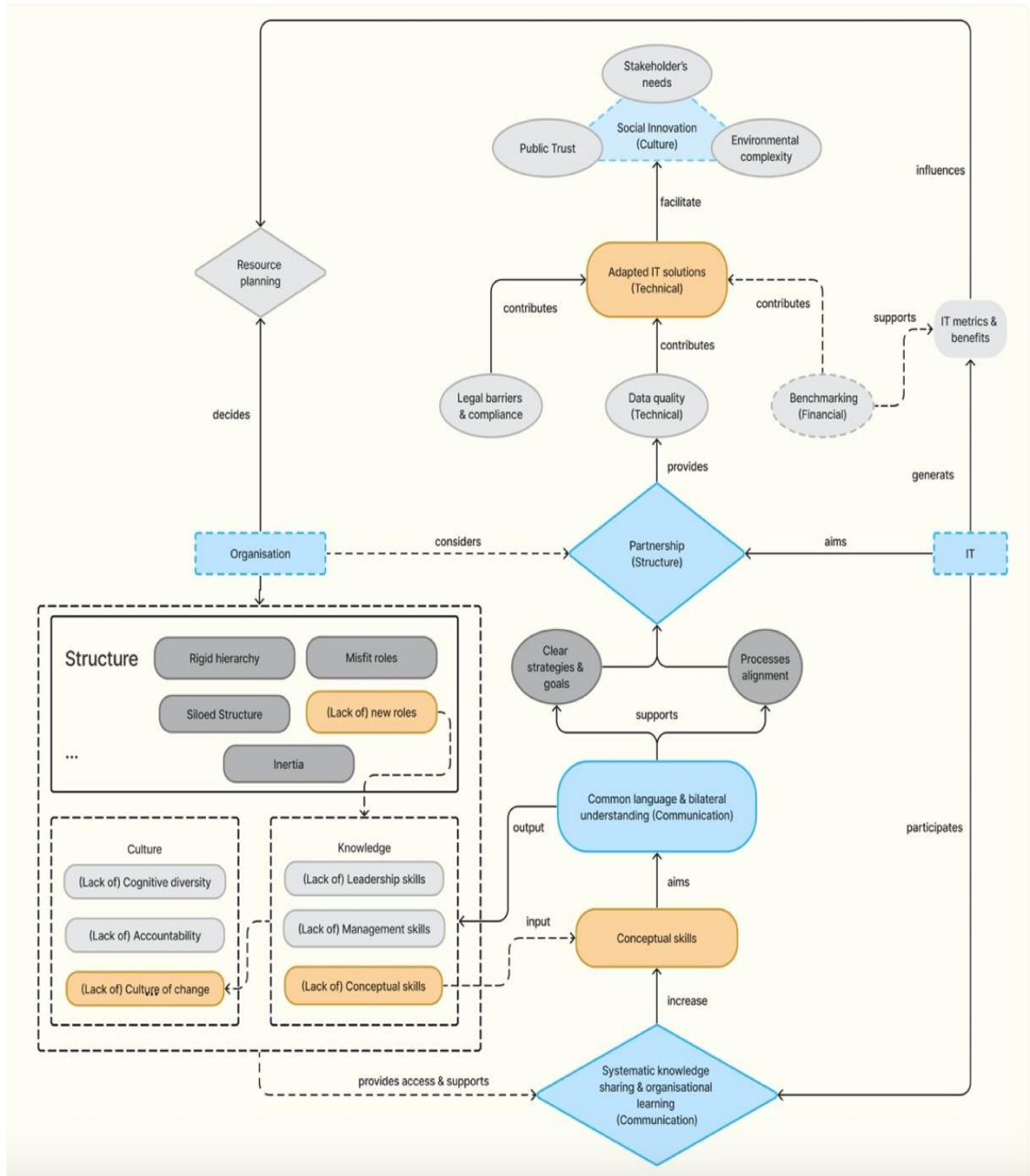


Figure 3: Barriers to digital transformation in NPOs (Radu et al., 2024, p. 10)

Though we acknowledge that this framework is not a finished end-product due to the limited amount of literature on the topic – specifically, only 8 articles – it proved highly helpful for understanding and conceptualizing the specific challenges we faced during our interactions with our case organization and the design and development of our artifact. By recognizing these potential barriers, we could take somewhat well-coordinated measures to inform both the definition of the problem (activity 1), the objectives of our solution (activity 2) and design and development (activity 3) of the adopted DSR-approach.

Yet, the focus of this study is not to confirm, deny or nuance these barriers. Instead – in accordance with the prescriptive focus from the DSR-approach – we aim to provide concrete guiding principles on how to support development of a *common language* beneficial for digital transformation in NPOs.

By focusing on supporting a common language through a DSR artifact, we attempt to also unveil some of the specific structural, cultural, and knowledge-related barriers identified in Figure 3 within Maternity Foundation. However, while these other barriers were also highly relevant throughout the study, our primary interest lies in facilitating a *common language* and *bilateral understanding*, which Radu et al. (2024) position as a central enabler of digital transformation. As depicted in Figure 3, it both shapes the wider organizational context and supports the development of clear strategies, goals, and process alignment. This makes it particularly relevant for a DSR study aimed at designing artifacts that can contribute to facilitating digital transformation in NPOs.

During our engagement with the organization, we found that they are currently developing a 5-year digital strategy plan, which made this focus – on shared language – highly relevant. Vogelsang et al. (2021) argues, that organizations in the NPO-sector “would benefit if they rise to the challenge and make having a digital strategy a long-term corporate focus.” (p. 307). As Radu et al. (2024) further describes, researchers and practitioners can “leverage these insights to inform the formulation of effective planning and execution of their digital transformation initiatives.” (Radu et al., 2024, p. 1). In accordance with the overall purpose of DSR as addressing “important unsolved problems in unique or innovative ways” (Gregor & Hevner, 2013) we thus develop an artifact that attempts to lay the foundation for bilateral understanding to support clear strategies and process alignment going forward.

2.1.3. The Complex Value-Creation Structure in NPOs

Though there is a broad acknowledgement that NPOs face difficulties in relation to digital transformation outcomes due to lack of resources and IT skills, and resource fluidity (Brink et al., 2020) a more precise discussion regarding the strategic approach to digital transformation in NPOs and NGOs is still needed (Cipriano & Za, 2024). A key argument is that organizations from the non-profit-sector differ from for-profit and public organization as their value-creation isn't centered around financial aspects or competitive advantage (Laurett and Ferreira, 2018). Instead, their value-creation structures are centered around social impact and missions. This gives rise to questions

regarding if it is meaningful to view digital transformation processes in NPOs through a framework mainly oriented towards for-profit and public organizations.

In the article by Cipriano & Za (2024), they identify characteristics describing DT processes in an NPO-context and in addition discuss if it makes sense for NPOs to adopt strategies for digital transformation centered around non-NPS organizational structures – such as for-profit and competitive advantages. They argue for a better understanding of how digital transformation alters the value-creation structure as they dynamically evolve and mutate as organizations engage with DT. Based on their taxonomy-analysis they introduce five dimensions relevant to consider when developing a strategy for digital transformation in NPOs.

- Digital scope (what technology)
- Organizational scope (the organizations main ambition in exploring a DT initiative)
- Organizational involvement (position of the organization in its business environment)
- Organizational element (business element involved in the DT initiative)
- Operational level (everyday practices and workflow)

These dimensions are described as fundamental for developing a strategy that drives the DT process, as they highlight the importance of *“aligning, integrating, and harmonizing all components of an NPO, as they could simultaneously generate synergetic stimuli and effects”* (Cipriano & Za, 2024, p. 750).

This framework informs this assignment as it specifically highlights the complex value-creation structure of organizations within the NPS-sector. Subsequently, this informed us of the importance of balancing operational efficiency with the social mission – the *raison d'être* - during the design and development of a meaningful artifact.

2.2. Concept 2: Process automation and RPA

Building on the broader discussion of implementing digital technologies in NPOs, process automation has emerged as a practical mechanism for translating digital ambitions into operational change. Among the technologies associated with this shift, Robotic Process Automation (RPA) has gained increasing attention due to its ability to automate repetitive and rule-based tasks across existing IT

systems, making it particularly relevant to the initial understanding of the organizational problem addressed in this study. As Blomkvist and Hansen (2025) argue, RPA should not merely be understood as an efficiency-enhancing tool, but also as a potential catalyst for broader digital transformation through the strategic alignment of technological and organizational factors.

With this point of departure, the following sections draw upon the existing literature to conceptualize RPA as a concept within the broader concept of process automation, while simultaneously examining the organizational implications of RPA implementation within NPOs and its broader role in facilitating digital transformation processes.

2.2.2 Robotic Process Automation

According to Dumas et al. (2018), process automation can broadly be understood as the automation of routine work embedded within business processes, ranging from the automation of simple tasks to the execution and coordination of more complex workflows. Within this broader understanding, Robotic Process Automation (RPA) represents one among several technological approaches to *process automation* (Hofmann et al., 2020).

Drawing on multiple contributions within the literature, this thesis conceptualizes RPA as a software-based technology that interacts with the user interfaces of existing information systems and employs predefined business rules and activity choreography to autonomously execute rule-based processes, activities, transactions, and tasks across one or multiple software systems (Aalst et al., 2018; Willcoks et al., 2015; Hofmann et al., 2020). Willcoks et al. (2015) further emphasize that RPA is particularly suited for so-called “swivel-chair” processes, in which employees manually extract data from one system, process the information according to predefined rules, and subsequently transfer the data into another system. Consequently, the primary ambition of RPA is to automate repetitive, time-consuming, and highly standardized tasks traditionally performed by human actors within organizations. Our initial understanding of their problem matched this description of an ideal RPA use-case, however as we unfold in the analysis the identified problem evolved to a less ideal scenario for an actual RPA implementation.

Unlike traditional “inside-out” approaches to information systems development, RPA approaches automation in an “outside-in” manner (Aalst et al., 2018, p. 269). The “outside-in” feature benefits

from existing infrastructure, rather than requiring substantial infrastructural redesign, RPA operates on top of existing digital infrastructures by interacting directly with graphical user interfaces. However, this also means that RPA is highly dependent on the quality, stability, and standardization of the existing infrastructure and underlying processes on which it operates. A dependency that can have potential challenges for RPA implementations, especially in NPOs where data is often unstandardized (Radu et al., 2024).

At the same time, this infrastructural orientation also differentiates RPA from more backend-oriented automation technologies. For example, the Microsoft Azure ecosystem enables process automation through ETL (Extract, Transform, Load) pipelines using technologies such as Azure Data Factory (ADF) (Kromer, 2022). In contrast to RPA, ADF operates closer to the infrastructural and data architecture layers of organizational systems and can therefore be characterized as a more “inside-out” automation approach to *process automation*. While both technologies support process automation and data handling, the primary distinction lies in their infrastructural orientation: ADF relies on backend integrations and data architecture, whereas RPA automates processes through user-interface-level interactions.

This distinction is analytically important because it reflects two fundamentally different approaches to process automation. Whereas backend-oriented automation often requires greater infrastructural maturity and technical integration, RPA is frequently framed as a more lightweight automation alternative associated with lower implementation costs and limited infrastructural requirements (Herm et al., 2022; Dumas et al., 2018; Hofmann et al., 2020). However, while some literature highlights the lightweight features of RPA, RPA may come at the cost of scalability and robustness, as it remains dependent on user-interface interactions rather than deeply integrated system architectures (Nunes et al., 2026). The distinction between “inside-out” and “outside-in” will serve as a central analytical lens for understanding infrastructural challenges to process automation within Maternity Foundation.

2.2.3 Implementing RPA

The implementation of RPA has received increasing scholarly attention due to its potential to improve operational efficiency and reduce manual workload across organizational processes. However, despite the growing popularity of RPA technologies, existing research consistently highlights that

successful implementation depends on a complex interplay between organizational, technological, and managerial factors. Indeed, several studies estimate that a substantial proportion of RPA initiatives fail due to inadequate implementation strategies, poor organizational alignment, and insufficient process maturity (Herm et al., 2022; Nunes et al., 2026).

Across the literature, one of the most frequently highlighted causes of unsuccessful RPA implementation concerns inadequate process selection during the early analysis phases of automation projects (Hofmann et al., 2020; Nunes et al., 2026). Thus, we spent much effort on process selection in the early phases of the DSR methodology - aimed at identifying a feasible process. Since RPA is particularly effective when applied to repetitive, rule-based, and highly standardized processes, implementation challenges are likely to emerge when organizations attempt to automate processes characterized by ambiguity, inconsistency, or weak process formalization (Nunes et al., 2026).

These findings suggest that successful RPA implementation depends heavily on organizational process maturity. Consequently, several scholars argue that RPA implementation should not merely be understood as a technical exercise, but rather as an organizational transformation process requiring process formalization, governance structures, and cross-functional coordination (Hofmann et al., 2020; Herm et al., 2022). Therefore, in this assignment RPA will not only be discussed and analysed through technical requirements but in relation to the concept of digital transformation – to address potential barriers, such as a lack of common language and bilateral understanding, when implementing RPA in the organization.

Overall, these implementation challenges and opportunities may become even more pronounced within NPOs, where limited technical capabilities, constrained financial resources, and heightened accountability requirements often shape organizational decision-making processes in ways that differ from private-sector contexts.

2.2.4 RPA implementation in the context of NPOs

While the implementation of RPA has been widely explored in private-sector organizations, its application within NPOs remains significantly underexplored. In the context of NPOs, lightweight and cost-efficient digital technologies such as RPA seemingly represent an opportunity for NPOs to streamline repetitive processes without requiring major changes to existing IT infrastructures

(Godefroid, 2021). However, research also highlights how NPOs continuously face trade-off decisions between timely and cost-effective operational performance and the need to fully serve society (Haddad et al., 2016). This tension adds a degree of complexity to the implementation of RPA and other process automation technologies in NPOs, as such technologies should not only improve operational efficiency but also clearly contribute to the NPOs mission to justify their cost.

Despite the growing body of research on RPA implementation, limited attention has been devoted to how RPA can be implemented within NPO contexts. Consequently, the tension between operational efficiency and mission contribution in relation to RPA implementation remains underrepresented in the existing literature. Godefroid explicitly notes that RPA solutions, although increasingly adopted in the private sector, “do not seem to appear” in current NPO research (Godefroid, 2021, p. 24). However, recent nonprofit literature has begun emphasizing the potential of automation technologies to optimize resource-intensive administrative tasks such as grant proposal development and donor management (Imediegwu & Elebe, 2023; Godefroid et al., 2023a). Notably, these specific use-cases differ from our focus on *accountability* and *impact reporting* by not being directly connected to the social missions of the NPO.

Furthermore, there is limited research investigating how scalable and lightweight automation solutions – such as RPA – can potentially support NPO with impact reporting. Prior research highlights the need for more empirical and practice-oriented studies examining the implementation of RPA solutions and process automation within NPO contexts (Godefroid, 2021). Thus, there remains a lack of design-oriented research exploring how process automation and RPA solutions can be practically designed and implemented within NPO environments to improve impact reporting.

2.3. Concept 3: Accountability in NPOs

The third concept that emerged as highly relevant during the development and evaluation of our artifact was the concept of impact reporting and accountability in NPO. NPOs are central actors in many societal affairs. They manage billions of dollars and work around the world to address humanitarian, social, health, and environmental issues. It is therefore no surprise that the concept of accountability within NPOs has attracted increasing scholarly attention (Kaba, 2021; Rana & Cordery, 2024).

Although there seems to be broad agreement on the value of exploring accountability in NPOs, Kaba (2021) describes the concept as “chameleon-like and ambiguous.” By reviewing 217 articles on accountability in NPOs, Kaba finds that there are 113 different conceptualizations of the term. He further explains that the most common conceptualization of accountability revolves around an “obligation to explain and justify conduct to stakeholders” (Kaba, 2021, p. 1). He also argues that a general distinction within the concept lies between *upward accountability* (toward funders, donors, and governments) and *downward accountability* (toward beneficiaries and communities). Elaborating on this distinction, upward accountability often dominates because funding relationships create strong pressure for measurable performance metrics that can legitimize the funding (Kaba, 2021).

Furthermore, accountability in NPOs is often conceptualized relationally, meaning that the organization is accountable to someone (donors, funders, beneficiaries, etc.) for *something* (finances, ethical conduct, mission fulfilment) through particular *mechanisms* (reports, evaluations, performance metrics, etc.). Cordery et al. (2023) describe how digital transformation in NPOs creates both opportunities and challenges that affect and transform how accountability is exercised. This will be elaborated on in the following section.

2.3.1 Accountability for NPOs in the Digital Era

Scholarly focus on accountability in relation to digital transformation in the context of NPOs and NGOs has only marginally been explored – with much of the focus being oriented towards the public and private sector (Cordery, et al., 2023). By drawing on the inherent complexity of accountability in NGOs they argue that there exist a “pressing need to provide answers to the questions of whether and how digital transformation can facilitate improvements in NGO accounting” (Cordery, et al., 2023, p. 2). They further argue that the vast increase in the use of digital systems have created stronger expectations from donors, governments and funders for measurable impact and report performance. As a result, NPOs often has the ambition to rely on predefined and standardized metrics and reporting systems. Through Vial’s (2019) conceptualization of *digital transformation* (figure 1), this can be seen as a *disruption* in stakeholder expectations within the sector which *trigger strategic responses* that increasingly *relies* on the use and implementation of *digital technologies*. In other words, a self-reinforcing cycle. While the implementation of digital technologies offers the potential to optimize accountability through automation techniques, real-time reporting and enhanced organizational

learning this also creates a challenge in relying significantly on quantifiable metrics and a loss of contextual and qualitative understandings and dimensions of the social impact (Cordery, et. al., 2023).

2.3.2. Performative Calculative Practices

Expanding on this challenge, Rana & Cordery (2024) argue that there is no such thing as a neutral technological upgrade aimed at improved efficiency and performance. Different digital technologies and accounting systems are not merely descriptive. Instead, they argue that they are inherently *performative* and that they also actively shape and reshape what “counts” as a social impact or a successful mission (Rana & Cordery, 2024). Thus, specific metrics do not simply represent and measure impact. Rather, they shape organizational behaviour and redefine priorities by actively shaping what counts as a successful mission. They never simply ‘observe’. Instead, they actively participate in the creation of organizational realities by defining and constructing what appear as legitimate. Accordingly, the notion of social impact – the very *raison d’être* of NPOs– risks becoming subordinated or secondary to digital calculative practices and its inherent economically and comparable logics. This is where they introduce the concept of *calculative practice* (Rana & Cordery, 2024). This term critically conceptualizes what happens when NPOs simplify complex social realities into measurable outcomes and standardized metrics which reshape and organize the behaviour of the organization. In other words, NPOs adapt their practices and structure to what the digital systems reward and can measure. The performative aspect lies exactly herein – the systems help create the reality they claim to measure. A central challenge thus present itself when central aspects of what NPOs do cannot necessarily be quantified. During the different activities of our DSR-study we try to reflect upon these complexities as often as possible to avoid subjecting ourselves and our solution to the alluring potentials of process automation.

The digital era complicates accountability by embedding NPOs within data-driven and calculative infrastructures that privilege measurable forms of performance, potentially marginalizing qualitative, relational, and context-dependent dimensions of social missions.

2.4. Literature Synthesis – Situating our Contribution

Research on digital transformation is dominated by private-sector assumptions regarding value creation, competitiveness, and organizational performance, while NPO-specific approaches to digital

transformation remain underexplored (Cipriano & Za, 2024; Vogelsang et al., 2021). Existing studies therefore provide limited guidance regarding how NPOs can approach digital transformation initiatives while balancing operational efficiency with mission-oriented value creation and the inherent complexity of accountability requirements. Additionally, although RPA and process automation technologies have received substantial scholarly attention, the existing literature remains heavily focused on private-sector implementation contexts (Godefroid, 2021). Consequently, there is limited empirical and design-oriented research examining how lightweight automation technologies – such as RPA – can be practically designed and implemented within resource-constrained NPO environments. Furthermore, the implications of implementing RPA's (Nunes et al., 2026; Willcocks et al., 2015; Hofmann et al., 2020) to achieve operational efficiency and the potential trade-offs remain underexplored in NPO and NGO contexts (Haddad et al., 2016). This is particularly interesting in relation to impact reporting and the extent to which standardized automation infrastructures can support capturing and communicating NPOs goals, mission fulfilment, and social impact.

While frameworks such as Radu et al. (2024) provide valuable conceptualizations of the interconnected barriers challenging digital transformation initiatives in NPOs, they remain mostly explanatory of what barriers exist and how they relate to one another. In particular, the framework (Radu et al., 2024) highlights how technical initiatives are deeply entangled with organizational structure, communication practices, conceptual competencies, and process alignment. Yet, limited research translates these interconnected barriers into concrete guidance for how to address and overcome these barriers within the context of NPOs.

Finally, while recent literature increasingly highlights the risks associated with performative calculative practices and metric-driven accountability systems in NPOs (Cordery et al., 2023; Rana & Cordery, 2024), limited research provides practical guidance regarding how automation initiatives can be designed and strategically governed in ways that balance scalability and impact reporting efficiency with contextual, mission-oriented, and beneficiary-centred forms of accountability.

The theoretical perspectives elaborated above, motivates the present study, which adopts a Design Science Research approach to design and evaluate how an artifact can contribute to both illuminating

NPO-specific tensions in the context of impact reporting while simultaneously deriving design considerations and principles.

3. Methodology & Methods

The following section will outline the DSR framework as well as relevant complimentary methods. The paradigm of DSR will guide us throughout the analysis as an approach embedded in the problem-solving paradigm oriented towards the creation of artifacts aimed to solve real-world problems (Vom Brocke, et. al, 2020). To clarify, the DSR framework is understood as a *methodology* while Process Portfolio, BPMN, Devil’s Quadrangle etc. are understood as *methods* useful to inform the DSR-methodology.

3.1. Introduction to Design Science Research

The DSR approach seeks to enhance human knowledge through the creation of “innovative artifacts” aimed at solving real-world problems (Vom Brocke, et. al., 2020). It originates from engineering and sciences of the artificial but has since increasingly been applied in research within information systems (IS) (Vom Brocke, et. al, 2020). This research paradigm has gained interest in the past years due to its potential in fostering tangible transformational change in organizations as well as in society (Vom Brocke, et. al, 2020). An overall argument for the strength and relevance of DSR as a framework lies in the pragmatist viewpoint that “scientific research should be evaluated in light of its practical implications.” (Hevner et. al, 2004, p. 77). In other words, the core foundational idea behind the DSR-approach lies in its close practical connection with the “real-world” and its inherent ambition to create knowledge that “aids in the productive application of information technology to human organizations and their management.” (Hevner et. al, 2004, p. 76).

Subsequently, the DSR-approach was adopted in this thesis due to its pragmatic orientation towards understanding *real-world* problems and interacting with these – instead of theorizing over these from afar – while also encouraging a focus on people, organizations and technology in combination (Hevner et. al, 2004). And not just an isolated focus on technology. Thus, the DSR-approach was deemed a beneficial perspective to adopt in order to actually try to solve the presented problem by designing an artifact and use this as a stepping stone for generating productive knowledge about the complexity concerning how process automation can be implemented in NPO-contexts.

The following section outlines how the DSR approach was operationalized within this study. The organizing of the following section is inspired by the activities represented in the DSR-process model (figure 4) (Vom Brocke, et. al., 2020). The description of these six activities is followed up by a section on the concept of *design knowledge*.

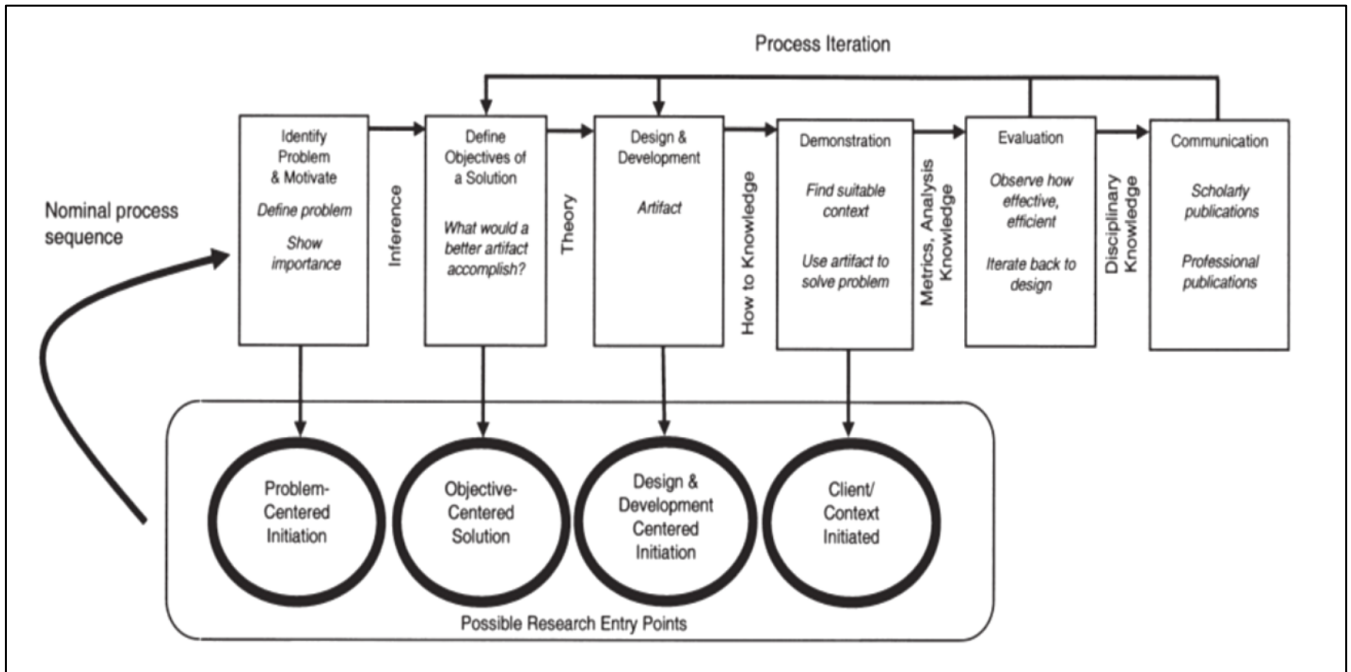


Figure 4: DSR Methodology Process Model (Adapted from Peffers et al. (2007))

It is important to again note the iterative nature of the DSR-approach. This is elaborated by Sonnenberg & Vom Brocke's concept of *Concurrent evaluation* (2012) through which they argue of the importance of continuously evaluating and assessing each of the activities – to refine both the artifact and the problem-understanding.

Before we describe our specific DSR-inspired approach, it is relevant to highlight that all the different activities are informed by both relevant elements from the literature – *Knowledge Base* – and the empirical data, technological possibilities and organizational setup – *Environment*. This is conceptualized in figure 5 (Hevner et. al, 2004).

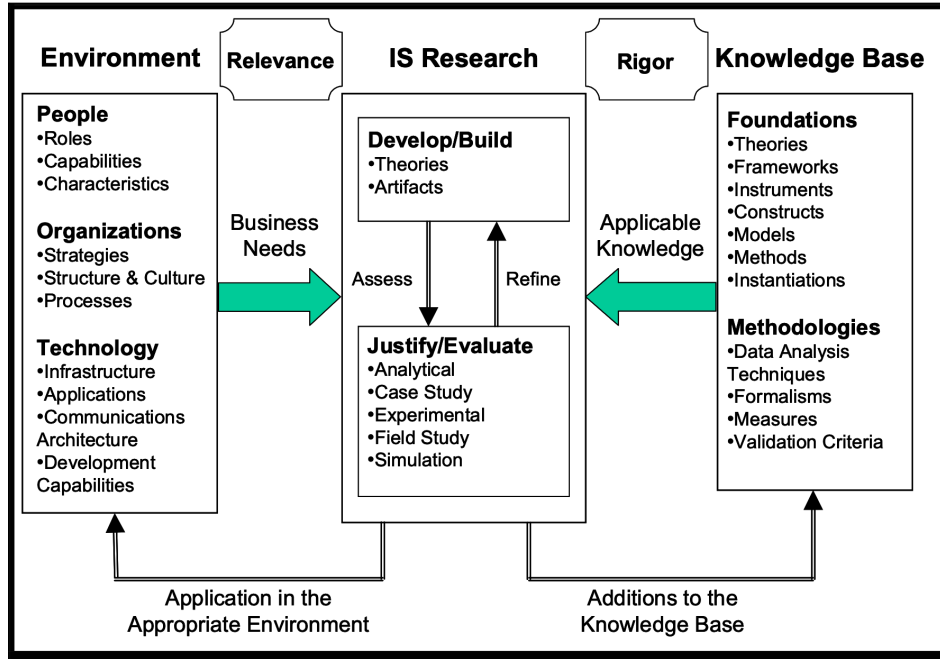


Figure 5: DSR Activities

As such, we tried to balance relevant and important insights from both the *Knowledge Base* and from the *Environment* to ensure both *relevance* and *rigor* during the development of the artifact (Peppers et. al., 2007).

3.1. Activity 1 – Problem Identification & Motivation

The first activity of the DSR-framework is centred around identifying a problem and simultaneously motivating a solution (Vom Brocke, et. al, 2020). In this activity we explore *how* to frame the problem and *why* this problem is important to address. Additionally, we expand on why an artifact is needed and what an artifact might accomplish in relation to the problem. During this activity we sought to get a thorough and comprehensive understanding of the organization and its structures and processes as well as the different perspectives of relevant stakeholders. As Peppers et. al., (2007) argue, the problem identification is central for the latter development of an artifact, and it is therefore often useful to “atomize the problem conceptually so that the solution can capture its complexity.” (p. 52). To make sure to grasp as much of the complexity surrounding the problem we conducted a BPM-analysis and a Process Portfolio Matrix (Dumas, et. al., 2018) to inform this activity. Due to the

inherent complexity of understanding the problem we had to revisit our understanding of the problem several times before deciding on a feasible and relevant problem-identification.

3.1.2. Activity 2 – *Define Objectives of a Solution*

Importantly, identified problems do not translate directly into objectives for the artifact to achieve (Peppers, et. al., 2007). Therefore, the second activity is centred around defining objectives of a solution in combination with knowledge regarding what is actually possible, realistic and feasible. During this activity we found that in order to define relevant and feasible objectives we needed knowledge about how the problem is perceived by different stakeholders, how their current IT-infrastructure is organized, why previous attempts have failed and how the general operation of the organization functions.

These objectives can both be qualitative and quantitative – and though we initially had a quantitative focus on improving time and quality of the process we ended up shifting focus towards more qualitative objectives centred around using the artifact as a ‘proof-of-concept’-artifact aimed at facilitating a critical dialogue with the organization that both address organizational alignment, their inherent IT-infrastructure, the complexity of addressing both impact reporting and social impact (mission-driven). We hereby address the question regarding “what the artifact needs to accomplish?”.

3.1.3. Activity 3 – *Design & Development*

This third activity deals with the creation of the artifact and includes both determining the desired functionality as well as the actual creation of the artifact – in accordance with the findings in activity 1 and 2. Seemingly, this raises the question regarding what an artifact can be? As Hevner broadly describes, an artifact can be constructs, models, methods, or instantiations as well as “new properties of technical, social, and/or informational resources” (Peppers et al., 2007, p. 49). Furthermore, he argues that – from a conceptual perspective – an artifact can be any designed *object* in which a research contribution is embedded in the design (Peppers et al, 2007).

In our case, the artifact was at first intended to be oriented towards being an *instantiation* but after encountering both technical, IT-infrastructure and broader social and cultural challenges we decided that our artifact should be a ‘proof-of-concept’ rooted in their natural IT-environment aimed at

facilitating a cross-departmental discussion regarding how to solve the barrier of (lack of) common language and bilateral understanding.

3.1.4. Activity 4 – *Demonstration*

The fourth activity concerns the *demonstration* of the artifact. This activity encourages us to *demonstrate* how the artifact can solve elements of the identified problem (Hevner, 2004). It shows how the artifact *can* be used and how it could be applied to address specific problems by running scenarios and explaining workflows. Hereby it differs from the *evaluation* as not being formalized and explicit assessment oriented.

3.1.5. Activity 5 – *Evaluation*

The activity of *evaluation* was the root of much reflexion in this study. As Gregor & Hevner (2013) describes, it concerns the assessment of how well the artifact supports a solution to the problem it is oriented towards. The challenge for our study regarding evaluation centred around deciding on relevant measurement and analysis techniques for assessing our ‘proof-of-concept’-artifact. We struggled quite a lot with finding a meaningful way of evaluating the artifact as the problem it set out to solve progressively changed over time. In other words, how can we assess if the artifact manages to resolve some of the tensions related to the specific problem? What counts as success and improvement? Inspired by the Framework of Evaluating DSR (FEDS-framework) by Venable et, al. (2016) we decided that our artifact should be evaluated through a qualitative, formative, naturalistic evaluation revolving around whether it facilitated reflections, enabled discussions and help stakeholders reason regarding the complexity about scaling and standardization in impact reporting.

To assess this, we decided to do a workshop-style discussion where we first demonstrated the artifact and highlighted its potentials and challenges followed up by a group-discussion. Since the artifact addressed organizational and cultural tensions rather than a purely technical problem, the evaluation thus focused on its ability to facilitate discussion, reflection, and *common language* among stakeholders. As Gregor & Hevner (2013) describes, an evaluation “may draw from many potential techniques, such as analytics, case studies, experiments, or simulations” (Gregor & Hevner, 2013, p. 351). As such, there is no single correct way to evaluate an artifact as it is dependent on the specific context and problem it sets out to solve.

Following Venable et al.'s (2016) extended DSR evaluation framework, the evaluation strategy adopted in this study can be characterized as a predominantly naturalistic and formative ex post evaluation. The evaluation was conducted after the development of the artifact and focused on assessing a proof-of-concept rather than a fully implemented solution. Consequently, the evaluation did not seek to measure operational performance through quantitative indicators. This approach was considered appropriate given the exploratory nature of the project, the developmental stage of the artifact, and the practical constraints associated with implementation within the timeframe of the study.

In our case, we incorporated elements of the two descriptive evaluation methods *informed arguments* and *scenarios* (Hevner et al., 2004). The informed argument method refers to the use of information from the knowledge base (e.g. bilateral understanding and communication, calculative performance, RPA, accountability etc.) to build a convincing argument for the artifact's utility, whereas scenarios refer to constructing detailed scenarios around the artifacts to demonstrate utility (Hevner et al., 2004). Additionally, the evaluation will also include the concepts of validity and efficacy to discuss our findings.

We acknowledge the challenge when we – as researchers - “interpret” the artifacts ‘impact on the field’. We try to mitigate this challenge by letting the empirical data speak for itself. This will further be explored in the discussion where we discuss our evaluation strategy.

3.1.6. Activity 6 – *Communication*

In the final activity of DSR, the need for *communication* is highlighted. This refers to communicating the problem and its importance to both practitioners and the relevant area of research. We argue that the communication activity is performed through this thesis which will be handed over and communicated to both ITU and Maternity Foundation.

3.2. The Artifact & Design Principles

The concept of artifacts has been the centre of some confusion within the field of DSR. As Gregor and Hevner (2013) argues, this confusion has risen “because the abstract knowledge contributions that are created in DSR [...] can also be treated as a type of artifact.” (Gregor & Hevner, 2013, p. 341). As such, the design principles that we derive from the evaluation of our proof-of-concept-

artifact in this study can legitimately be conceptualized as an artifact. To prevent this misunderstanding, we adopt the distinction proposed by Gregor & Hevner (2013) where the artifact is understood as “a thing that has, or can be transformed into, a material existence as an artificially made object” (Gregor & Hevner, 2013, p 341) whereas broader theoretical reflections – such as design principles – are “more abstract, has a non-material existence, and contains knowledge additional to the description of a materially existing artifact.” (Gregoy & Hevner, 2013, p. 341). As such, even though our proof-of-concept-artifact – *the artifact* – also has quite some abstraction in it, it is the possibility of it being converted to a material existence that distinguishes it from our design principles – which has a *non-material existence*.

Deriving prescriptive design principles from the empirical findings related to our artifact is thus a central step in the process of developing more “comprehensive bodies of knowledge and design theory” (Gregor & Hevner, 2013, p. 341) and will be elaborated on in section 3.4.

3.4. Design Knowledge – Deriving Design Principles

A further important element in adopting the DSR-approach lies in the concept of *Design Knowledge*. vom Brocke et. al., (2024) argue that the knowledge created in DSR-studies does not lie in the designed artifacts. Rather they argue that the extended knowledge and understandings derived from designing and evaluating artifacts is the true contribution of DSR-studies (vom Brocke et. al., 2024). Design Knowledge can be represented in various ways but the bottom-line is that the derived knowledge from a DSR-study should be *prescriptive* rather than *descriptive*. That means, it should concern *how to*-knowledge and not theoretical and descriptive knowledge of *what is*. Hereby, it contributes to the practice of solving real-world problems (vom Brocke et. al., 2024). In this study we chose to represent our findings through *design principles* and *design considerations* (figure 33 & 34).

After the DSR-activities, we thus strive to derive specific *design considerations* and *design principles* regarding how NPOs can support a *common language* that enables the development of a *strategy* and *process alignment* (Radu et al., 2024) towards *digital transformation* that acknowledge their specific value-creation structure, operational set-up, impact reporting necessities and challenges. This is again performed through critical dialogue with existing concepts from the literature to increase the rigor of the design principles. This will be elaborated on in the analysis.

3.5. Structure of the Analysis – *Organized through Three Iterations*

Even though the above 6 DSR-activities are represented through a nominally sequential order, the iterative nature is an inherent part of the DSR-approach (vom Brocke et al., 2024). In practice, insights obtained during stakeholder discussions, interviews, evaluations and demonstrations continuously informed refinements to both the artifact design and the understanding of the problem. This was also highly the case in our DSR-analysis. So even though the structure of the analysis might look linear and organized, the actual research process was far from it. In reality it was quite messy and overlapping. In the analysis, we strive to present the iterative elements of interacting with the complexity of the case as honest as possible.

The analysis will be organized through three iterations where we circle through relevant activities of the DSR-approach. Each iteration is initiated by a shift in understanding of the problem, triggering new objectives of a solution and the subsequent design & development of an artifact.

3.6. Methods to inform the DSR-approach

In this section we will give a brief introduction to the analytical tools used to inform our DSR framework. Together with our literature review, these tools constitute the *Foundations* of our *Knowledge Base* of the DSR Framework and aims to ensure rigorous research.

3.6.1. BPM – Process Landscape Model

As defined by Dumas et al. (2018), Business Process Management (BPM) “is the art and science of overseeing how work is performed in an organization to ensure consistent outcomes and to take advantage of improvement opportunities” (p. 1). Broadly speaking, business processes encompass all the activities and processes that enable an organization to deliver services or products to its customers. In this thesis, we employ Porter’s Value Chain to categorize processes into three overarching categories within the *process architecture*: Core Processes, Support Processes, and Management Processes (Dumas et al., 2018). The process architecture therefore represents both the relevant organizational processes and the interrelationships between them.

In this thesis, the process landscape model is employed for two primary reasons: (1) to reduce the risk of selecting an unsuitable process for automation by improving process documentation, and (2)

to better understand the interrelationships between processes and their connection to the NPOs mission. The mapping of the process landscape model is subsequently followed by a *process selection* through the *process portfolio matrix* aimed at defining appropriate criteria for selecting processes suitable for improvement. These two steps are informed by interviews and process documentation provided by the NPO and will inform the first DSR activity: *Problem Identification and Motivation*.

3.6.2. Process Portfolio Matrix

Dumas et al. argue that “not all processes are equally important, and not all processes can receive the same amount of attention” (Dumas et al., 2018, p. 56). Therefore, Dumas et al. (2018) propose different selection criteria to guide this evaluation and present the process portfolio matrix as a useful tool for visualizing and selecting different business processes.

Specifically, Dumas et al. (2018, p. 56) propose three primary selection criteria:

1. *Health* – Which processes are experiencing the greatest challenges?
2. *Strategic Importance* – Which processes have the “greatest impact on the strategic goals of an organization, for example considering profitability, uniqueness, or contribution to competitive advantage”?
3. *Feasibility* – How susceptible is the process to BPM initiatives when organizational culture and politics are taken into account?

These concepts inform Activity 1 (Problem Identification and Motivation) within the DSR framework, as vom Brocke et al. (2020) argue that “resources required for this activity include knowledge of the state of the problem and the importance of its solution” (p. 4). Consequently, these three criteria were used as guiding parameters during the process selection phase conducted as part of the first DSR activity.

3.6.3. BPMN – Business Process Model Notation

Business Process Model and Notation (BPMN) is a widely used technique for modelling business processes. While process modelling can serve multiple purposes, an essential first step is to develop a thorough understanding of the process in question (Dumas et al., 2018). Consequently, it is important to establish the purpose of process modelling before initiating the modelling activity, as the

intended purpose significantly influences the design of the models. Although BPMN models can vary, there are established modelling rules, such as ensuring that only message flows – and not sequence flows – can occur between pools. Additionally, BPMN represents business processes through a set of core elements, primarily events, activities, gateways, and sequence flows (Dumas et al., 2018).

In this thesis, BPMN is applied to develop both *AS-IS* and *TO-BE* models. The AS-IS models represent the current state of the processes based on collected empirical evidence. Dumas et al. (2018) refer to this as the process discovery phase, which entails developing a detailed understanding of a selected process. Throughout this thesis, the AS-IS BPMN models evolve across each iteration through an iterative interaction between empirical evidence, theoretical insights, and deeper knowledge about their digital infrastructure.

In contrast, TO-BE models represent redesigned versions of the processes and are used to illustrate potential process improvements (Dumas et al., 2018). In this thesis, the TO-BE models are developed on the basis of the AS-IS understanding of the selected processes. Consequently, each TO-BE BPMN model takes outset in a corresponding AS-IS model representing the current understanding of the process from which the redesign is initiated.

Within the DSR methodology, BPMN models constitute a key method supporting both problem understanding (AS-IS model) and solution development (TO-BE model). As such, they function as a bridge between empirical findings, theoretical insights, and technological capabilities.

3.6.4. Devil's Quadrangle and Redesign Heuristics

Dumas et al. (2018) argue that if redesign is approached as an ad-hoc activity – without the inclusion of relevant methods – interesting opportunities are often overlooked. Therefore, to establish objectives of a solution and guide the design and development of an artifact we – in the first iteration – utilized Devil's Quadrangle and Redesign Heuristics to determine objectives.

The Devil's Quadrangle determines business processes potential weaknesses and strengths across four interrelated performance dimensions: *time*, *cost*, *quality*, and *flexibility* (Dumas et al., 2018). In the first iteration, these dimensions are used to assess the AS-IS performance of the selected process

however the expected TO-BE performance was not evaluated as we stopped the development phase of the artifact before being able to evaluate it. This will be elaborated upon in the analysis. The *time* dimension concerns the duration required to complete a process, while the *cost* dimension relates to the resources consumed during process execution. The *quality* dimension refers to the extent to which process outputs satisfy stakeholder expectations and requirements, whereas the *flexibility* dimension concerns the ability of a process to adapt to changing conditions, requirements, or organizational needs (Dumas et al., 2018).

After the assessment of the AS-IS process, the next step is to determine on which dimensions to improve. To do this we select redesign heuristics that are specific redesign principles proven to increase performance across certain dimensions (Dumas et al. 2018). The selection of heuristics is done with outset in pain points and needs expressed by stakeholders. However, selecting a heuristic typically impacts some dimensions positively and others negatively, hence the name “Devil’s Quadrangle” (Dumas et al., 2018). Thus, redesign heuristics is a systemic and analytical redesign approach that draws upon empirical and tested redesign principles such as, activity elimination (removing unnecessary activities) and activity automation (automating an activity partly or entirely) (Dumas et al., 2018). The interplay between AS-IS/TO-BE BPMN models, Devils Quadrangle and redesign heuristics help us: “*rethink and re-organize business processes to make them perform better*” (Dumas et.al., 2018, p. 297). Importantly, this was only done in the first iteration.

3.6.5 MoSCoW

While we utilized the Devil’s Quadrangle and redesign heuristics during the first iteration, we chose to include the MoSCoW framework in the second and third iterations. This decision resulted from a change in our problem identification and motivation for a solution, which required a new approach for defining the objectives of the solution. MoSCoW is a structured framework that helps manage and prioritize requirements while remaining adaptive, continuous, and receptive to change, particularly within agile environments (AbdElazim et al., 2020). The MoSCoW framework allows for reprioritization across iterations as stakeholder needs and pain points shift or evolve. The method categorizes requirements into four clear and structured priorities: Must have, Should have, Could have, and Won’t have (Kravchenko et al., 2022).

In this thesis, MoSCoW is used to categorize both functional and non-functional requirements, enabling us to capture the conceptual aspects and requirements of the empirical data. This differs from the Devil's Quadrangle, which focuses on evaluating the performance of AS-IS and TO-BE processes in relation to process improvements prior to implementation. Consequently, the focus of the *Design and Development* DSR activity shifted from an intended *instantiation* toward developing a *proof-of-concept* aimed at enabling facilitating and supporting a common language.

4. Data Collection – Strategy, Challenges and Themes

In this section we will describe how we collected data for our thesis and our strategy behind this. We interviewed several different actors to ensure a rich and comprehensive understanding of the phenomena-in-practice and how it appears to different stakeholders. The concept of availability largely affected who we could interview and at what time. Due to these – among other things – these practical/logistical challenges, we decided to conduct several group interviews with the aim of collecting “large amount of data on a specific topic in a short time.” (Smithson, 2019). This will be elaborated on later. Figure 6 summarises the data collected for the project. Each interview was transcribed and pseudonymized in accordance with the signed NDA:

Data name & focus	Location	Stakeholder participant(s)	Date	Data output format
<u>Introductory informal interview</u> – no recording	Physical meeting at MF location.	Ileen and Eisha	February 2nd 2026.	Informal meeting. Notes from meeting accessible in
<u>1st interview – (individual)</u> Introductory interview with Ahsa. Focus on her daily work, pain points and broad ideas for future.	Online interview, via Teams.	Ahsa	February 9th, 2026.	Transcription (Appendix A)
<u>2nd interview (individual)</u> Interview with manager of department. Focus on management, ambitions for the future and getting access to the different platforms.	Online interview, via Teams.	Tian	February 13th, 2026	Transcription (Appendix B)
<u>3rd interview and observation (individual)</u> Follow-up questions + watching her perform key processes.	Online interview, via Teams.	Ahsa	February 23rd, 2026.	Transcription (Appendix C)
<u>4th interview (Focus group interview)</u> Focus: identifying cross-departmental workflows.	Physical interview – at MF location	Yvette, Ahsa, Eisha, Ulrika, Eke	March 2nd, 2026.	Transcription (Appendix D)
<u>5th interview (individual interview)</u>	Physical interview – at MF location	Ileen	March 2nd	Transcription (Appendix E)
<u>6th interview (workshop-interview)</u>	Online interview.	Ahsa + Ileen.	March 23rd	Transcription (Appendix F)

7 th interview Artifact evaluation (presentation + workshop) We present our initial solution.	Online + physical interview.	Ileen, Ahsa, Yvette, Eisha, Atana	29th of April	Transcription (Appendix G)
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Figure 6: Data Collection Overview

4.1. Qualitative Data Collection & Strategy

Before conducting the interviews, we made interviewguides in accordance with Kvale & Brinkmann's (2015) guidelines for the semi-structured interview. Inspired hereby, we formed both open and closed questions in the interview-guide. We divided our interview-guide into different *themes* ranging from the individuals understanding of the purpose of the organization to the digital architecture and processes within the organization (Kvale & Brinkmann, 2015, p. 144). However, the approach to each interview changed and was adapted to our knowledge at the time, our phase in the DSR-activities and to the specific informant's role in the organization. Each theme consisted of different questions – both open-ended and closed. We modified each interview-guide in relation to the role and position of the informant, but the overall themes were largely consistent. Though it is quite prevalent for DSR-studies to also collect quantitative data for the various evaluation phases, we chose to only collect qualitative data in alignment with the ambition of our DSR-informed purpose of our artifact. This will be elaborated on in the analysis and in the discussion.

4.2. Online Interviews & Language Barriers

Seeing that two of the central stakeholders for our study were located within Maternity Foundation's offices in India, the best way to conduct interviews with them was by conducting online interviews through Teams – as this was their preferred tool. Specifically, all interviews with Ahsa and Tian were conducted online via Teams. The other informants were interviewed in-person at their office in Denmark. As Chiumento et al., (2018) argue, online interviews raise some specific practical and methodological considerations. It is – however – important to note that both Ahsa and Tian are quite familiar with the use of online communication tools as they often have online meetings with their colleagues around the world.

Chiumento et al., (2018) argue, that the use of digital tools to conduct online interviews is not neutral – it actively shapes meanings, tones and the overall interaction. On a practical level, unstable internet

connection, technological interruptions and challenges can influence the interviews. All these practicalities also affected our interviews in varying degree. A reoccurring issue was that the informants forgot to “unmute” themselves before speaking which led to the information being lost. We also encountered “delayed speech”, which we critically reflected upon as having the risk of preventing informants of speaking impulsively and thus making the interaction less “natural”. Conducting online interviews also reduce access to body language, facial expressions and silences. By being aware of these challenges, we tried to adjust and adapt our interview-strategy to these challenges by being less rigid in our interview-approach. However, this sometimes led to a more “rushed” interview-flow where the informant hurried to get to explain what they saw as the problem without the needed contextualization. As Chiumento et al., (2018) further argue, online interviews can often lead to “question-and-answer exchanges” rather than dynamic and organic discussions. This question-and-answer exchange was highly prevalent in several of the interviews. We have tried our best to both reflect upon this and limit it in practice during the assignment.

Another challenge centred around the language barriers that we encountered while interviewing Ahsa and Tian. This often led to us not completely understanding what was being said by the stakeholders. For example, we initially thought that they said “HOBO” instead of KOBO when referring to the digital tool KoboToolBox. Whether this was due to the online format or different dialects is hard to determine. But the combination of both online-interviews and a different pronunciation definitely influenced the interviews more than first anticipated. Seen in retrospect, this can also have influenced our problem with activity 1 of the DSR-approach centred around *problem identification* – as both the online and disembodied format combined with the language barriers could have hindered us in getting the ‘full picture’ of the stakeholders’ perspective on the processes and their relation to the overall organizational challenges and ambitions.

Another challenge we encountered during the online interviews was when the informants shared their screen to visualize how their data is structured and how a workflow is executed. This is difficult to “transcribe” and capture all the details. We have tried to describe what was being shown by taking specific screenshots and add nuance to these with words, when relevant.

By reflecting on these challenges and making them explicit we hope to prevent it from being a blind spot – and instead foster a critical awareness by acknowledging the interconnectedness of language

and the complex creation of meaning as situated in the specific interview-situations (Resch & Enzenhofer, 2018).

4.3. Conducting Group Interviews

Due to limited access to the relevant stakeholders, we found it necessary to conduct three so-called “group-interviews” (appendix D, F and G) - where the concept of “group interview” is understood as having more than 1 informant (Smithson, 2019). While we also saw explicit benefits of these group interviews – with regards to both practical concerns and to elicit “general viewpoints” - we acknowledge that they also entail some methodological challenges. It is similarly important to note that the final group interview had both online and in-person informants, which provided more methodological challenges.

An important methodological viewpoint regarding group interviews is that it offers the potential for the data to be more “naturalistic” as participants are encouraged to discuss, argue with and interrupt each other with less interviewer intervention (Smithson, 2019). This happened in both interview 5 and interview 7 where significantly interesting internal discussions unfolded which could not have been captured as well through individual interviews. On the other hand, group interviews often lead to an unequal balance between how much different informants contribute. Additionally, unequal power balances and pressure towards conformity often also impacts this interview-form (Smithson, 2019). In the second group-interview, we found that all stakeholders contributed somewhat equally. However, in the final group interview, the new Interim Head of IRL Atana had a lot to say, making it somewhat unbalanced. This can be interpreted both as her way of introducing herself and assert herself. Also, the lesser contribution from the other stakeholders can be seen as a reluctance to express views that might conflict with their new colleague – and for some – manager, and the related power imbalance (Smithson, 2019). Still, this interview fostered fruitful discussions amongst informants – in alignment with our ambition for the interview.

4.4. Thematic Analysis of Qualitative Data

We then adopted the six-step thematic approach conceptualized by Braun & Clarke (2006) to organize and code each interview. As stated in the article, “Analysis involves a constant moving back and

forward between the entire data set, the coded extracts of data that you are analysing, and the analysis of the data that you are producing.” (Braun & Clarke, 2006, p. 15).

Informed by the iterative nature of our DSR-approach, it was challenging to organize the interviews through common broader themes, as some interviews were oriented towards technical elements while others were oriented towards processes and cultural/organizational understandings. Still we tried to organize the data through the first 5 steps as presented by Braun & Clarke (2006). The final step regarding “writing the report” was deemed less relevant and therefore excluded. In the following we will follow their proposed structure:

Phase 1: familiarising yourself with your data

In this first phase, we tried to become “familiar with the depth and breadth of the content.” (Braun & Clarke, 2006). In practice, this was done by reading through the interviews before we started organizing and coding them. Prior to this, we transcribed the interviews through Teams. Naturally, we checked “the transcripts back against the original audio recordings for ‘accuracy’.” (Braun & Clarke, 2026).

Phase 2: generating initial codes

We then generated our initial codes. This helped us organize and structure the collected data. To do this we used the software-tool called “Ethnote” as this allowed us to add specific codes and sub-codes in a shared document. This has several affordances such as enabling us to have an overview of the codes and themes and making it possible to search for codes across interviews. At this initial phase we ended up having 28 different ‘codes’.

Phase 3: searching for themes

In phase 3 we started analysing out codes to consider how different codes could combine to form an overarching theme. In this phase we ended up having the following codes as presented below.

- (Lack of) standardization
- (Lack of) strategic alignment
- Resource constraints
- IT-set up (what technologies do they use and why?)

- Organizational purpose and goals
- Barriers to digital transformation
- Accountability & impact reporting
- 5-year futuristic plan (AI)

Phase 4 and 5: reviewing themes and defining name and themes

After reviewing and critically revising the themes in phase 4, we ended up with the following overall themes the themes in phase 5, where we also renamed the themes:

- Standardization & automation
- Strategy for digital transformation (5-year plan incorporating AI)
- Cultural resistance towards standardization (cultural misalignment)
- Social impact vs. impact reporting (accountability to whom?)
- Local IT-infrastructure and IT-skills in Maternity Foundation (affordances)

5. Case Organization: Maternity Foundation

Maternity Foundation is a Danish NGO and NPO that focus on improving maternal and newborn health in +70 low- and middle-income countries (Maternity Foundation, n.d.). The organization was founded in 2005 with specific focus on helping local health-care professionals in Ethiopia through clinical trainings. Here, Maternity Foundations' professional midwives and partners train local health-care professionals in topics such as birth, labour, abortion, contraception etc. In 2015, these clinical trainings were complimented by Maternity Foundations' Safe Delivery App developed in collaboration with University of Copenhagen and University of Southern Denmark. The app contains educational material such as videos, illustrations, quizzes wherethrough everyone can learn about maternal and newborn health. This digital transformation changed the set-up of the organization making it more scalable while also requiring digital maintenance. Seemingly they expanded to several new countries making it a global organization with partners in Africa, Asia, USA and Latin America and offices in Ethiopia, India and Denmark. In 2023 they expanded their work to Latin America and the Caribbean and as of 2025 they have reached more than 500.000 midwives and other healthcare professionals.

Organisationally, we engaged with Maternity Foundations following departments:

Insight, Research and Learning Department (IRL): This department works with collecting, processing, analyzing, and visualising data for clinical trainings, app development, internal processes, fundraising, impact reporting etc. The department is also responsible for writing yearly -, grant-, and impact reports. The 3 interviewees of Ileen, Atana and Ahsa work in this department.

IT Department: The IT Department takes care of all technological, infrastructural and digital aspects of the organisation. They own the Safe Delivery App and works on improving it continuously. Any changes to MF's infrastructure needs to be approved by them as they are very concerned with containing the monetary costs. The interviewee Tian works as Head of the IT Department.

Program Development Department: Responsible for scaling the Safe Delivery APP, strengthen health systems and expand the network with partners and collaborators. Here, interviewee Ulrika works.

Business Development Department: Responsible for fundraising, grant management and impact reporting. Here, interviewee Yvette works.

Clinical Department: Responsible for preparing, executing and developing clinical trainings. They travel to clinical trainings to train local healthcare professionals. Here, interviewee Eke works.

Communications Department: Responsible for MF's national and international communication activities, with a core focus on visual communication and branding. Here, interviewee Eisha works.

We made following organigram for better overview of the departments we engaged with:

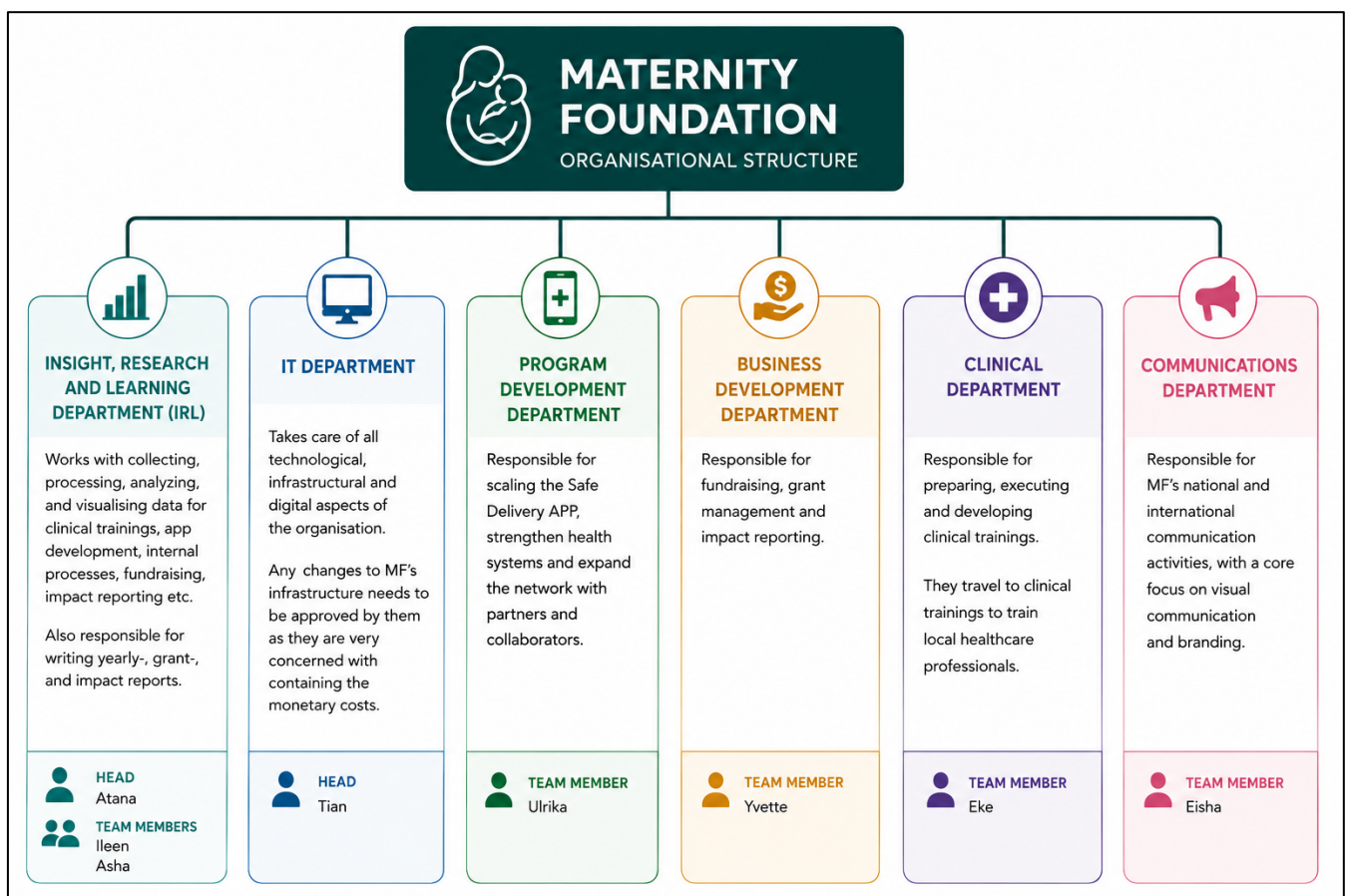


Figure 7: Maternity Foundation Organigram (Visually generated using AI)

6. Analysis – Revolving around Three Iterations

Depicting the iterative nature of a DSR-study is difficult given a thesis' inherently linear structure. To address this, the following analysis is organized around three overall DSR iterations. For each artifact, we have described the corresponding DSR-activities and methods used to inform the specific activities. As additional empirical data was collected and analyzed, our understanding of the organization's processes, stakeholder perspectives, technical infrastructure, and data ecosystem became increasingly nuanced. Consequently, the activities related to problem identification and motivation (Activity 1 – see figure 4), defining objectives of a solution (Activity 2), and design and development (Activity 3) were not conducted as strictly sequential phases, but rather as interdependent and iterative activities that continuously informed and reshaped one another throughout the research process.

6.1. Iteration 1 – Fail Fast Approach

This section presents the first iteration of the DSR framework, covering the activities of Problem Identification and Motivation, Define Objectives of a Solution, and Design and Development. This first iteration was inspired by the “fail fast” approach, whose goal is to quickly identify new and productive design paths (Abraham et al., 2014; vom Brocke et al., 2024).

The first iteration was characterized by our initial understanding of the problem as a manual and time-consuming process that could potentially be solved through a technical implementation. However, as will be demonstrated, empirical evidence and engagement with the literature – both on RPA and Digital Transformation – shifted this understanding toward the broader organizational barriers and challenges faced by MF. We found that these barriers should be addressed before any successful implementation could be achieved.

6.1.1. Activity 1 – Problem Identification & Motivation

Process Landscape Model (BPM) of Maternity Foundation

This section presents an analysis of the empirical data collected from interviews and organizational documents to develop an initial understanding of MF's process landscape which is the highest abstraction of the process architecture in an organization (Dumas et al., 2018). In this sense, the process landscape informs the first DSR activity, *problem identification & motivation*.

Our goal with constructing a process landscape model is to understand the various processes that exist in M F on an overall level and how these processes interact. The first step of the analysis was to identify MF's core processes, understood as the activities underpinning the organization's essential value creation (Dumas et al., 2018). Across the interviews, it became evident that MF's overall mission is centred around strengthening the capacity of midwives and healthcare professionals in remote and resource-constrained contexts in order to reduce maternal mortality rates. As the CTO Tian explained, "the goal of Maternity Foundation is to reduce the maternal mortality rate, especially in LMIC [Low- and Middle-Income Countries]" (Tian, Appendix A, p. 14).

A central discussion during the mapping of MFs process landscape concerned the relationship between the Safe Delivery App, the training programmes, and the organizations impact reporting activities, and how these collectively contributed to the organizations broader mission and value proposition as well as their interrelations. As Tian puts it:

"although safe delivery app is one of the most important, you know, at the centre of the Maternity Foundation's goal... but it's still the programmes and the capacity building through the various initiatives that we take, you know, that makes the overall ship glide through to attain our goal as such."

(Tian, Appendix A, p. 14)

Capacity building through field trainings – often referred to as training programmes – constitute the core process of MF. Simultaneously, the Safe Delivery App emerged as a central digital infrastructure supporting several organizational activities. In this sense, the Safe Delivery App appeared to extend MFs existing value propositions by digitally supporting capacity-building activities beyond in-person trainings programmes. Tian provided us with figure 10 that visualizes MFs core digital infrastructure that is built up around the Safe Delivery App.

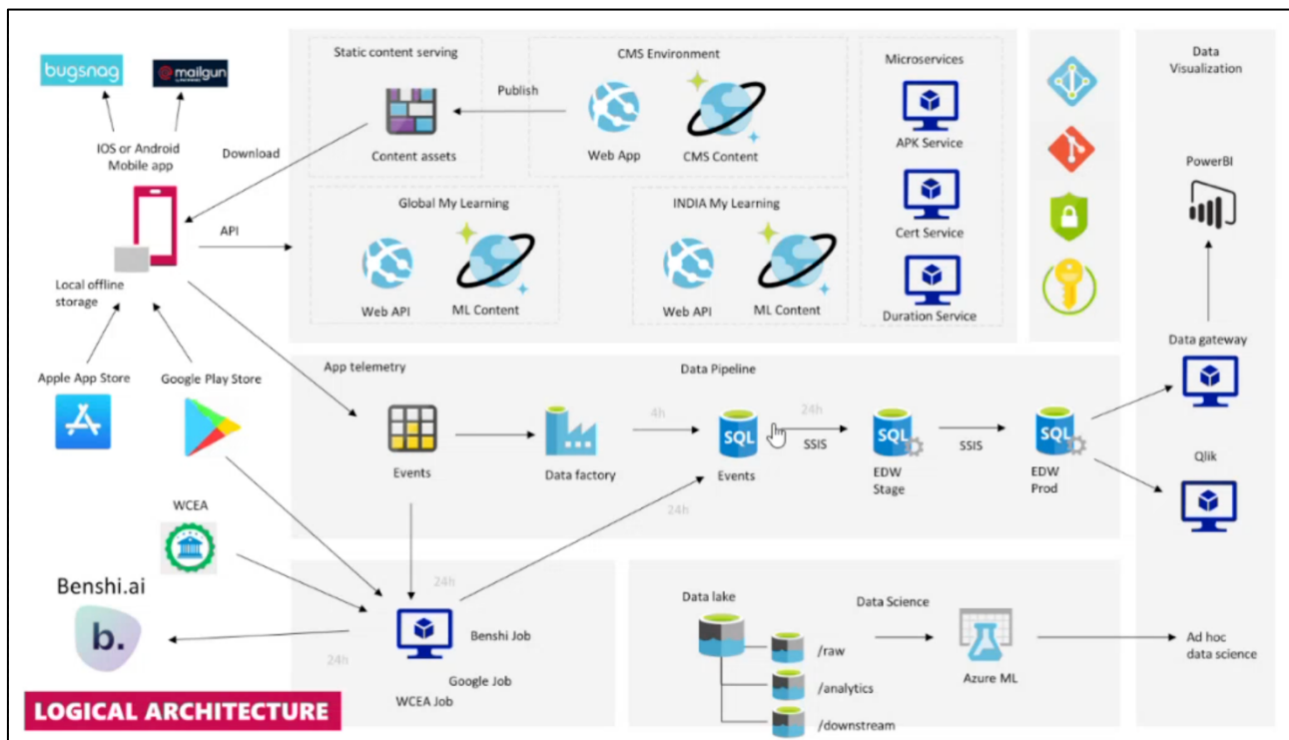


Figure 10: Safe Delivery App Ecosystem

Figure 10 depicts a thorough digital infrastructure that also indicates a high level of process awareness in regard to MF’s Safe Delivery App. The use of SQL databases and Azure Data Factory were some of the key infrastructural elements we focussed on, as they constitute the data pipeline that supports impact reporting through Power BI dashboards and “ad hoc data science”. However, while the Safe Delivery App ecosystem appeared relatively well-structured and technologically integrated, other organizational activities were instead articulated in terms of operational pain points, technological systems, and infrastructural needs rather than as interconnected end-to-end business processes.

A clear example of this is the technological system KoboToolbox – a survey data collection tool – used by IRL-department to collect data from trainees of a training program to support impact reporting. The primary pain points expressed by Ahsa was largely connected to the fragmented and isolated state of KoboToolbox from the Safe Delivery App Ecosystem (Appendix A). Subsequently, the processes and systems that exists outside of the Safe Delivery App Ecosystem, were much more ambiguous to delimit as there were no formal overview of the processes related to the KoboToolbox IT-system.

Maternity Foundation: Business Process Landscape

Limited process formalization is not uncommon within NGOs and NPOs, where operational practices are often shaped by pragmatic and mission-driven concerns rather than explicit process-oriented management structures (Haddad et al., 2016). This is also highlighted by Radu et al., (2024) through the barrier of *fragmented processes*. In our case, the lack of process formalization around IRL-department and KoboToolbox complicates interventions, as we first need to understand how their processes relate to other processes and MFs value proposition. Additionally, Nunes et al. (2026) highlight how the high failure rates of RPA initiatives are directly linked to insufficient understanding and governance of processes. Therefore, BPM and process mapping becomes a necessary step before we can select an *important* and *feasible* process for redesign and process automation. While Haddad et al. (2016) argue that BPM can be a key factor in formalizing processes within NPOs, they also highlight how NPOs exists in a tension between operational efficiency and their broader social mission. Thus, we visualized the interdependencies between MFs processes and their goal of reducing maternal mortality rates in LMICs (figure 11). This enabled us to identify strategically important support processes while not losing sight of their connection to the broader social mission.

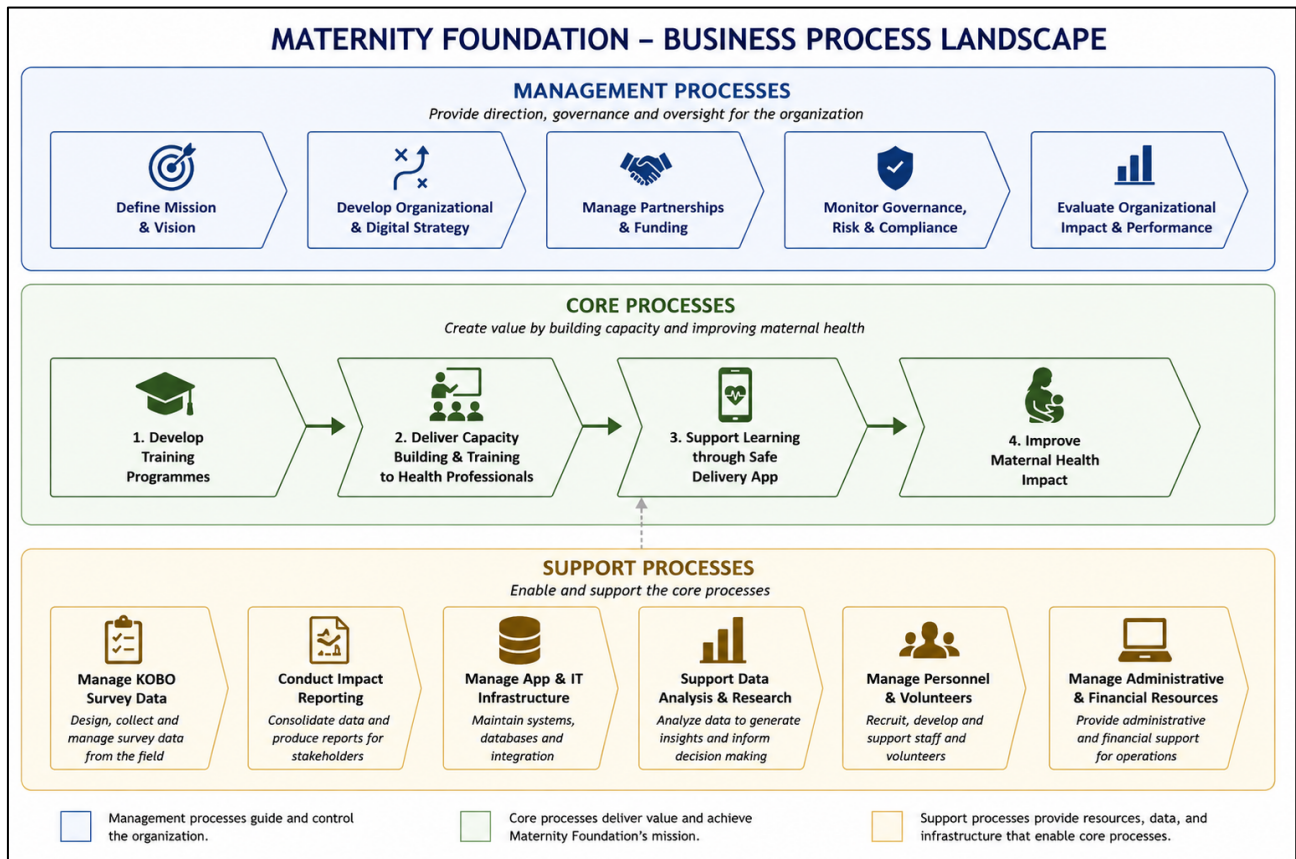


Figure 11: Maternity Foundation Business Process Landscape (Visually generated using AI)

While we had an initial understanding of the process landscape (figure 11) from the first two interviews (Appendix A), we continued to return and refine the process landscape further contextualizing the relationships between processes we gathered more empirical insights. This high-level *process architecture* supported the next step of the *process portfolio matrix* by outlining how the different processes are inter-related in Maternity Foundation.

Process Portfolio Matrix

Having contextualized MFs process landscape, the next question centred around: “what business process do we aim to improve?” (Dumas et al., 2018) and what the problem in this process is. To ensure a systematic approach that avoids focusing on the apparent problems and solutions presented by MF, the Process Portfolio was enacted as a guiding framework to identify and prioritise business processes. Empirically, the identified and selected process was initially a product of the thematic analysis of our interview 1 & 2 (Appendix A and B). The process portfolio matrix was continuously

revisited and refined throughout the DSR phases as our understanding of existing processes evolved and new process interdependencies emerged. Figure 12 represents the final process portfolio matrix.

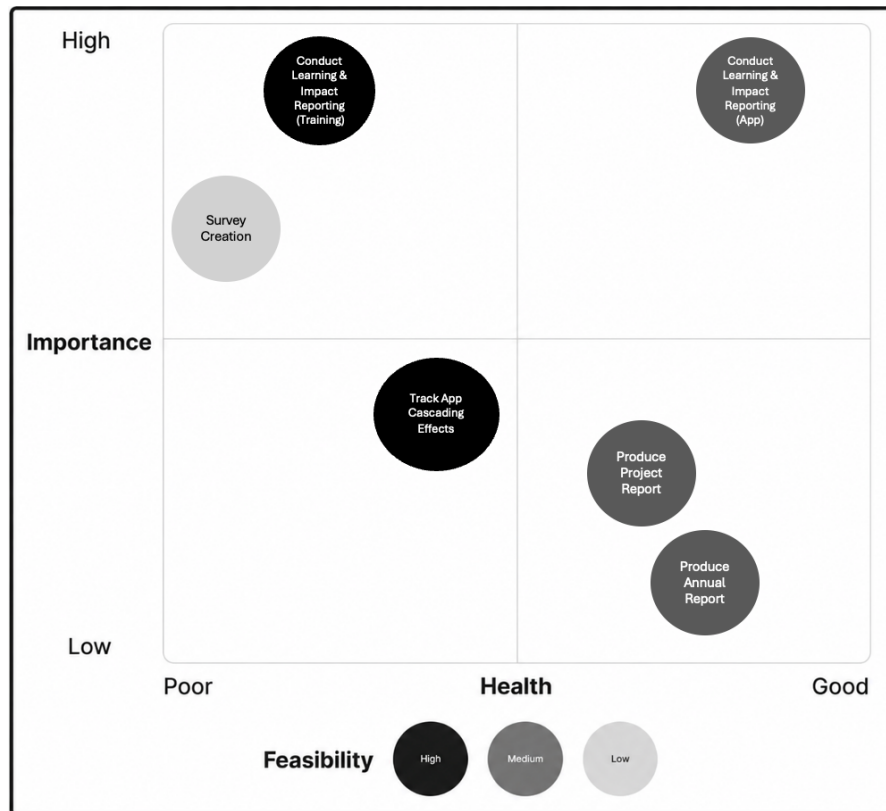


Figure 12: Maternity Foundation's Process Portfolio Matrix

In the following sections, we will focus on why “Conduct Learning & Impacts Reporting (Training)” (hereinafter referred as: CLIRT-Process) was prioritised as the chosen process for our DSR-study activities based upon its importance, feasibility and health. The process and its name refer to the set of activities expressed by Ahsa that are conducted following the completion of a capacity-building training initiative, including data collection from the field, data storage, data extraction, data processing and the subsequent communication of insights to both internal and external stakeholders (Appendix A). The CLIRT-process will be elaborated more in-depth and in dialogue with the empirical data in the AS-IS BPMN sections.

Drawing on our Business Process Landscape (figure 11), the CLIRT-process can be categorized as a supportive process that interacts with several of the other support processes identified in figure 11, particularly: *Manage KOBO Survey Data*, *Conduct Impact Reporting*, and *Support Data Analysis &*

Research. This support process has been given relatively high strategic *importance*, because it directly supports improving training programmes and securing partners and funds (Ahsa, Appendix A). Furthermore, both Ahsa & Tian, brought up recurring pain points regarding time-consuming manual workload associated with data extraction and processing (Appendix A & B). This implies that the specific process may become increasingly strained and difficult to scale as the number of capacity-building trainings and associated data collection activities continue to grow. Additionally, the fragmented relationship of the training data with the organization's broader digital infrastructure was frequently brought up by stakeholders (Appendix A & B). Taking this into consideration, we find the *health* of the current process relatively poor. Moreover, stakeholders expressed explicit support and engagement towards rethinking the current process (Appendix A & B). Additionally, because the training-related data was already collected and stored digitally within KoboToolbox, the process initially appeared highly *feasible* for redesign and automation initiatives. In particular, KoboToolbox provided accessible REST API functionalities, creating opportunities for automated data extraction, processing, and integration within the broader Safe Delivery App ecosystem. This infrastructural setup suggested that RPA-based solutions could potentially support the automation of repetitive data-handling activities while simultaneously enhancing analytical capabilities related to impact reporting and internal learning within MF. Furthermore, Tian – CTO in MF – also stated that “I would love to understand from you guys, how can we automate the overall process, because that's the maintainability, you know, effort reduction.” (Appendix B, p. 11). This statement contributed to the initial impression that process automation was not only a strategic goal but also a feasible idea.

Additionally, we want to clarify the distinction shown in figure 12 between our chosen process and its counterpart “Conduct Learning & Impacts Reporting (App)”. The distinction between app-related and training-related learning and impact reporting processes reflects the finding in the BPM analysis that these activities were supported by partially fragmented infrastructures and data flows. While both processes contributed to organizational learning and impact reporting, they relied on different technological systems and degrees of integration within MFs broader digital ecosystem.

In combination, the strategic importance, poor health and high feasibility of the specific process drove our decision to focus on this.

Problem Identification Formulation & Motivation

Having identified and selected a process – following the DSR approach – we now formulate our initial problem understanding and motivate why the identified process was seen as important and worth improving. Therefore, after the first and second interview we decided on the following problem identification formulation:

“The selected process – Conduct Learning & Impact Reporting (Training) – is characterized by extensive manual work practices, fragmented data flows, and limited integration between reporting activities and existing app data. Consequently, the current process creates significant operational inefficiencies and constrains the organization’s ability to conduct scalable and reliable impact reporting.”

As the CLIRT-process exists entirely outside the Safe Delivery App ecosystem, relying on Excel sheets and manual extraction and processing, the value of a solution to this problem is threefold: Firstly, by addressing the fragmentation of the impact reporting processes the aim is to unlock the potential for new analytics that can empower both impact reporting and internal learning. Secondly, a new infrastructure enables scalability as data processing and manual tasks are reduced by leveraging process automation and RPA. Thirdly, by integrating training-related data into the broader Safe Delivery App ecosystem, analytical outputs could be more effectively visualized, shared, and utilized across organizational departments. This integration could reduce the siloed nature of the existing process and support greater cross-functional access to insights and evidence generation stemming from the field.

6.1.2 DSR-Activity 2 - Define Objectives of a Solution

First AS-IS BPMN

The next step was to examine the process in depth to get a thorough understanding of it. The BPMN in figure 13 maps of our initial understanding of the CLIRT-process and served as a starting point for defining *objectives of a solution* that addresses the concrete pain points raised by Ahsa and Tian.

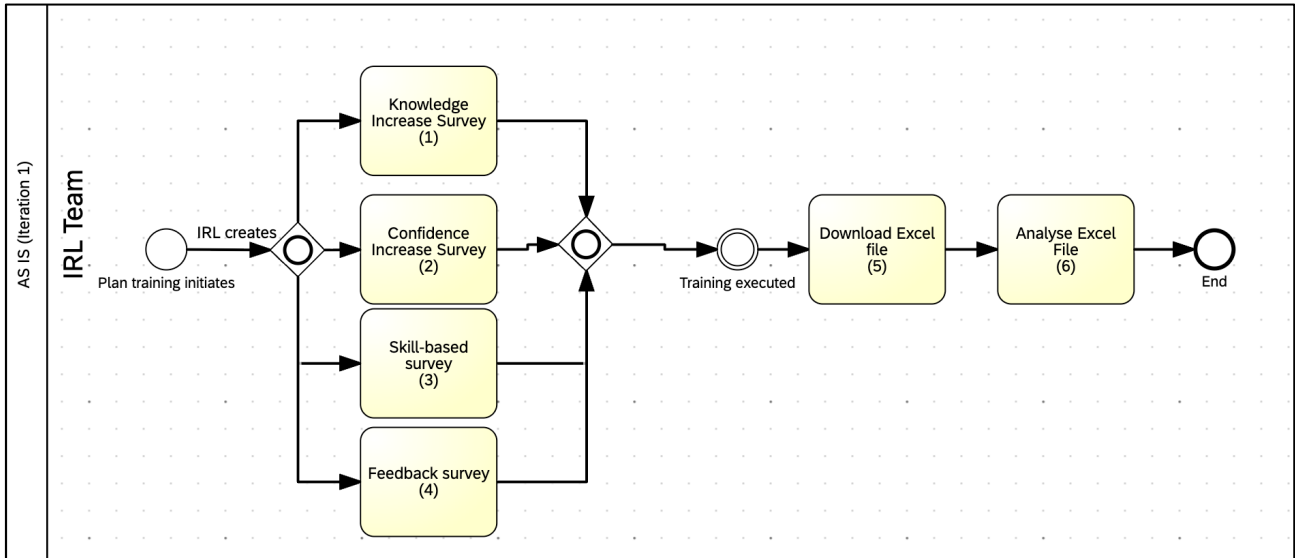


Figure 13): First iteration CLIRT-process AS-IS BPMN

Ahsa had expressed that they were designing a master form for the surveys seen in Figure 13. This led us to believe that surveys used to collect data were somewhat standardized (1-4). Thus, the first AS-IS BPMN captures our initial understanding of the manual load of extracting each excel file each time a new survey is conducted. This is visualized with the *parallel gateway* exemplifying that multiple types of surveys can be created when a training is initiated. This creates a bottleneck because an IRL employee must download each excel file individually and analyse it. Ahsa who is currently engaged with the AS-IS process elaborates on this: “This has been a time consuming process for all the team members is... getting the data from Kobo” (Ahsa, Appendix A, p. 10) and after the time consuming data extraction the team has “to do a lot of manual analysis and then we try to gather some insights together. So that is a main challenge I would say.” (Appendix A, p. 2).

Furthermore, the AS-IS BPMN also highlights the infrastructural limitation of the data being stored isolated in excel. Consequently, the data remains isolated from the broader IT infrastructure, limiting opportunities for integration and further data analysis to support impact reporting. Ahsa underlines the main reason for this pain point as “the two major aspects from where we get all the data is one... is from the app and other is from when we are conducting trainings. So we are not able to merge both.” (Appendix A, p. 2). Ultimately, the first AS-IS BPMN is designed around these insights and constitutes the outset from which we will define the *objectives of our solution*. Additionally, at this point we lacked information regarding what the element of “knowledge increase” (1) referred to and how it was calculated. This was explored more in-depth in a following interview.

Using Devil's Quadrangle to Analyse the AS-IS Process

To conceptualize the *objectives of our solution*, we employ the framework Devil's Quadrangle (Dumas et al., 2018). The framework is based on the four performance dimensions (time, flexibility, cost, and quality). We thus analysed the interviews through these four performance dimensions, enabling a structured assessment of the CLIRT-process' characteristics and weaknesses. This analysis subsequently informed the selection of redesign heuristics aimed at addressing identified shortcomings in the CLIRT-process. Based on interview 1 & 2 (Appendix A & B) we analysed the following Devil's Quadrangle of the AS-IS CLIRT-process (figure 14):

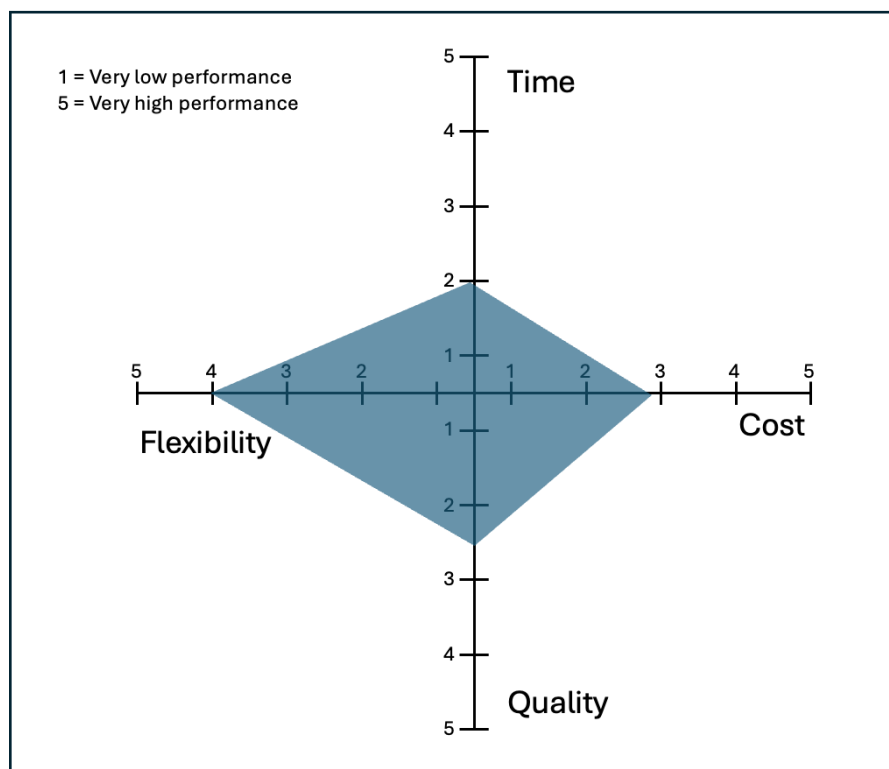


Figure 14: First AS-IS Devil's Quadrangle

Centrally, figure 14 visualizes how the *time* dimension performs very low. This was made very clear by Ahsa throughout the interview: “Yeah, so this is something that I have been mentioning from the beginning of the call and this has been a time-consuming process for all the team members is... getting the data from Kobo.” (Ahsa, Appendix A, p. 10).

The *flexibility* dimension was at this stage somewhat ambiguous to determine. On one hand, the process could be argued to have high flexibility considering it seemed IRL had control over how

surveys are created and what types of analysis they do and thereby easily adapt to change. On the other hand, the process was rigid and inflexible as it was incapable of being combined with other data sources and could break down if the trainings and subsequent surveys scaled excessively. We decided that the process' ability to adapt to change – specifically in survey design and possible analytical output – should result in a relatively high flexibility score.

The *cost* dimension proved somewhat difficult to assess, but we knew from our interview with Tian that cost containment was important for the organization (Appendix B). On one hand, the existing technological setup with KoboToolbox was not inherently associated with substantial financial costs. On the other hand, the AS-IS process was highly time-consuming, but we could not reliably quantify the indirect costs related to the additional manual labour required from IRL employees to extract, process, and manage the data. Consequently, the cost dimension was assessed as moderate relative to the other dimensions within the Devil's Quadrangle.

Finally, the *quality* dimension was placed below moderate in the AS-IS process due to the limitations imposed by the fragmentation of the AS-IS process from the Safe Delivery App Ecosystem. Consequently, both Ahsa & Tian expressed the limitations this imposed on potential insights from the data (Appendix A & B). Furthermore, the manual aspect of the AS-IS process also limits the amount of analysis that can be conducted to the available resources in the IRL department.

Drawing on our stakeholder's pain points of the AS-IS process, we were able to select redesign heuristics to guide our redesign of the AS-IS process. Specifically, the stakeholders were very concerned with the aspects of manual time-consuming workload and the constrained quality of the process as a consequence of the isolated data and its format. Therefore, our main focus during the redesign phase was to address the quality and time dimensions. However, we also had to consider how improving the time and quality dimensions impact other dimensions, especially the cost dimension as NPOs constantly must consider trade-offs between operational efficiency and actual impact in society (Haddad et al., 2016; Godefroid et al., 2021). While Tian had expressed willingness to invest in new technologies to improve the process – such as RPA – Tian also underlined that it had to make sense from a cost perspective (Appendix B). Consequently, this balance added a complexity to the redesign process as these trade-offs should be considered when designing a first iteration TO-BE process.

Redesign Heuristics: Defining Objectives of a Solution

Ultimately, the design focus of the artifact was oriented towards improving the *time* and *quality* dimensions of the Devil's Quadrangle, while simultaneously seeking to ensure that any increases in the cost dimension remained controlled and proportionate to the anticipated organizational benefits within impact reporting. To support these objectives, we carefully selected the following two redesign heuristics as they specifically address processes that are underperforming in *time* and *quality*: *Activity automation* and *Integral technology*.

The *activity automation* heuristic seeks to improve processing time and provides predictable results (Dumas et al., 2018). In our case, this heuristic guided the redesign of the manual data extraction and data processing in the first iteration AS-IS process. Additionally, the *integral technology* heuristic entails introducing a new technology to elevate physical constraints in a business process (Dumas et al., 2018). Drawing on the affordances that RPA has (Godefroid et al., 2021; Hofmann et al., 2020; Herm et al., 2022), we aspired to introduce RPA as a new technology to address both the manual workload of first iteration AS-IS process and the fragmentation from the Safe Delivery App ecosystem.

TO-BE BPMN of Desired Solution

Based on the identified *objectives of our solution* and the selected redesign heuristics, the next step involved developing a TO-BE BPMN model of the desired process. The purpose of the TO-BE BPMN was to translate stakeholder pain points, process inefficiencies, and redesign objectives into a visual representation of a redesigned future-state process. With outset in the two selected redesign heuristics – Activity Automation and Integral Technology – the redesign focused on reducing manual and repetitive process activities and improving data integration across systems. This focus translated into the following the first iteration TO-BE BPMN as visualized in figure 15.

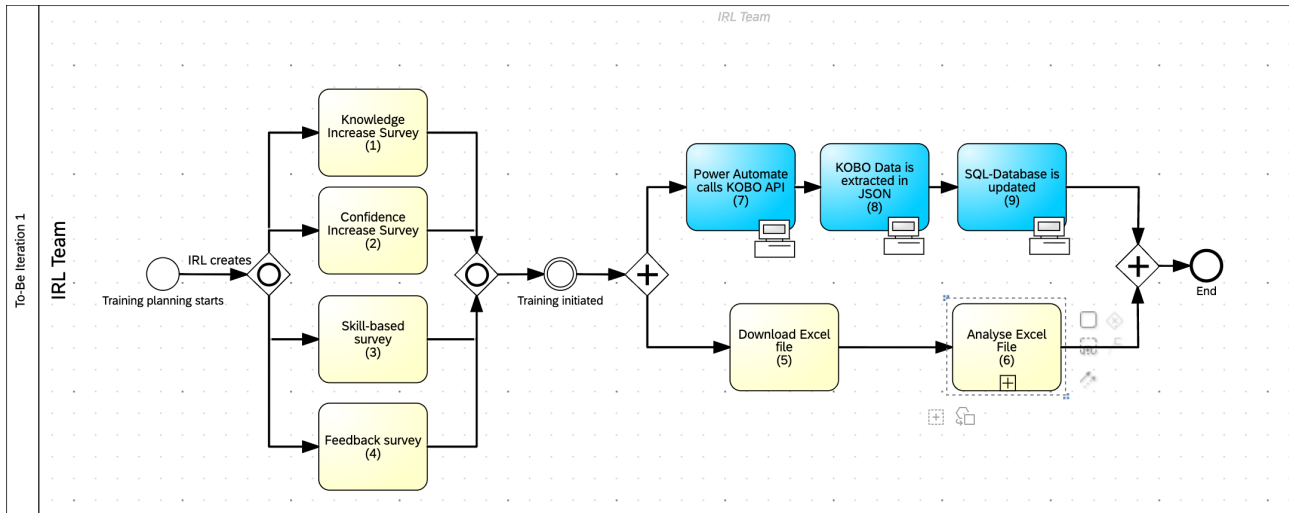


Figure 15: First TO-BE BPMN

The TO-BE BPMN in figure 15 is revised with 3 new activities (blue). The redesign heuristic of *activity automation* is addressed by the use of RPA technology Power Automate. Additionally, the introduction of an RPA technology also addresses the heuristic *integral technology*. Furthermore, the RPA technology connects the data to the Safe Delivery App ecosystem as visualized in the BPMN activity called ‘SQL-Database is updated’ where the survey data now is stored in the same infrastructural environment as the Safe Delivery App data.

6.1.3. DSR-Activity 3 - Design & Development

Our first design decision was to explore the use of RPA as a mechanism for automating the extraction of data from the KOBO platform. To support this objective, we utilized Microsoft Power Automate, an RPA and workflow automation tool within the Microsoft ecosystem (figure 16). More specifically, the solution leveraged an HTTP GET action capable of retrieving data from external REST APIs, thereby enabling automated access to data generated within the KoboToolbox platform. This was first tested with fictional data placed in SharePoint that we extracted to a Microsoft list. After confirming the functionality, we finally got access to a test environment in the KoboToolbox platform with examples of real data. We then adapted our artifact to extract data from KOBO and storing it in an Azure SQL database that corresponds to their existing infrastructure used for the app data. The core data extraction executed by a scheduled HTTP call to KOBO can be seen in figure 16:

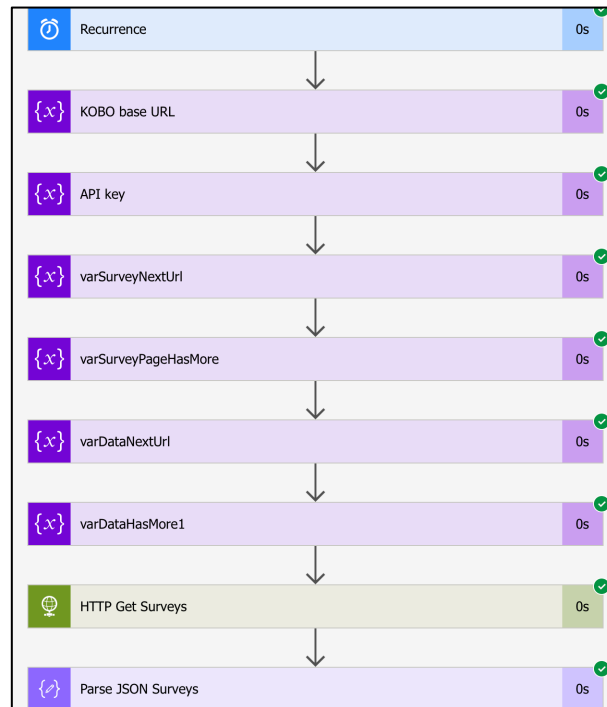


Figure 16 – Power Automate Flow 1

As can be seen in figure 16, there are various variables being initialized before the API call itself. The two first variables (KOBO base URL & API key) are initialized so the RPA calls the correct base URL with the correct credentials. The rest of the variables are used for capturing the state of the data after each extraction cycle. As the RPAs trigger is set to run 1 time each week, we need variables to determine what data has been extracted and if new data has been submitted so the flow does not extract duplicates. These variables are stored in an SQL table that tracks the KOBO survey states between each extraction as seen in figure 17:

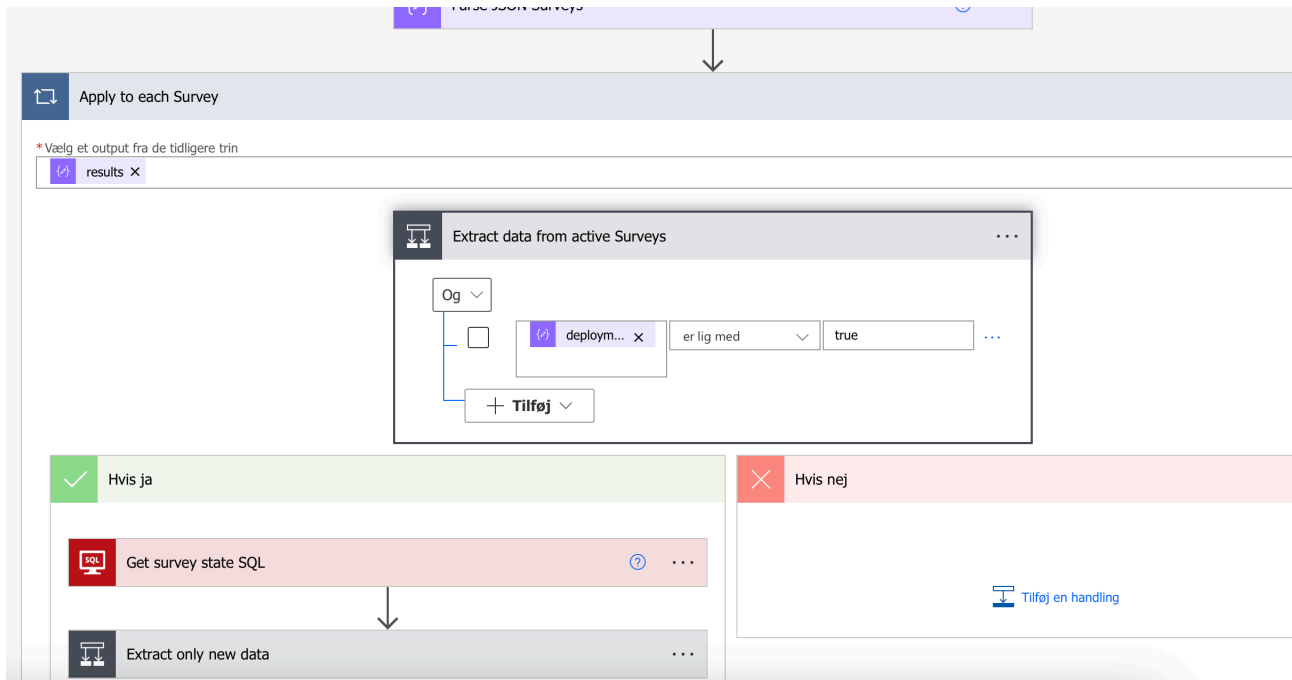


Figure 17: Power Automate Flow 2

What this RPA essentially does is if new surveys are deployed or new data is inserted into existing surveys, the flow only extracts and inserts the new data into an Azure SQL database. The “Extract only new data” action and the loops that follow can be seen in more detail in appendix H by uploading the appended file to a Power Automate account. At this point in time the Power Automate inserts raw JSON into the SQL database.

Having extracted every survey from KoboToolbox we began to examine the data of the 50+ surveys provided to us. Crucially, it was at this point that we realised that the survey data was much more complex than we first anticipated. While some commonalities existed within survey types, we found the overall survey designs to be considerably more heterogeneous than initially anticipated. Some surveys were distributed across multiple countries and languages, while others varied substantially in terms of question structures, formats, and content (See appendix I and J). The master form mentioned by Asha (section 6.1.2) turned out to be much less developed and used which was different from our initial understanding. This finding was problematic, as RPA technologies generally perform best in standardized, repetitive, and structured process environments. Consequently, integrating and processing the survey data would require substantial manual preprocessing and standardization efforts. Furthermore, the variation across questionnaires constrained comparability between surveys

and limited the possibilities for unified reporting and automation. This realization prompted a temporary pause in the redesign process, as it became clear that our understanding of the AS-IS process and its underlying data complexities was not yet developed to support continued progress along a productive design trajectory.

6.1.4 Reflections over First Iteration

Our approach for our first iteration was to engage in rapid design cycles to “fail fast” and to identify new productive paths towards a satisfactory solution” (Abraham et al., 2014). This pragmatic approach encouraged active engagement with the practical and operational aspects of the problem space. Furthermore, it enabled us to formulate more targeted and informed questions for informants. Consequently, the primary objective of the first iteration was not to develop a finalized or optimal solution, but rather to deepen our understanding of the process – particularly the technical constraints and barriers related to the data structure and lack of standardization.

6.2. Iteration 2 – Towards a ‘Proof of Concept’-artifact

As described in the first iteration, we realized that the survey data was significantly more complex than initially anticipated. The surveys differed in terms of question structures, formats, data types, and languages, resulting in limited possibilities for standardization and comparability. This challenged our initial ambition of developing an RPA-inspired process automation artifact – and shifted a move towards a proof-of-concept-artifact less oriented towards being instantiated and more oriented towards illuminating other potential tensions within the organization.

The findings in the first iteration thus encouraged us to conduct additional interviews to gain both a deeper and broader understanding of the process and how it is perceived by other departments.

6.2.1. Reassessing Activity 1 - *Problem Identification*

To investigate these challenges, we conducted interview 3 (Appendix C) and 4 (Appendix D). Interview 3 was a walkthrough interview with Ahsa from the IRL-department, where we encouraged her to provide a step-by-step explanation of how survey data was handled, processed, and analysed in practice. Interview 4 was a cross-departmental group interview to gain a broader organizational understanding of the challenges associated with standardizing the data collection and analysis and thereby enabling scalability and comparability across datasets through process automation. The ambition in this interview was also to explore if there were other tensions across the departments.

Walk-Through Interview of the CLIRT-process

A particular methodological challenge related to this interview was that Ahsa shared the screen throughout large parts of the session to demonstrate step-by-step how survey data was handled, cleaned, analysed, and visualized in practice. While this walkthrough provided highly valuable process insights, many of the observations were difficult to fully transcribe due to their visual nature. We aim to contextualize the most significant observations as clearly as possible.

The walkthrough interview was primarily conducted to gain deeper insights into the following questions:

- How is data currently stored, processed, analyzed, and visualized?

- What is the current status of the proposed “Master Form” initiative?
- To what extent are increases in *confidence* and *knowledge* treated as key impact metrics within this process?

During the walkthrough, it became increasingly apparent how dependent the existing process was on extensive manual data cleaning and validation. As Ahsa explained: “There is a lot of manual work. First of all, obviously we have to clean the data because there are many repetitions, many do multiple submissions.” (Ahsa, Appendix C, p. 3). Following this initial cleaning phase, demographic filtering and manual validation procedures were conducted to ensure data completeness and consistency (see Appendix I and J for examples of cleaned datasets).

Although these process pain points initially appeared well-suited for RPA implementation due to their repetitive and rule-based characteristics, the walkthrough simultaneously revealed a deeper organizational challenge related to the lack of standardized survey infrastructures. This became particularly clear during the discussion of the proposed “Master Form”, which aimed to create greater comparability and consistency across surveys. When asked about the current status of the Master Form, Ahsa replied:

“So we have tried in the past also to have one single form, but because of multiple trainings happening in different countries, it has been somewhere we failed to use the same form because of a lot of changes and a lot of different end movement changes. So we are... but now when we see like individually, we have like 50, 60 forms in our Kobo files. So we have identified as a big challenge for all the teams and for all of us in the team that we have to analyse it. All the time a training is happening. So we thought of having a master form. So that will be useful for all of us and save a lot of time doing all the manual calculations and manually designing it every time.”

(Ahsa, Appendix C, P. 6)

The quote highlights how the continuous modification of survey forms constituted a major organizational pain point, resulting in the existence of nearly 60 different forms. Furthermore, the ambition of standardization was complicated by regional adaptations and evolving clinical requirements. As Ahsa explained: “But with clinical team, it happens that they come up with some new update, maybe some government guidelines have changed. So accordingly, this change needs to

be updated.” (Ahsa, Appendix C, p. 6). This finding revealed an important tension between the organizational need for flexibility and the technical need for standardization to enable scalable automation. While adapting surveys to local contexts and updated clinical guidelines was central to maintaining the internal learning need for clinical department, the constant modification of forms simultaneously complicated data comparability and process formalization. Drawing on the barriers to digital transformation in NPO’s (Radu et al., 2024), this challenge both highlights the fragmented nature of the process, the silo structure, low data quality and a lack of adapted technical solutions.

Revisiting the core question of Activity 1 in the DSR process regarding the problem identification, the extensive manual effort spent cleaning datasets, removing duplicates, and continuously creating or modifying forms emerged as the most pressing operational challenge. However, the walkthrough also revealed that the lack of standardized survey data represented a significant barrier to implementing RPA within the existing AS-IS process, despite multiple attempts to identify technical workarounds within the current infrastructure. This will be elaborated further in the Design & Development section.

Simultaneously, the literature cautions against approaching process automation solely as an efficiency-driven initiative and technical issue, particularly within NPO contexts where accountability structures and social impact play a central role (Herm et al., 2022; Nunes et al., 2026). While informants from IRL strongly encouraged the development of an automation solution, the literature on accountability and impact reporting in NPOs highlights how an excessive focus on impact reporting and standardization may risk shifting attention away from the organization’s broader social mission (Rana & Cordery, 2024).

To further explore these tensions, we facilitated the fourth interview (Appendix D), a cross-departmental group interview with informants from multiple organizational units. The purpose was to gain a broader organizational understanding of the challenges related to standardizing survey infrastructures, while also exploring how different departments and stakeholders conceptualized legitimate forms of impact reporting.

Cross-Departmental Group Interview – Broader Organizational Focus

In the cross-departmental group interview relevant stakeholders from Business Development, Clinical Team, Communication, Programme Team and IRL attended. The overall purpose was to understand what data they find relevant from the CLIRT-process, how they “use” the IRL-department and how they view the issue concerning the standardization of forms.

When asked broadly about who owns the CLIRT-process Ulrika answered “it's actually a good question. I'm pretty sure that originally it was Clinical Team who owned it, because you're the ones who do the training. Yeah, I'm not sure if that's still something we need to decide, because I think it's a mixture together...” (Ulrika, Appendix D, p. 6). The CLIRT-process thus appears both unorganized, unstructured and as situated between departments.

Regarding what data-points they find relevant and important, Eisha – from Communication – described the following when she and her colleagues report to donors and funders: “the number of people reached and number of trainings and then also if available, the skills, knowledge and confidence increase. It is always very nice to display for us” (Eisha, Appendix D, p. 5). Yvette from Business Development describes, that her department mainly focus on the “high-level global data” (Yvette, Appendix D, p. 3) by focusing mainly on number of trainings executed and how many midwives and partitioners there were in each training – as well as the average increase in knowledge, confidence and skills. Seemingly, both Communication and Business Development focus almost exclusively on the “bigger numbers” as they report to funders and donors – through *upward accountability*.

Eke – from Clinical Department – describes that “I guess knowledge and confidence that's the most important for the clinical team... yeah, especially when we are training, but of course also like which module are mostly used” (Eke, Appendix D, p. 6). Interestingly, Eke adds that Eke also uses the Kobo-data to “go back and see and reverse, like, what went good and what went didn't go so well, and yeah, [...] how can we improve?” and “why didn't they score good here? What happened? Like, what could we do differently?” (Eke, Appendix D, p. 9). Informed by the literature regarding the discussion between social impact and impact reporting (Cordery et al., 2023) this hints at a crucial discussion point as Clinical Department actually uses the more specific and complex insights from the data to explore how to improve the specific trainings – ultimately aimed at improving the core

value-creation of the organizations which is the capacity building of midwives and trainers. On the other hand, Business Development and Communication focus on the *upward accountability* and the bigger numbers.

When we asked if they would find it meaningful for us to explore the possibility of designing an interactive Power BI Kobo-dashboard across trainings and countries, all the attending informants – including Ulrika from Clinical Department – agreed that it would be beneficial and hugely potential. Ulrika from Programme Team argues that it would both free up a lot of time while also satisfying donors:

“If we do have a big initiative where partners are actively engaged and they want to know what's happening and how many trainings have happened and whatnot, there I definitely see potential in having this dashboard. Because that would mean that we don't have to be in there or Ileen or whoever, sending every 5 minutes updated information. Yeah, because it takes...A lot of time, yeah.” (Ulrika, Appendix D, p. 13-14)

When asked about the “Master Form”, Ulrika from Programme Team highlights that “it also depends on who are like, who are we training, what project is the, yeah, which language, and stuff like that” (Ulrika, Appendix D, p. 14). Here it is highlighted that the specificity of the surveys is necessary as it depends on what project it is, which country it is in and what language etc. Eke from Clinical Team further adds that “It's never the same clinical questions and stuff like that, [...] that has to be obtained, because most likely it will always be different modules” (Eke, Appendix D, p. 14). Eke hereby further argues that it has to be highly specific and non-standardized as the clinical questions are never the same seeing that they depend on the specific trainings.

This tension between flexibility required for making context-specific surveys highly contrasts with the ambition and desire for standardization, scalability and an interest in the ‘bigger numbers’ as expressed by IRL, Business Development and Communications. This tension further complicates our initial idea of using RPA to automate the process.

Final Problem Identification Formulation

After the third and fourth interview we decided on the following problem identification formulation:

“The organization lacks a *common language* and *bilateral communication* for critically discussing and assessing the trade-offs involved in standardizing survey data to enable process automation and scalability compared to the potential loss of flexibility and contextual insights.”

We decided on this broad formulation as a way of capturing the complexity of standardization vs flexibility in the specific context. Though IRL-department presents the problem as being manual and time-consuming, Clinical Department seemingly rejects the idea of standardization as it would not enable them to analyse the specific trainings in depth while also highlighting the importance of developing context-oriented surveys for the specific trainings. As we perceive it, the problem thus cannot be framed simply as *time-consuming* and *manual* or as an isolated technical issue that can be solved through the implementation of RPA – as we thought in iteration 1.

Additionally, it is important to highlight that this first DSR-activity also must consider how a potential artifact can solve or contribute to the problem-identification. In relation to this, it seems illegitimate for us to decide whether standardization is the right way to go or not. But by choosing this broader problem identification formulation we aim to develop an artifact that can facilitate important and relevant discussions regarding the potentials and challenges of *process automation* within the distinct characteristics of MF to – ideally – both optimize impact reporting, social impact and operational effectiveness in their specific context.

This conceptualization of the problem was informed by central findings from the three concepts from the literature review. Firstly, by drawing on the concept of communication (common language and bilateral understandings) as a barrier to digital transformation (Radu et al., 2024) we argue that the organization lacks a common cross-departmental understanding of what needs to be done, and how digital technologies can make this happen. The concept of *communication* is defined as being “intricately connected with organisational structure, knowledge resources and organisational culture” (Radu et al., 2024, p. 7). Simultaneously, it is described as having the potential to *inform process alignment* and *clear strategies and goals*. This focus on *communications* hereby also aligns with

something Tian stated in the second interview which we didn't give much attention in the first iteration: "we are also moving to a new platform. And in the new platform, we will see if we can really integrate the two data together." (Tian, Appendix B).

Several interesting insights appear from this quote in combination with the insights gained from interview three and four. Firstly, the CTO of the organization seems unaware of the conflicting understandings regarding flexibility and standardization and how the flexibility - for Clinical Department – is highly important for the essential value-creation of the organization. This further highlights the silo structure of the organization and lack of *bilateral understandings*. The idea is, that the above problem identification formulation could potentially support process alignment and the development of a clear strategy – that doesn't compromise important value-creation processes. Inspired by the emphasis on *digital transformation strategy* in NPOs (Cipriano & Za), 2024 we therefore decided to define the *problem* at hand as this broader and more communication-oriented problem rather than just a question of improving time, quality and cost through process automation tools.

Secondly, the literature on impact reporting and *accountability* in NPOs further encourages this broader problem-identification as a way of avoiding transforming the organization and its social missions into *calculative performance objects* (Rana & Cordery, 2024). If we implemented a process automation that automatically extracted, analysed and visualized data this could over time perform *funder-centric accountability* and thereby removing focus from the beneficiary-centred social impact. However, we still acknowledge the importance of standardized and scalable data to inform upward accountability as this is an essential element for NPOs. We simply find that the social impact and impact reporting must be balanced. Through this communication-oriented problem-identification we aim to facilitate an internal discussion through which a common cross-departmental understanding of the potentials/challenges of process automation can be reached.

Thirdly, the literature on process automation and RPA further supports this broader problem conceptualization by emphasizing that successful automation initiatives depend heavily on process standardization, stable workflows, process maturity, and cross-functional organizational alignment (Hofmann et al., 2020; Herm et al., 2022; Nunes et al., 2026). This perspective became increasingly relevant as our empirical findings revealed tensions between the organizational need for flexibility

and contextual adaptation on the one hand, and the technical need for standardized and formalized processes to enable scalable automation on the other.

6.2.2. Activity 2 - Redefining Objective(s) of a Solution

In alignment with our new problem understanding, we moved on to redefining the objectives of our solution. While this new problem understanding significantly changed the objectives of our solution, it remained our ambition to design an artifact as closely aligned with their organizational environment as possible, thereby making it meaningful and feasible for them to understand the specific trade-offs of process automation within their context. But instead of designing an artifact that can be implemented in their IT-infrastructure, we decided to move towards a proof-of-concept-artifact. The aim was to develop this as realistically and true to their IT-infrastructure and digital technologies as possible – to highlight the potentials, challenges and trade-offs of process automation.

We chose to include the MoSCoW method to inform this activity, as it provided a beneficial way of demonstrating structured prioritization regarding what our artifact must, should, could, and won't accomplish – thereby helping navigate competing organizational concerns. Additionally, we used the BPMN method to organize and map our new understanding of the specific process. These BPMN models will also be used in our demonstration as a way of visualizing what a standardized and automated workflow could look like compared to the current one.

Using BPMN to Analyse and Map the AS-IS Process

The third and fourth interviews greatly deepened our understanding of the AS-IS BPMN by adding activities occurring both prior to and after a training session (figure 18). We have added numbers to each activity in the BPMN, which we will refer to throughout this section. The added elements are marked in blue from Interview 3 and orange from Interview 4.

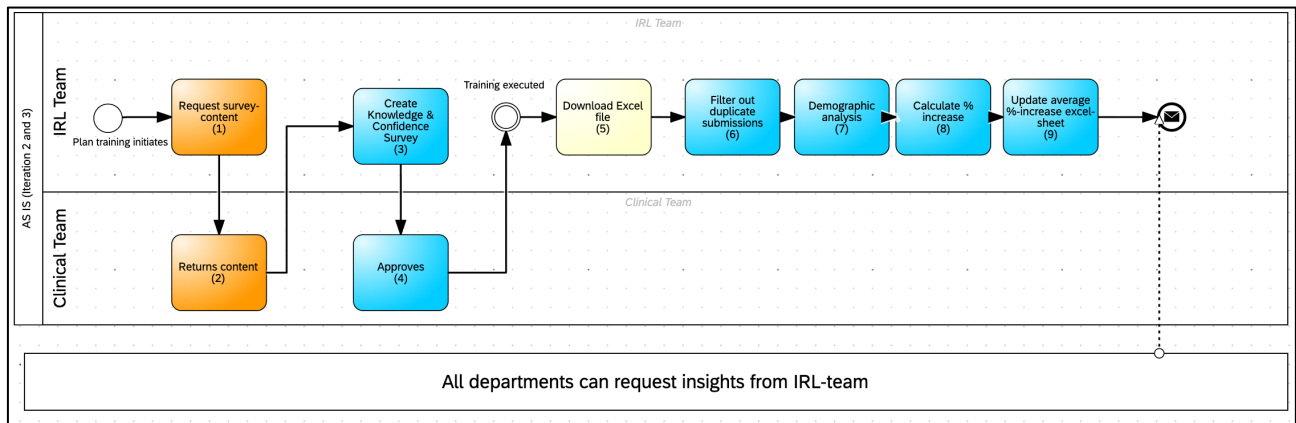


Figure 18 – Second AS-IS BPMN

In the interview with Ahsa, we expected her to walk us through the analysis of all the different types of surveys (activity 1-4 of figure 13). However, as the Knowledge and Confidence survey (3) was presented as the most important, Ahsa only walked us through this specific survey type. By gaining a deeper understanding of this survey, we decided to focus on it in the BPMN model.

During the interview, Ahsa explained and demonstrated that the Clinical Team must approve surveys (4) before they are created and used in trainings. This led us to create a separate lane for the Clinical Team in the AS-IS BPMN model. As Ahsa described it: “But with clinical team, it happens that they come up with some new update [...]. So accordingly, these changes need to be updated. So yeah, we still need to get the go ahead from the clinical team.” (Ahsa, Appendix C, p. 6).

The data-cleaning and analysis elements of her work also became clearer during the walkthrough. First, Ahsa cleans the data by removing duplicate submissions (6). Then, Ahsa conducts demographic analyses across variables such as age, district, and profession (7). After this, Ahsa calculates the increase in knowledge and confidence (8) and inserts the results into an Excel sheet (9). Whereas the first DSR iteration primarily explored the activities occurring before a training session, this interview shed important light on the post-training activities related to cleaning, processing, and analyzing the data.

The focus group interview further deepened our understanding of the process. We learned that it is the Clinical and Program Team that provide the survey content (2). Eke explained:

“Before we start training, we have handed out the content, the clinical content that we want to do a test in, a pre-test and a post-test. So we’ve handed that to the IRL team.”

(Eke, Appendix D, p. 4)

This finding was important because it helped explain the unstructured nature of the surveys. The creation of survey content was not managed by the more analytically oriented IRL team, but rather by the Clinical team. This flexibility enabled them to tailor surveys to individual training sessions, which appeared to be an integral part of their work practices.

Regarding impact reporting, the Program, Business, and Development Teams were asked how they retrieved data. Eisha from the Communications Team explained that: “I usually just discuss it with the IRL team [...]. And so usually we ask them, can you highlight some of the most important numbers?” (Eisha, appendix D, p. 9). Similarly, Yvette from Business Development stated: “I’m not always sure that I’ve got the right parameters in there, so I would always check with IRL, is this correct?” (Yvette, Appendix D, p. 13). This indicates that impact-reporting data was not directly accessible to other departments, as they depended on the IRL team to retrieve and organize the data for them. In the BPMN model, this dependency is represented in activities (1) and (2).

To summarize, these interviews significantly deepened our understanding of the AS-IS process by identifying new activities occurring both before and after the execution of trainings. Furthermore, we identified a possible explanation for the non-standardized nature of the surveys and found that other departments were highly dependent on the IRL team for impact reporting. This more comprehensive understanding of the process informed the objectives of our solution by structuring and visually organizing the workflow, while simultaneously identifying potential entry points for how our artifact could intervene in and improve the process.

Using MoSCoW to Define Objectives of a Solution

One of the main challenges related to defining the objectives of a solution suited to the more organizationally and culturally oriented problem identification concerned how to design an artifact capable of addressing such a complex problem. As our artifact had now taken the role of a “proof of concept,” we found the *MoSCoW prioritization method* to be a beneficial approach for categorizing the functional requirements of the solution on a more conceptual level.

This method facilitated productive internal discussions regarding how we should prioritize the technical aspects of the solution in relation to the more organizational, operational, and cultural aspects. After several rounds of discussion, we decided on the prioritized requirements outlined in figure 19. These priorities were based on insights from all four interviews conducted thus far and therefore encompassed both technical and organizational elements.

Must	Should
- Support cross-departmental discussion regarding standardization trade-offs - Visualize operational implications of process automation through BPMN - Include a visual representation of a “KoboToolbox-dashboard”	- KOBO-data should be stored in a format allowing comparability with APP-data - KOBO-data should get extracted out of KOBO - Preserve interpretability of local survey contexts
Could	Won’t
- Support a rethinking of the current KOBO-survey data-infrastructure. - Touch upon the performance dimensions of cost, time and quality - Present our example of a standardized “Master Form”	- Full organizational implementation/integration - Be categorical and argue for what is the right solution moving forward

Figure 19: Iteration 2 MoSCoW

Importantly, the prioritization outlined above was also informed by insights generated during Activity 3 – Design & Development. As such, the outlined priorities were not only empirically informed, but also technically informed. Although the method may appear somewhat rigid and simplistic, we found that it greatly helped clarify what objectives our artifact must accomplish – the so-called “non-negotiables” of the artifact within the organization’s complex operational landscape. At this stage, we identified three non-negotiable “must-haves” for the artifact to address.

Firstly, our artifact must *support cross-departmental discussions regarding standardization trade-offs*. This requirement was not explicitly raised by any individual stakeholder but rather emerged from

our broader analysis of the problem through critical dialogue with the literature. We considered this requirement crucial in addressing the lack of a shared basis for critically discussing and assessing the trade-offs involved in standardizing surveys to enable scalability at the potential expense of flexibility and contextual insights. The following “must-haves” more concretely elaborate on how this objective could be achieved.

Secondly, our artifact must *visualize operational implications of process automation through BPMN*. By this, we mean that the implications of implementing process automation must be made visible and understandable. In practice, this is achieved through both a visual walkthrough of how our specific proof-of-concept artifact could automate the extraction, processing, and visualization of survey data, as well as through a TO-BE BPMN model illustrating which activities the artifact could automate within the process. This will be elaborated on in the following section.

Thirdly, our artifact must include a visual representation of a “KOBO dashboard.” This was conceptualized as a “must-have,” as all informants – including those from the Clinical Department – expressed interest in the idea of a survey-data dashboard accessible to all employees, with possibilities for filtering across countries, regions, and training sessions.

Thus, our artifact must not only demonstrate that process automation is technically possible, but also visually illustrate how it can be implemented, what this requires from the underlying data structure, and what it enables in terms of data accessibility and visualization through a dashboard. Furthermore, the artifact must facilitate critical discussions regarding the consequences and possibilities associated with standardization. Overall, these requirements were selected as central objectives for the artifact at this stage of the process, with the ambition of facilitating organizational discussions that could help synthesize the affordances of process automation in alignment with the organization’s value-creation structure centred around the “capacity building of midwives and trainers.”

TO-BE BPMN Incorporating Identified Objectives of a Solution

Before turning to Activity 3 of DSR – *Design and Development* – we present our TO-BE BPMN, which was developed through iterative dialogue between the technical possibilities and the requirements of our solution. This helped translate identified pain points and solution objectives into a visual representation of a redesigned future-state process. This time, we focused on reducing manual

work and presenting a solution that allows employees to access data themselves — without necessarily relying on the IRL department to retrieve and organize the data. This resulted in the revised TO-BE BPMN shown in figure 20.

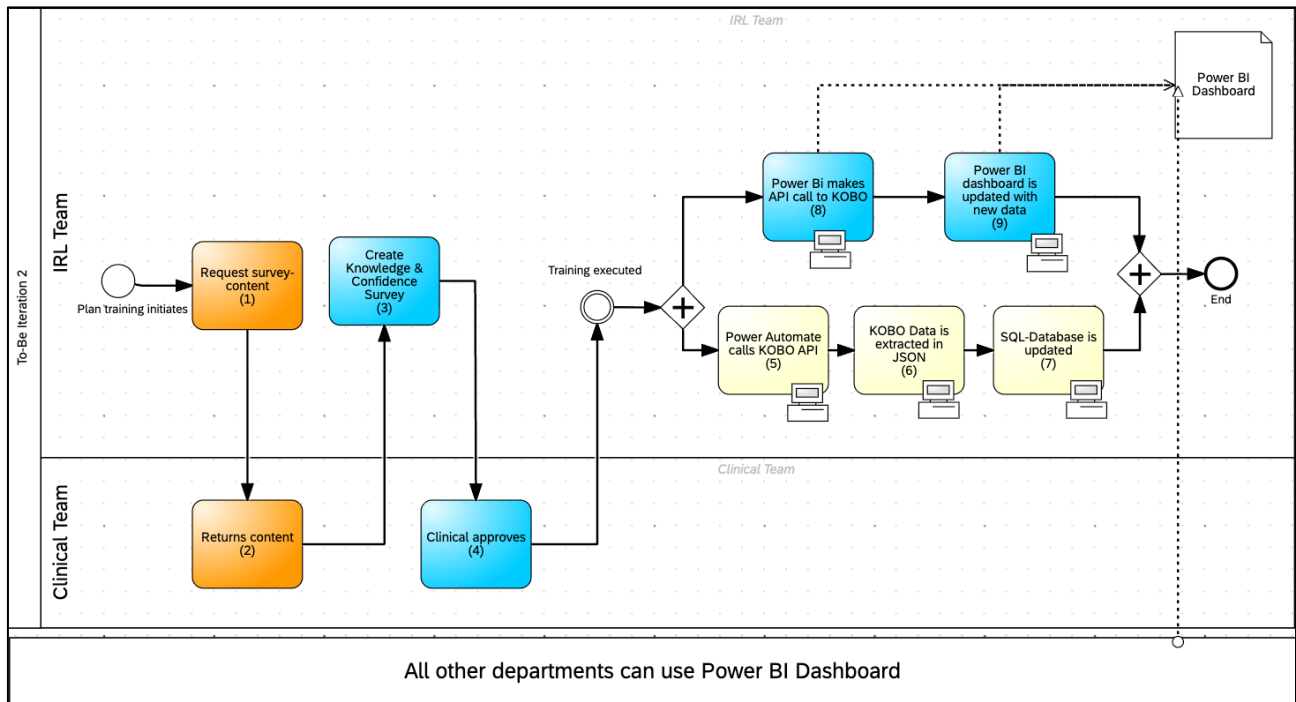


Figure 20: Second iteration TO-BE BPMN

Only the last two activities (8 and 9) have been added since the TO-BE BPMN from the first iteration (figure 15), encompassing two main redesigns. All activities related to data processing and analysis after a training session are now automated (8), and the data is visualized in an interactive Power BI dashboard accessible to everyone within MF (9). These additions complement activities (5), (6), and (7) in the TO-BE BPMN from the first iteration.

Since MF already uses Power BI within its Safe Delivery App infrastructure (figure 10), we decided that Power BI could equally serve as a useful platform for communicating insights from trainings. Power BI utilizes the same REST APIs as our first artifact to fetch data directly from KoboToolbox. Additionally, Power BI supports the “must-have” requirement of a KOBO dashboard. Through its accessibility and dynamic interface, it affords employees across MF the possibility to analyze, filter, and compare data independently. In this way, departments within MF no longer need to rely on the IRL team to retrieve data for impact reporting purposes.

Thus, this TO-BE process addresses several of the key operational pain points and thereby serves as a foundation for facilitating broader cross-departmental discussions regarding the possibilities and implications of process automation and survey standardization.

6.2.3. Readdressing Activity 3 - Design & Development

Having reassessed the objectives of our solution, we now move on to the activity of designing and developing an artifact that corresponds to these objectives. Firstly, we found that our artifact had to take on the role of a proof of concept aimed at illustrating the *concrete* possibilities and challenges associated with standardizing workflows and survey data. Through this, we aimed to facilitate an important discussion among key stakeholders grounded in a technical solution embedded within their existing IT infrastructure.

As we had discovered that the different survey types were not standardized, the designed artifact was developed using only data from the Knowledge & Confidence survey rather than multiple survey types. This decision was made for two reasons: (1) stakeholders had identified this dataset as the most important for impact reporting (Ahsa Appendix B), and (2) focusing on a single survey design allowed us to demonstrate actual value using real organizational data. The revised artifact is illustrated in figures 21, 22 and 23 below.

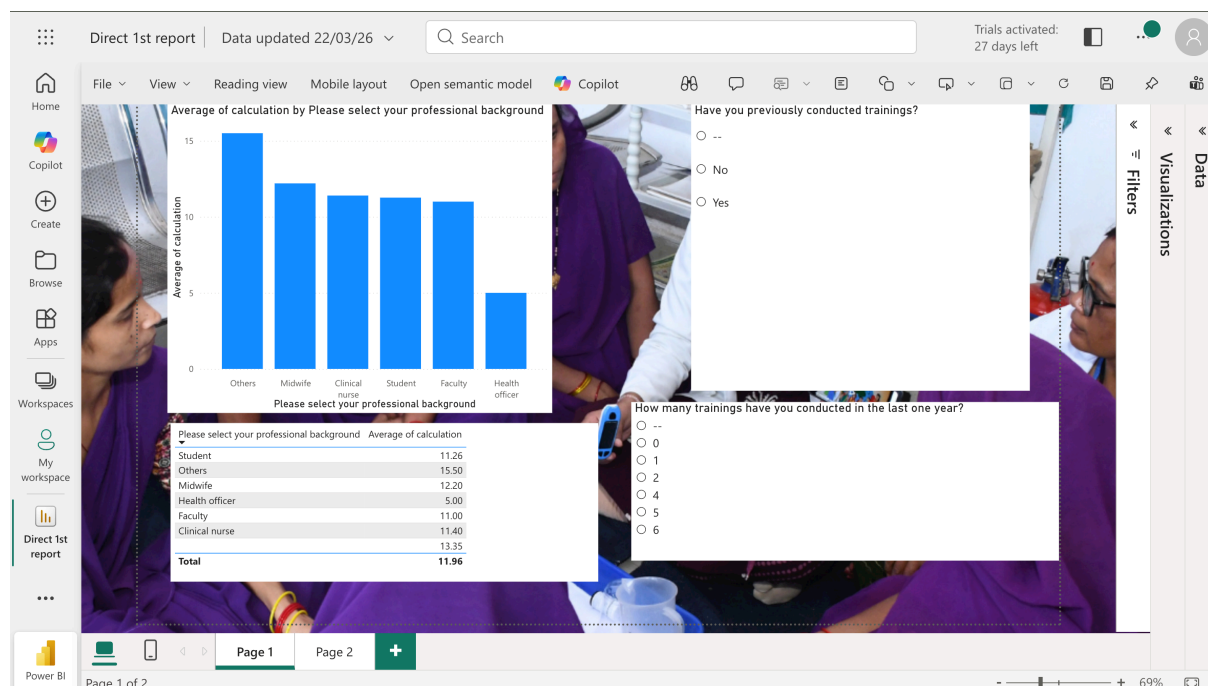


Figure 21 – Power BI Dashboard

The Power BI dashboard (figure 21) is centred around a graph displaying the average knowledge score across different professional backgrounds. The dashboard supports both the Clinical Team in internal learning processes and the Business Development Team in impact reporting activities. Employees are now able to filter and slice the graph data across various drop-down parameters, such as previous training experience. These parameters are highly customizable, as users can filter across all survey questions and dynamically modify the graph's x-axis.

Power BI's low-code/no-code functionality makes it both dynamic and particularly suitable for NPOs operating with limited technical resources. Essentially, employees would now be able to independently extract and analyze data for impact reporting purposes without relying on the IRL team.

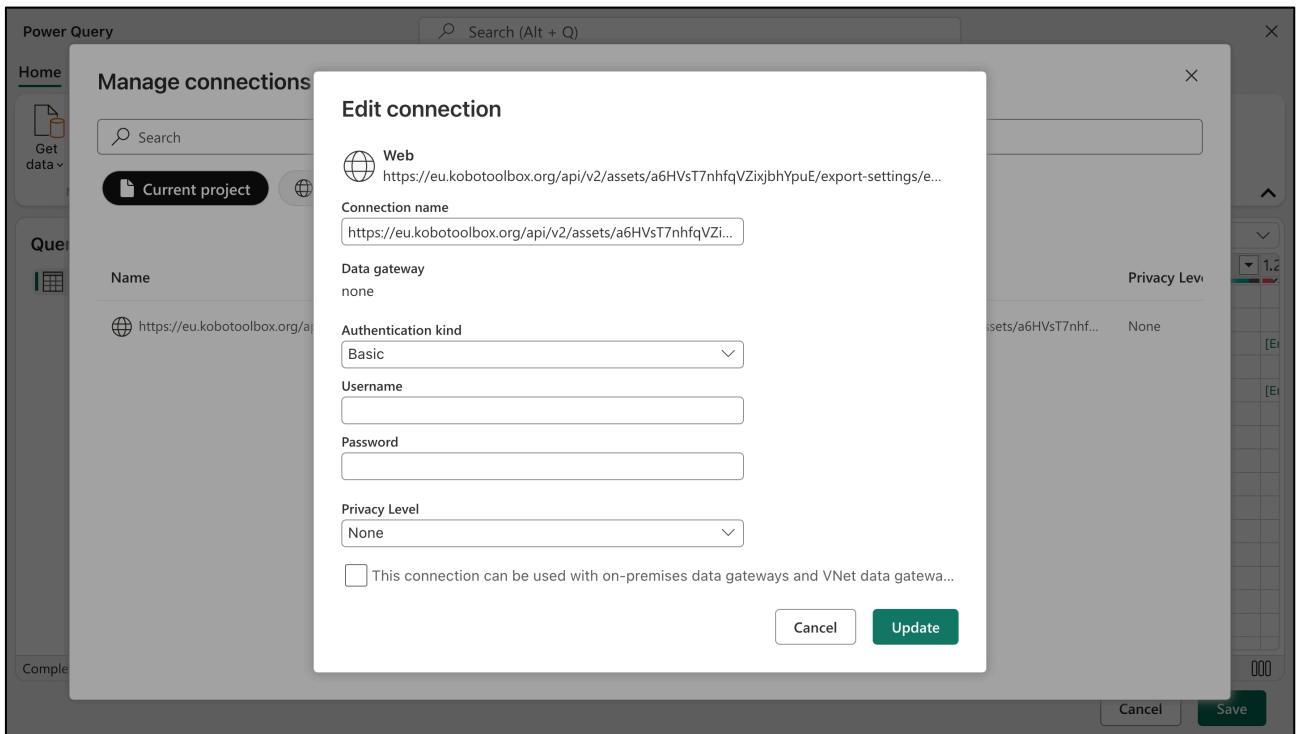


Figure 22: Power BI calls KoboToolBox's Rest API

Beyond providing accessible data outputs to relevant departments within MF, the artifact also addresses the pain point of manual data processing and extraction, as Power BI directly calls KoboToolbox's REST APIs to continuously retrieve updated data (figure 22). The data analysis is conducted automatically, and when new survey responses are submitted during a training session, the dashboard automatically updates with demographic data as well as knowledge and confidence scores.

Power Query

Home Transform Add column View Help

View ribbon: Data view, Schema view, Script view, Diagram view, Query settings, Go to column, Always allow, Advanced editor

Formula bar: Table.TransformColumnTypes(#"Removed columns", {{{"Please enter your age", type number}}})

Queries [1]	Is this a Pre...	Please enter your full na...	1,2 P...	Ple...	Plea...	I...	What is your job title?	Pleas...	I...	How many years of exp
1	PRE-training		39	Female	Faculty	null	Ms.	Masters	null	More than 10 years
2	PRE-training		53	Female	Midwife	null	Lecturer	Masters	null	6-10 years
3	PRE-training		[Error]	Female	Midwife	null	Midwifery lecturer	Masters	null	More than 10 years
4	PRE-training		44	Female	Midwife	null	Senior Clinic Midwife	Bachelors	null	More than 10 years
5	PRE-training		[Error]	Female	Midwife	null	Midwifery lecturer	Masters	null	More than 10 years
6	PRE-training		39	Female	Midwife	null	Midwife	Bachelors	null	1-5 years
7	PRE-training		37	Female	Midwife	null	Clinical midwife	Bachelors	null	More than 10 years
8	PRE-training		36	Female	Midwife	null	Midwifery Educator	Bachelors	null	6-10 years
9	PRE-training		65	Female	Midwife	null	Nurse Educator	Bachelors	null	More than 10 years
10	PRE-training		44	Female	Midwife	null	Midwife/Deputy Director Nurs...	Bachelors	null	More than 10 years
11	PRE-training		39	Female	Faculty	null	Ms.	Masters	null	More than 10 years
12	PRE-training		46	Female	Midwife	null	Midwife	Bachelors	null	More than 10 years
13	PRE-training		28	Female	Clinical nurse	null	General Nursing Officer	Diploma	null	1-5 years
14	POST-training		null	null	null	null		null	null	
15	POST-training		null	null	null	null		null	null	
16	POST-training		null	null	null	null		null	null	

Status bar: Completed (3.26 s) Columns: 89 Rows: 96 Column profiling based on top 1,000 rows

Buttons: Cancel Save

Figure 23 – Power BI data foundation – names are hidden

However, an important limitation of this aspect of the artifact concerns the underlying data infrastructure. Although the Power BI dashboard appears to function as an independent data environment (figure 23), it does not actually extract and permanently store data outside of KoboToolbox. Instead, Power BI continuously re-fetches the most recent version of the data directly through KoboToolbox’s REST APIs (figure 22). Consequently, if data were deleted or corrupted within KoboToolbox, these changes would similarly propagate into the Power BI environment. While the data appears to “live” within Power BI, it therefore remains highly dependent on the KoboToolbox infrastructure.

Thus, unlike the first iteration – where the Power Automate flow extracted and stored data within a SQL database – the Power BI artifact does not fully address the siloed and infrastructurally fragile nature of the training data. However, this limitation should be understood in relation to the MoSCoW prioritization developed for the second iteration. While permanent extraction and storage of KOBO data represented a “Should” requirement, the primary focus of the iteration was centred around the “Must” requirements of facilitating cross-departmental discussions regarding standardization trade-offs, visualizing the operational implications of process automation, and demonstrating the possibilities of a shared KOBO dashboard.

Consequently, the second iteration intentionally prioritized the communicative and organizational value of the artifact over full infrastructural redesign and implementation. Rather than attempting to provide a finalized technical solution, the Power BI dashboard functioned as a proof of concept aimed at concretely visualizing how standardized and automated survey infrastructures could support organizational learning, impact reporting, and cross-departmental accessibility to data. Simultaneously, the artifact also made visible some of the infrastructural limitations and requirements and trade-offs associated with relying on lightweight “outside-in” automation approaches.

In this regard, the artifact does not simply demonstrate what process automation could technically accomplish. It also enables broader organizational discussions regarding what type of value the process should ultimately deliver within the distinct accountability structures and operational conditions of MF. In the next section we will demonstrate both the first iteration data extraction through Power Automate and the Power BI dashboard data visualization. Demonstrating these artifacts supports the third and final iteration design of an artifact from which we can initiate a cross-departmental discussion.

6.2.4. Activity 4 – Demonstration of ‘proof-of-concept’-Artifact

Having designed and developed an improved artifact, we facilitated an interview with Tian and Ahsa aimed at *demonstrating* the artifact. Importantly, the activity of demonstration differs from *evaluation* by being less formalized and more oriented toward illustrating how the artifact could be used in practice. Our ambition with the interview was to gain more in-depth feedback on the technical aspects of the artifact and to assess whether we were designing a solution that was either too distant from their current IT infrastructure or misaligned with their “futuristic plans.”

Again, the ambition of DSR is to address “real-world problems” rather than merely theorizing about them from afar. Accordingly, the demonstration was considered important in determining whether the artifact was meaningful and operationalizable in practice from the stakeholders’ perspective (Hevner, 2004). Our demonstration therefore focused on whether the artifact was designed in a meaningful and practically relevant way, as perceived by Ahsa and Tian. Through this activity, we moved away from the more abstract design phase toward a more concrete discussion of usability and practical applicability.

Demonstrating the Artifact to Ahsa and Tian

The demonstration was conducted online and took place through online screensharing. We began the interview by contextualizing the challenges we had identified before sharing our screen and demonstrating the solution. The presentation largely entailed the elements described in the previous section regarding Activity 3 – *Design and Development* and will therefore not be repeated here. Instead, we turn to the feedback provided after the demonstration, as this informed the third and final iteration.

However, before presenting the feedback, it is important to clarify that the purpose of the demonstration was primarily to improve the technical aspects and infrastructural alignment of the artifact. Consequently, the activity was not directly concerned with the broader objective of facilitating cross-departmental discussions regarding the possibilities and challenges associated with process automation in impact reporting. Rather, the design of a proof-of-concept-artifact aligned with their existing IT infrastructure, future-oriented plans, and real-world organizational conditions was viewed as a crucial foundation for enabling meaningful discussions regarding the implementation of process automation and RPA.

Their initial reaction was positive, and they said that “this is really nice. You've done a wonderful job, no doubt about that” (Tian, Appendix F, p. 2) and “Yeah, no, yeah, and like Tian said, everything looks very great” (Ahsa, Appendix F, p. 4). Specifically, Tian sees potential in our artifact as it has the potential to connect into the SQL-database where the app-data exists:

“I'll tell my preference here and why I'm saying that. If I get this data, probably I can create more advanced use cases from the data that we have at KOBO right now. If everything comes into SQL Server, I can actually create more AI use cases around this. Because I can then understand the behaviour of the users as well, or understand, do the implicit and explicit profiling of the user based on the question that has been asked. So from my technical perspective, I would love to see the data in the SQL Server for sure.”

(Tian, Appendix F, p. 5-6)

Accordingly, the most important element for Tian was that the data should be extracted from KoboToolbox and inserted into the SQL Server, and “not the Azure SQL” (Interview 6, Tian,

Appendix F, p. 3), as the current artifact does. Tian elaborated on this by explaining that Data Factory (a data-processing tool from Microsoft) could be used as a middle layer to convert the data into a relational SQL format: “[...] using the functionality of Data Factory, we can actually pull the data from Kobo and put it into the SQL Server DB, that would really help.” (Tian, Appendix F, p. 3). Additionally, Tian highlighted the potential for using the data in future “AI use cases.” Naturally, our artifact needed to align with their future-oriented plans in order to increase the fidelity of the artifact and facilitate not only discussions regarding how process automation could improve current workflows and reduce manual work, but also discussions concerning the future possibilities such infrastructures could enable.

Tian further expanded on the future-oriented AI perspective by stating: “Yeah, so we are exploring different AI use cases and I see that there is a value of the data that we are capturing here. That would be really helpful in profiling a user, right?” (Tian, Appendix F, p. 7) Here, it is explicitly described that the envisioned AI use cases may revolve around profiling users, in which case Tian viewed the automatic extraction of survey data into a SQL database as particularly valuable. Ahsa similarly described that the automation process “will be really helpful if it is something more easy to calculate at the end of the training.” (Ahsa, Appendix F, p. 9). Ahsa further elaborated by explaining that:

“But connecting to the SQL [database] directly would be an added advantage for us to visualise both the data set together somewhere down the line. [...] So I guess that will make the process more easy in terms of having all the data set together and doing the visualization.”

(Ahsa, Appendix F, p. 4)

To summarize, Tian and Ahsa perceived the artifact as fundamentally aligned with their organizational infrastructure. Furthermore, they highlighted how it could enable more flexible use cases and faster data processing. Moving forward, they specifically suggested: (1) converting the JSON format into a relational format within a SQL Server, and (2) having Power BI retrieve data directly from the SQL Server instead of KoboToolbox. The feedback from this demonstration therefore encouraged us to cycle back to the second activity of the DSR process and revise both our Objectives of a Solution and the design of our artifact. This initiated the third and final iteration, which will be described in the following section.

6.3. Iteration 3 – Final Iteration

Based on the demonstration with Tian and Ahsa and their suggestions, we cycled back to Activity 2 and Activity 3 and thereby embarking on our third and final iteration. The following section outlines how the feedback from the demonstration informed this iteration, which was oriented toward the final demonstration of our artifact.

6.3.1. Activity 2 - Readjusting the Objectives of our Solution

This section serves to translate the newly acquired knowledge regarding requirements, as well as insights into what was technically possible and realistically achievable, into concrete objectives for the artifact. As the demonstration was highly oriented toward the technical details of the artifact, we gained important new insights into how to prioritize its technical aspects. Which in turn will lay the foundation for cross-departmental discussions.

Using MoSCoW to Readjust the Objectives

After the demonstration-interview it was clear that we had to reassess our prioritization-schema from iteration 2. After careful assessment, we made this final MoSCoW-prioritization in figure 24:

Must	Should
- Support cross-departmental discussion regarding standardization trade-offs - Visualize operational implications of process automation - Include a visual representation of a “KOBO-dashboard” - Use Data Factory as a Middle Layer between KOBO and Power BI - Extract KOBO-data into SQL-server in relational data format	- KOBO-data should be stored in a format allowing comparability with APP-data - KOBO-data should get extracted out of KOBO - Preserve interpretability of local survey contexts
Could	Won't
- Support a rethinking of the current KOBO-survey data-infrastructure.	- Full organizational implementation/integration

- Touch upon the performance dimensions of cost, time and quality - Present our example of a standardized “Master Form”?	- Be categorical and argue for what is the right solution moving forward
---	--

Figure 24 – Final MoSCoW

Three elements have changed in this new and final prioritization schema. Firstly, we interpreted the use of Data Factory as a middle layer between KoboToolbox and Power BI as a “must-have” requirement for the artifact. Secondly, the KoboToolbox data must be extracted and inserted into the SQL Server in a relational data format. This also means that the earlier requirement concerning the extraction of data from KoboToolbox has now been moved from a “Should” to a “Must,” following the technical modifications and insights gained throughout the demonstration. Otherwise, the prioritization schema remains largely unchanged. In the next section, we explore how these changes affected our final TO-BE BPMN.

Readjusting the Final TO-BE BPMN to the Revised Objectives

During the interview we did not gain new understandings of the process. Therefore, the AS-IS BPMN (figure 25) remains unchanged from the previous (figure 18):

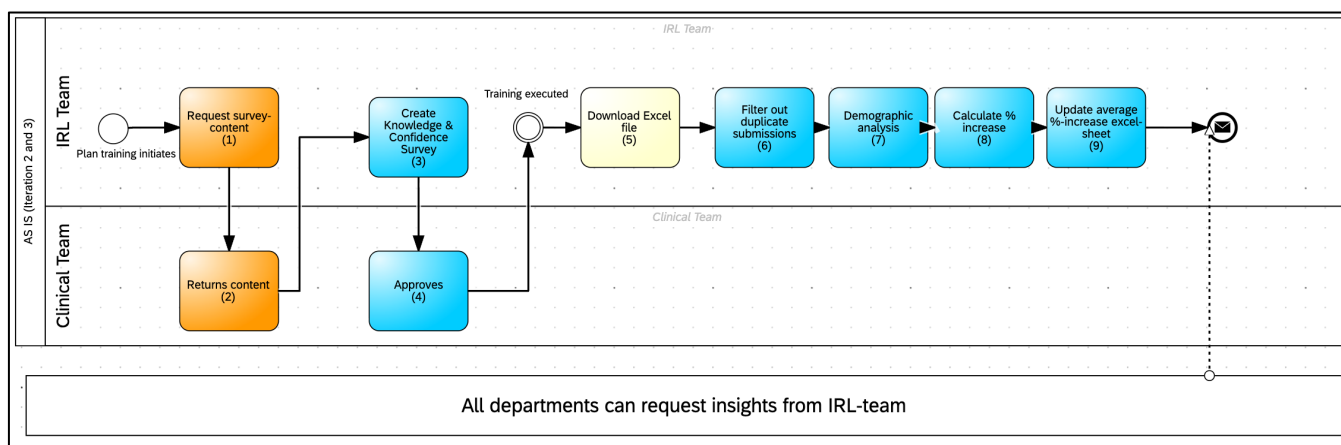


Figure 25 – Iteration 3 AS-IS BPMN

The question instead centred around how to reflect their technical requirements in the best possible way in our TO-BEBPMN. Through dialogue with the *design & development* activity we mapped this out in the below model (figure 26). This is the final BPMN-model that adheres to the new objectives – the new activities are highlighted by being blue:

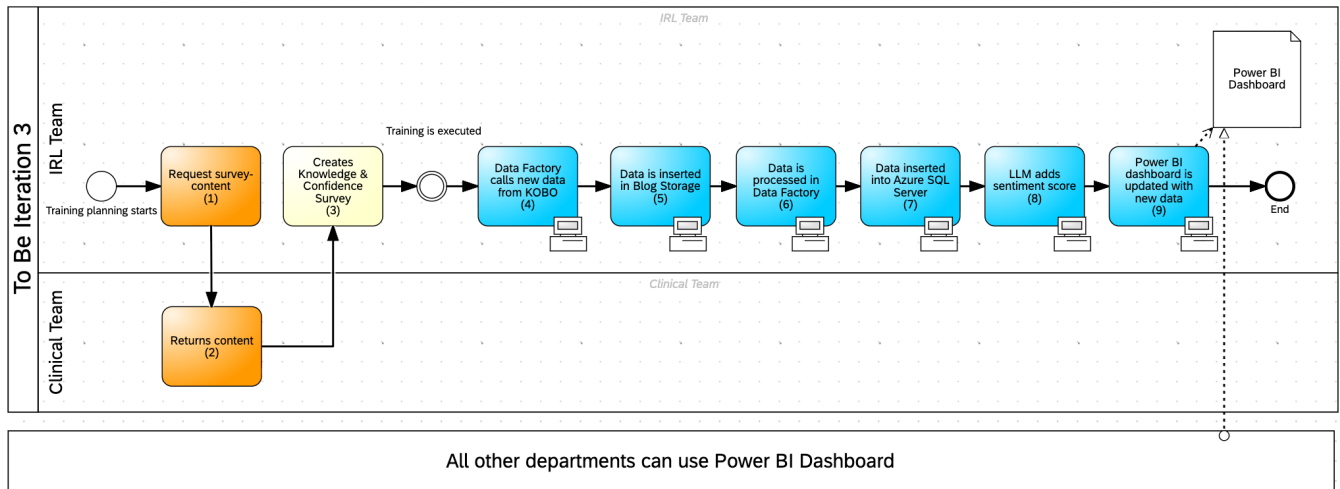


Figure 26 – Iteration 3 TO-BE BPMN

The idea was that the data extraction and visualization should be integrated into a single solution with the help of Data Factory. In the second TO-BE BPMN (figure 20), these components were separated. In the revised process, Data Factory is responsible for calling the KOBO API (4). It extracts the data and inserts it into temporary Blob Storage in Azure (5). Subsequently, Data Factory processes the data (6) and inserts it into a SQL database in Azure (7).

This infrastructure then enables the introduction of RPA in the form of a small Power Automate flow, in which an LLM analyses the qualitative survey feedback and assigns a sentiment score (8). Finally, Power BI now calls the SQL database to visualize the data (9), rather than calling KoboToolbox directly. The process is elaborated on in greater depth in the following *Design and Development* section.

6.3.2. Activity 3 – Final Reassessment of Design & Development

The third iteration artifact was designed with the purpose of enabling a broader organizational discussion regarding the future of MFs impact reporting infrastructure. The goal was to create an artifact that functions with real data and also highlights the trade-offs of a more standardized approach. Rather than focusing solely on technical optimization, the iteration sought to visualize how process automation would affect organizational workflows, data governance, and cross-departmental collaboration. The design process was therefore guided by the final MoSCoW prioritization presented

in figure 24, which functioned as a boundary-setting mechanism for what the artifact should and should not attempt to achieve.

First, the artifact *Must* support cross-departmental discussions regarding standardization trade-offs. Through the previous iterations and interviews, it became increasingly clear that the main challenge was not simply automating data extraction, but rather the fragmented and inconsistent structure of KoboToolbox-surveys. Consequently, the third iteration was not centred around a finalized technical solution, but as a tangible discussion artifact capable of illustrating the infrastructural implications and possibilities of different standardization choices and process automation in relation to impact reporting.

Additionally, the artifact had to visualize the operational implications of process automation. To support this, the infrastructure was redesigned around a layered architecture where Azure Data Factory acted as an intermediary between KoboToolbox, SQL and Power BI. This differed significantly from the previous second iteration, where Power BI directly fetched data from the KoboToolbox API. The new architecture enabled the visualization of how survey data could be extracted, transformed, standardized, and stored relationally before being visualized. In this sense, the artifact moved beyond a simple dashboard and instead illustrated a potential future data infrastructure for impact reporting. While also introducing training data into the Safe Delivery App Ecosystem.

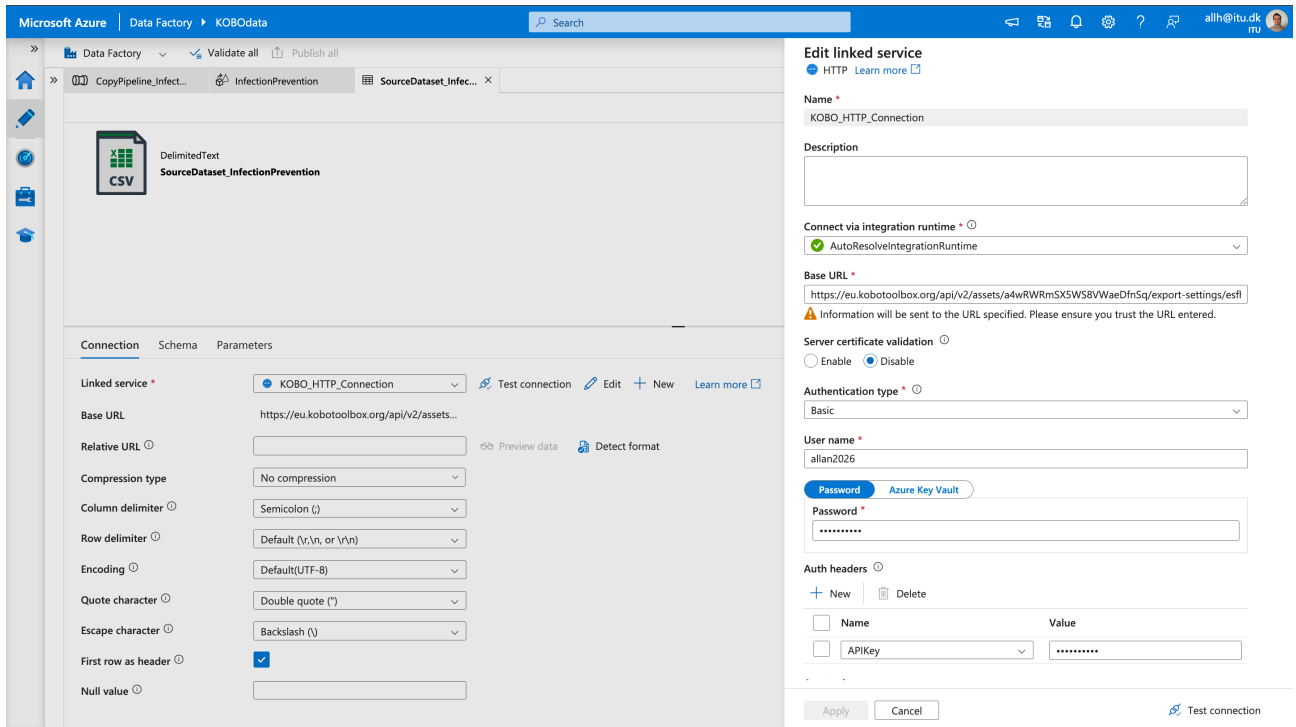


Figure 27: Data Factory calls KOBO REST APIs

As illustrated in figure 27, Azure Data Factory now calls the KoboToolbox REST API to extract survey data. This design choice directly addressed the *Should* requirement stating that KoboToolbox-data should be extracted out of KoboToolbox rather than remaining in isolated survey exports in excel. The extracted data is temporarily stored in an Azure Blob Container, shown in figure 28, as Data Factory cannot directly insert raw API responses into a relational SQL-database.

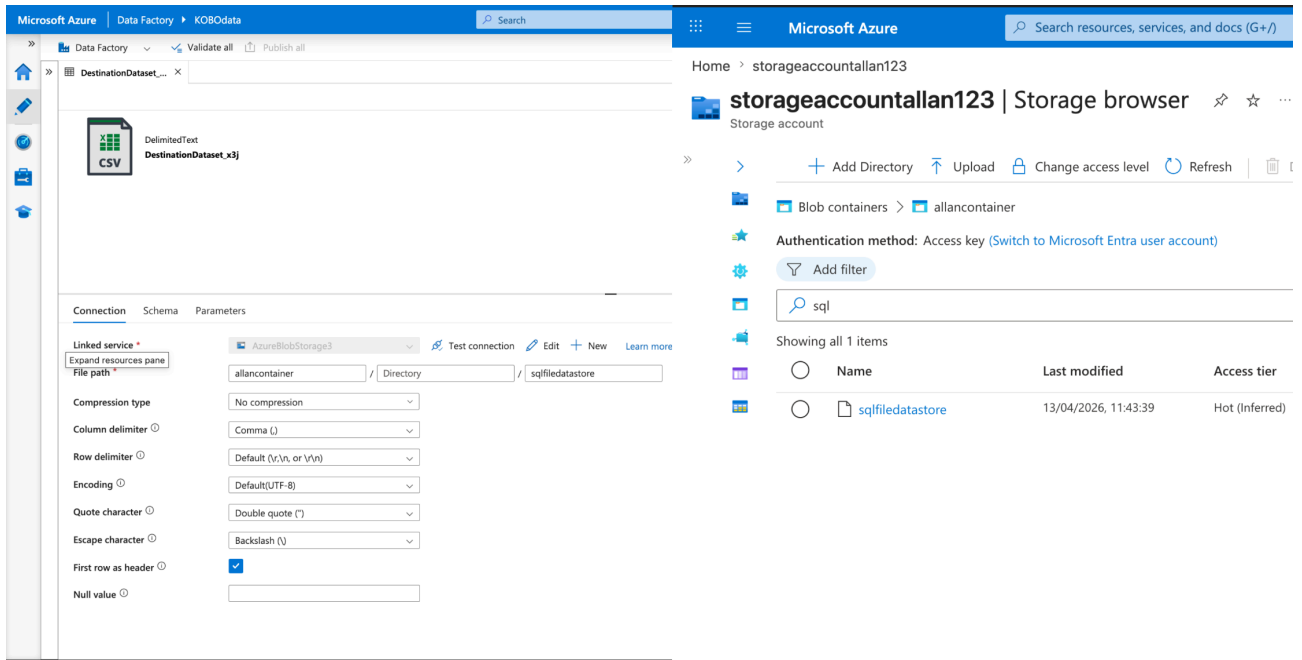


Figure 28: Data stored temporarily in Blob Container in Azure

Within Data Factory, the data undergoes a transformation process where columns are mapped, datatypes converted, and duplicate values removed before insertion into Azure SQL. This relational restructuring was an important design decision, as it operationalized the *Must* requirement of storing KoboToolbox-data in a relational format while simultaneously supporting the *Should* requirement of enabling comparability with APP-data. The relational model therefore represented more than a technical optimization; it visualized how standardized survey structures could potentially enable scalable analytics and future integration across organizational systems.

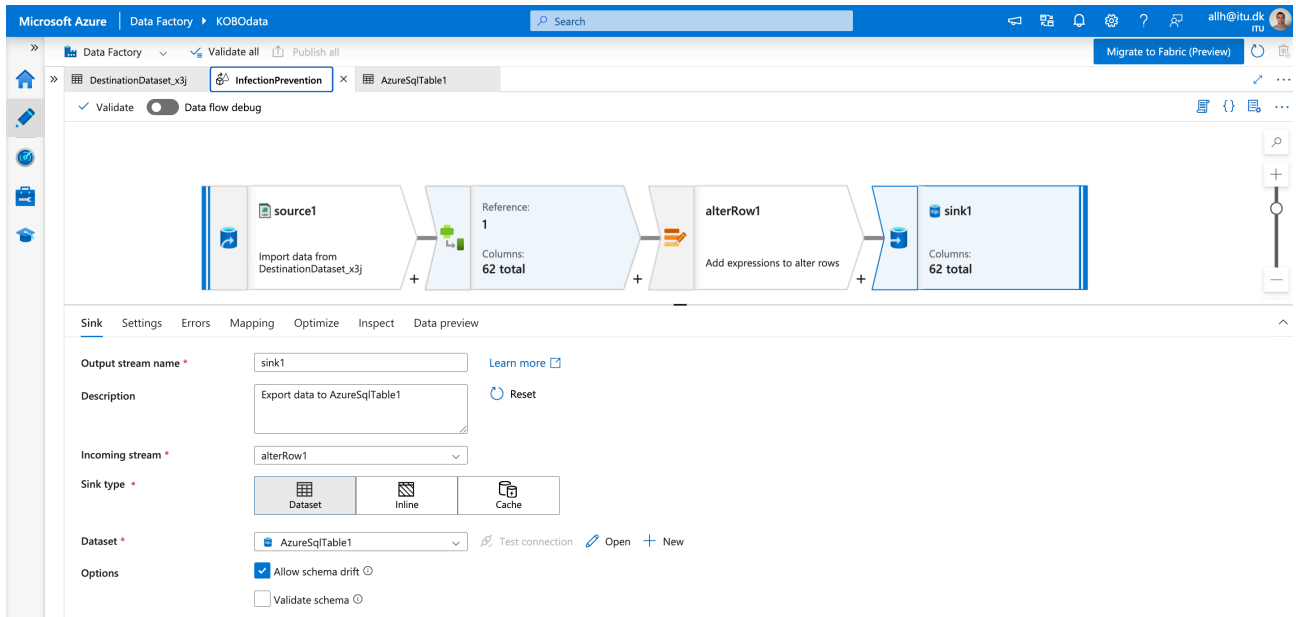


Figure 29: Data stored in Azure SQL-database

Figure 29 illustrates the resulting Azure SQL-database, which functioned as the centralized middle layer of the infrastructure. By introducing this layer, the artifact also intentionally supported the *Could* requirement of encouraging a rethinking of the current KoboToolbox-survey infrastructure. The infrastructure demonstrated how survey data could move from being isolated and project-specific toward becoming part of a more unified organizational data architecture. However, it also highlighted the trade-offs associated with increased standardization, particularly the reduced flexibility for the Clinical team in relation to survey design. At the same time, the artifact deliberately avoided becoming prescriptive. In line with the *Won't* requirements, the iteration did not attempt to argue for a singular “correct” infrastructural solution, nor did it represent a fully implementable system or process. Instead, the artifact functioned as a hypothetical, yet grounded design proposal intended to facilitate reflection, discussion and to support common language and bilateral understandings (Radu et al., 2024).

To further visualize the implications of process automation through RPA, a Power Automate flow was integrated into the infrastructure, as shown in figure 30 (see also appendix M). Whenever new data was inserted into the SQL-database, the flow automatically extracted qualitative feedback fields, submitted them to an LLM for sentiment analysis, and inserted the resulting sentiment scores back into the database. This addition was primarily included to illustrate how intelligent automation technologies (LLM model) could potentially complement traditional RPA approaches when dealing

with unstructured qualitative data (Nunes et al., 2026). But it was also included to showcase the need for a suitable and robust infrastructure to operate upon (outside-in automation) to enhance actual value delivered. The RPA inclusion also touches upon the *Could* requirement of addressing performance dimensions such as quality and time, as qualitative feedback could now be analyzed automatically rather than manually.

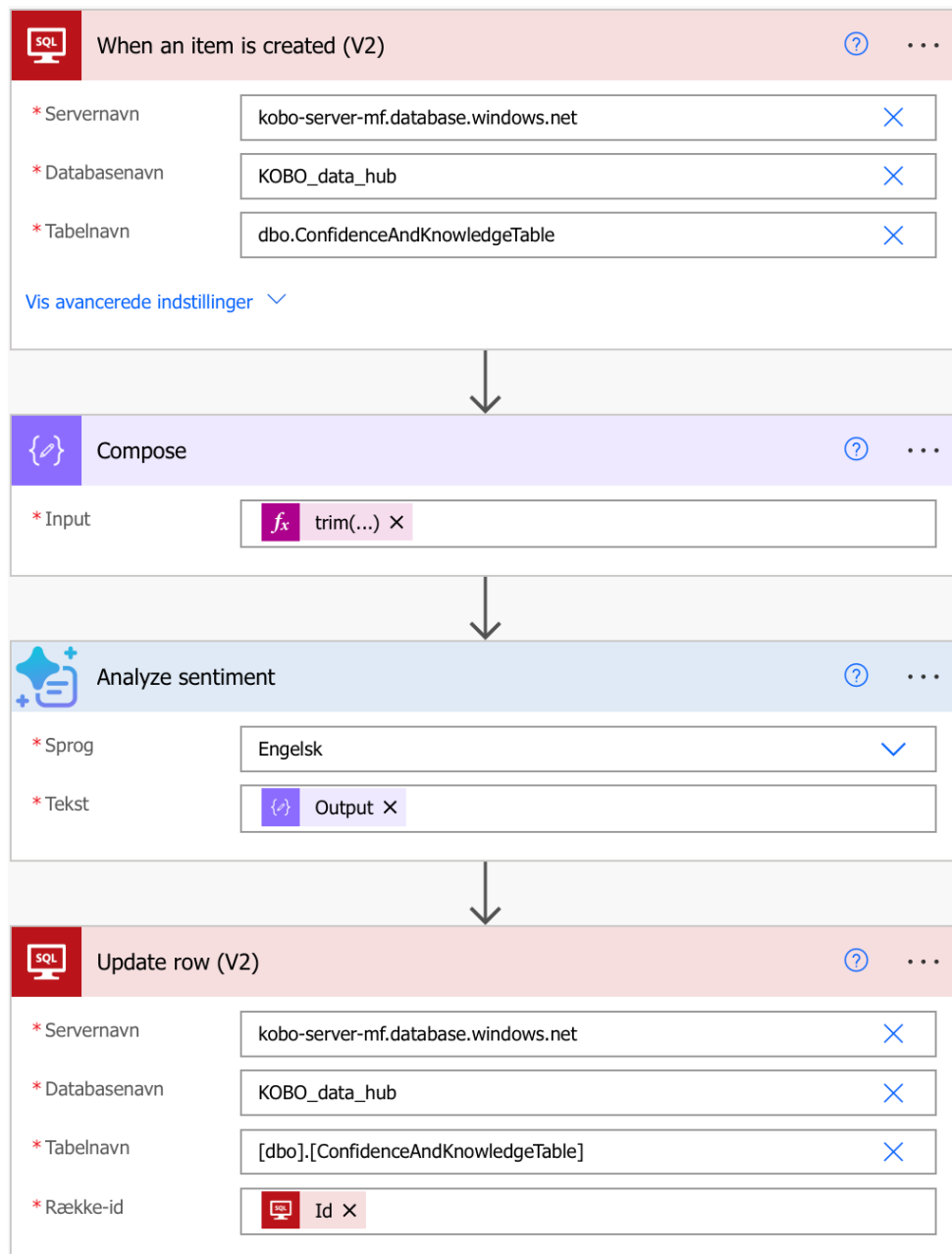


Figure 30: Power Automate adds sentiment score (Appendix M)

Finally, Power BI visualized the processed data from the SQL-database rather than directly querying KOBO, as illustrated in figure 31. The dashboard was expanded with additional visualizations and slicers related to gender, age, country, educational level, and confidence increase. These additions were introduced to increase the fidelity of the artifact and thereby better support the final evaluation with stakeholders. The dashboard also included a visual representation of a potential “KoboToolbox-dashboard”, fulfilling one of the central *Must* requirements.

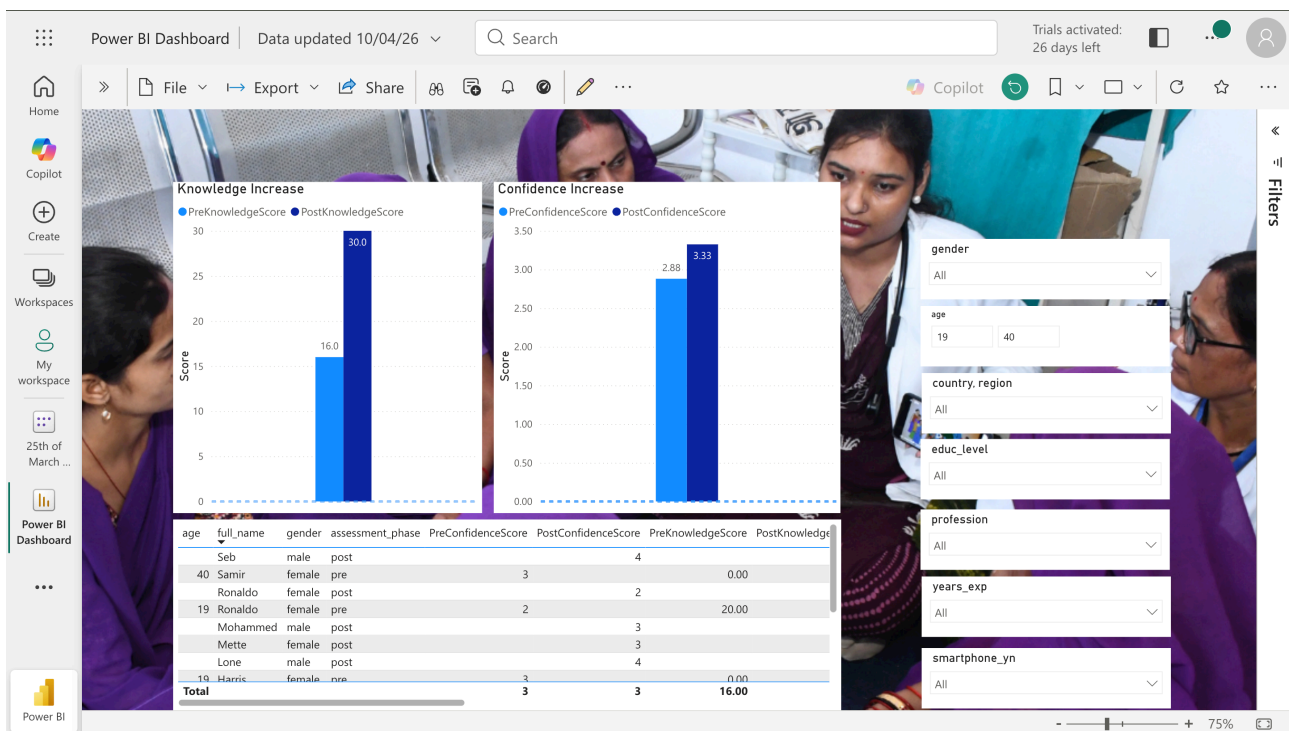


Figure 31: Final Power BI Dashboard

Overall, the third iteration represented a shift from designing a narrow automation artifact of the current AS-IS process, toward designing an infrastructural conversation artifact. The purpose was not only to demonstrate technical feasibility, but also to make visible the organizational trade-offs, governance implications, and strategic decisions embedded within process automation and digital transformation initiatives.

6.3.3. Activity 4 & 5 - Final Demonstration & Evaluation of Artifact

As described in the methodology-section, evaluation of artifacts serves as a key activity in Design Science Research by assuring rigor of the research as well as providing feed-back for further

development (Venable, et al., 2016). In accordance, after redesigning our artifact we set out to present it for the organization and evaluate to what extent it contributes to solving the *problem-identification*. This ambition however raised important methodological questions regarding how to evaluate it, who to invite and how to generally assess if our artifact accomplishes what it set out to? This section will sketch out our evaluation strategy.

Evaluation Approach and Criteria

The evaluation approach combined elements of scenario-based evaluation and informed arguments (Hevnet et al., 2004). First, the proof-of-concept artifact functioned as a *scenario* of a potential future automation architecture, allowing stakeholders to discuss its organizational implications and assess how it might influence impact reporting practices, standardization efforts, and future digital strategy. Second, the evaluation incorporated *informed argument* by grounding assessments of utility, efficacy and validity in prior literature on digital transformation and process automation initiatives in NPOs. This approach was chosen rather than relying on quantitative performance measures.

Grounded in the identified *Objectives of our Solution* and our evaluation approach we decided to organize the evaluation towards the following criteria (figure 32):

Criteria	Description
Common language - making process automation trade-offs clear	Does it clarify the potentials and challenges of implementing a process automation solution concretely?
Facilitation of concrete discussion on standardization and flexibility	Does it stimulate concrete organizational discussion and dialogue around this topic?
Stimulate a discussion around what counts as “impact”	Does it facilitate reflections regarding what counts as impact – and letting this guide the data-collection and data-analysis
Inform future digital strategy	Does it illustrate tensions relevant to for their future digital strategy

Figure 32: Criteria for Evaluation

This aligns with the cross-cultural, -organizational, -strategical and -technical focus of our artifact – oriented towards facilitating a *common language* of the trade-offs related to implementing process automation technologies to improve impact reporting. These criteria reflect what the purpose of our artifact was oriented towards solving. The limits and challenges embedded in these criteria will be

elaborated on. For now, it is relevant to highlight that we acknowledge limitations present when we – as researchers – interpret if the artifact meets these criteria based on stakeholders' descriptions, reflections and internal discussions. We tried to mitigate these challenges by being as truthful to the empirical data as possible.

Evaluation Attendees & Procedure

We invited all informants interviewed this far to the final interview – to get as many different perspectives and insights as possible. We also informed them, that anyone working in the organization was welcome to join. The ambition hereby was to ensure that we got feedback from different relevant perspectives; a strategic and organizational level, data- processing and IT-infrastructure level, impact reporting and social impact level and operational level. Due to several of the informants having a tight schedule and being anchored in the practicality of the real-world, some of the key informants could not attend. This meant that Tian, Eke and Ulrika were unable to attend, eliminating highly relevant insights from the IT-, Clinical and Programme department. This was unfortunate as our artifact had taken their requirements and perspectives into careful consideration. This will also be reflected upon in the discussion. The following stakeholders attended:

- Ahsa from IRL
- Atana from IRL
- Eisha from Communication
- Ileen from IRL
- Yvette from Business Development

Importantly, a new stakeholder, Atana, participated for the final interview – without us knowing in advance. Atana had just been hired into as interim Head of IRL-department.

We started the interview by introducing the agenda of the interview and highlighting that we were interested in hearing all their reflections and thoughts on the artifact – both from a technical perspective but also regarding impact reporting, social mission and standardization vs. flexibility aspects. Our PowerPoint slides from the meeting can be seen in Appendix K.

We then proceeded to situate our artifact by comparing the final TO-BE BPMN (figure 26) that the artifact sets out to facilitate and compare it to how it is – the current AS-IS BPMN (figure 25). This was done to *visualize the operational implications of process automation* – as the objectives of our solution prescribes. We then presented the IT-infrastructure as described above in *design & development*. (For further information about the artifact, see walk-through video in Appendix L). We also kept iterating that the “solution isn't as such finished, it cannot be implemented as of now. It's more like a framework for discussing the pros and cons of digital elements.” (Appendix G, p. 1). Both during and after the presentation we encouraged the informants to comment, ask questions and participate which fostered fruitful discussions and perspectives. These will be highlighted in the next section.

Evaluation Findings

We will now highlight the central findings from the evaluation and discuss these in relation to the criteria (figure 32) mentioned above.

Criterion 1 – Common language

The first criterion centred around if the artifact made it possible to discuss the problem more *concretely* and not just theoretically and abstract. In relation to this, the new interim Head of IRL was quick to point out how relevant it was that we facilitated this discussion:

“So I'm just going to interrupt. I think this is great. Perfect timing.[...] There's an opportunity because of the transition to a new platform for the app to do things a little differently.[...] and I'm like, we need the data stored differently so that we can do interesting analysis. This is a nice start for that.” (Atana, Appendix G, p. 1).

This quote entails several interesting elements. Firstly, Atana points out that it is really important to have this discussion to provide organizational clarity regarding how to use and store data in the organization. As such, we interpret that Atana agrees with our *problem-identification*. Secondly, Atana points out that only by storing the data differently can they do “*interesting analysis*” (Appendix G, p. 1). Atana further describes that she had just had discussions with director of External Relations (Mike) and the CEO (Anna) about the:

“idea of having data integrated so it can be analysed in a more bivariate way and, you know, perhaps even ultimately multivariate way is very appealing. It's clear that that's the next step that we need to take with all the interesting information that we collect and keep but have trouble using.”

(Appendix G, p. 5).

Atana further adds, that a realistic mapping of the possibilities could support her in her discussions with those who allocate the budgets:

“if we can...quite clearly map out the advantages, the potential analyses and value addition as we started. I think it will make a great conversation to have with people who make decisions about how to allocate budget, available budget. So, you know, we shouldn't just stop at... [cost], oh my God, that's the kind of IRL things that we need strategically as an organization.”

(Atana, Appendix G, p. 5)

In this quote Atana emphasizes just how important it is to have a foundation and shared basis for discussing the potentials and challenges of implementing process automation. This would also be a healthy starting point in discussions with regards to the financial aspect of the implementation – to enable resource allocation. And as such, we interpret that Atana finds that the artifact helps in creating supporting a common language and bilateral understanding by mapping out the “*potential analyses*” and “*value additions*”.

On a practical note, Ahsa points out that the proof-of-concept would be desirable because it caters to the pain points raised throughout earlier interviews regarding repetitive manual work and data cleaning:

“So I guess... In terms of data cleaning, that will be super important and time saving for us in terms of having that as part of the initial data cleaning process to get the duplicates removed. And in terms of dashboard, also, this is something that we want to have as an overall for all the projects”

(Ahsa, Appendix G, p. 5)

In this description, Ahsa emphasizes that the presented proof-of-concept artifact helped visualize how a process automation could potentially improve their process and the *time*-saving pain point. As such,

we interpret this as indicating that our proof-of-concept helped facilitate a *common language* and *bilateral understanding* of how process automation could be time-saving and improve quality by making an accessible dashboard for all projects. In the next section we will dive into how it – to some extent – also helped constitute a discussion around the highly complex topic of standardization vs flexibility.

Criterion 2: Facilitating Discussion Regarding Standardization and Flexibility

A highly challenging aspect of the process centred around the flexibility and standardization possibilities. A central question then was, if our artifact helped facilitate organizational discussion regarding this – with our artifact as a point of departure for how standardization and process automation could potentially be implemented. We will argue that the following descriptions from the interview expressed that our artifact did just this. Ahsa – for example – expressed that the set-up and process automation that we presented was really nice, but that: “we have to have the standardized forms to have that set-up because with the current system, it will be difficult to have that in place.” (Ahsa, Interview 7). As such, Ahsa finds the isolated process automation appealing but highlights the issue regarding the non-standardized forms and the related non-standardized data as a foundational issue. Atana – the new interim manager of IRL – then describes her view on how to solve this specific matter:

“It's been done for decades [in other contexts] where there are standardized questions and modules that are used everywhere, and then there are additional questions or additional entire modules that can be added. [...] So I think we absolutely can find that balance between the standardization that we need to make”

(Atana, Appendix G, p. 6)

This quote is interesting as it highlights that the current process has not found the right balance between standardization and flexibility, as it perhaps favours the flexibility-aspect too much. Ileen from IRL then shares her thoughts on the discussion by voicing concerns regarding a learning-curve and highlights the importance of having different forms – and thus flexibility – because “there are different types of trainings and that's why we have different forms as well” (Ileen, Appendix G, p. 6). Ileen elaborates by saying that IRL would be resistant towards the complete standardization: “I think that is something that they [Clinical Department] want to maintain, that if we are doing a country-

specific training, we want to tailor it to the country.” (Ileen, Appendix G, p. 7). This then turned into quite an interesting discussion between Ileen and Atana:

Atana: “But everything will be comparable moving forward, so it can be automated in terms of summarization.”

Ileen: “I understand that bit, but my question is if we have different types of forms.”

Atana: “We cannot allow that in the future.”

Ileen: “[...] I’m not sure if KOBO at this stage can handle something like that, because it is also a resource issue, again, and we don’t have everything that we want them to have. [...] So it’s just good to have the idea. If we cannot do it, then what? Of course, we all want to do it because it’s easier and it makes sense logically, but what if we are not able to?” (Appendix G, p. 7).

In this excerpt we see a highly interesting discussion between two central stakeholders unfolding, which is exactly what we hoped our artifact would accomplish. Interestingly, Atana points out that *we cannot allow for that [different types of forms] in the future*. Ileen then raises concern if KOBO can handle Atana’s specific proposal for a “master form” with the possibility for additional questions. Ileen’s reluctance and doubt regarding the possibility of standardization hints at a crucial tension point in the organization revolving around the barriers of *lack of culture of change*, *lack of conceptual skills* and *lack of processing alignment* (Radu et al., 2024). As such, we will argue that our artifact helped facilitate a discussion around what IT-tools to use in the future as perhaps the KoboToolBox cannot meet their specific demands for both flexibility and standardization.

Criterion 3: What Counts as “Impact”?

Regarding the criterion concerning the extent to which our artifact served as a foundation for broader reflections on what constitutes impact, the following statements suggest that the artifact facilitated several valuable reflections. The following remark by Atana indicates that standardization should not be pursued at all costs: “We need data points, but we also need to understand the how and the why behind those data points” (Atana, Appendix G, p. 14). This reflects a central discussion within the broader challenge of balancing impact reporting with the understanding of social impact. As such, Atana's statement can be interpreted as an awareness of the risks associated with succumbing to purely *calculative practices* (Rana & Cordery, 2024). Atana further elaborates on this perspective by stating:

“what I'm interested in is a level of standardization that facilitates automation to free up time, but also that we do it in a way that we can, we have trust in the data... that it's... it's actually representing what we mean it to represent.”

(Atana, Appendix G, p. 10)

Consequently, Atana acknowledges the importance of having data that represents what it is meant to represent, which is a central tension-point in NPO accountability. This specific discussion is rounded off by Atana with: “there's a desire to make progress as an organization and reaching a consensus-based conceptualization of impact that we can then try to provide data for.” (Atana, Appendix G, p. 13 Interview 7). Herein lies what Atana sees as a central challenge for the organization – to reach a broad conceptualization of what impact is – and then try to provide data for that, and not the other way around. We will argue that our artifact proved a beneficial starting point for discussing this element by highlighting the importance of agreeing on certain metrics of impact before implementing a process automation process that might just reinforce an unnuanced metric that does not actually represent what the organization aims at producing. Atana rounds off this discussion by saying:

“So the standardization and comparability issue also links with the trust and reliability, so I don't want standardization to make things rigid and not contextualizable, cause that's important, but I want there to be enough where we feel confident that these data points or answers to these questions are actually comparable and we can summarize them together. That's what I'm concerned about. And I think we probably have a little bit of cleanup to do. And this, you know, this is an important opportunity [...] So I just want to make that clear that, you know, that's kind of what I'm interested in”

(Atana, Appendix G, p. 10)

The new manager of IRL thus both wants standardized data in the future while also data that is reliable and trustworthy. The question is if it is realistic to have both. But having the reflection on all these elements is exactly what our artifact set out to do.

Criterion 4: Inform Future Digital Strategy

The final criterion concerned whether the artifact informed stakeholders about tensions that were particularly relevant to their future digital strategy. Drawing on the three criteria discussed above, we argue that the artifact helped identify three overarching tensions that should be acknowledged in their future digital transformation efforts.

First, the artifact helped visualize and concretize the tension between standardization and flexibility within the organization's current setup by highlighting both the opportunities and challenges associated with implementing process automation in the CLIRT-process. Second, it prompted a thorough discussion of what should count as impact and emphasized the importance of allowing these reflections to guide the digital strategy. In doing so, it challenged the notion that impact should be defined solely through quantifiable digital metrics and highlighted the risk of becoming overly reliant on performative calculative practices. Third, the artifact facilitated discussions regarding the benefits and drawbacks of maintaining a flexible and non-standardized organizational setup. Such an approach may have both positive and negative implications, and it is not for us to determine whether the overall balance is beneficial for the organization. However, our findings illustrated that the current fragmented process and flexible KOBO setup make meaningful process automation particularly challenging to implement.

Discussing the Findings: Efficacy, Utility & Validity

Following Hevner et al. (2004) and Gregor & Hevner (2013), the artifact is discussed through the concepts of utility, validity, and efficacy.

In relation to our specific problem identification, the utility of the artifact becomes a question of the extent to which it supported meaningful discussion and the development and supports of common language between stakeholders. Although this is challenging to either confirm or deny, we argue that by enabling a dialogue grounded in a technical proof of concept, the artifact brought forward strategic, organizational, standardization, and data quality concerns. This can be interpreted as helping bridge separate concerns into a more bilateral understanding between IT and organizational departments. Overall, we assess that the artifact generated value by highlighting underlying organizational tensions and facilitating discussion around them.

The efficacy of the artifact becomes a question of the extent to which it achieved its intended objectives. Did it produce the effects it set out to achieve? This is somewhat more questionable, as the artifact cannot be said to have directly created a common language and thereby a sense of shared understanding. On the other hand, given that the objectives were also oriented toward facilitating dialogue and discussion around key trade-offs, the discussions that emerged among the informants indicate a degree of efficacy. The identified challenges and conflicts were not resolved. However, simply illuminating these challenges and exposing the lack of common language and bilateral understanding can itself be interpreted as an indication of efficacy.

The concept of quality is somewhat difficult to operationalize in relation to our artifact and evaluation process. Instead, we focus on validity, which becomes a question of the extent to which we can legitimately conclude that the artifact helped address the identified problem. In other words, how trustworthy are our findings? This raises interesting questions regarding whether the discussions were caused by the artifact or whether they would have occurred even without it. Given that Atana – the new manager – appeared to have an interest in the topic prior to being introduced to the artifact, it is likely that a similar discussion would have taken place at some point regardless. However, the artifact grounded the discussion in concrete trade-offs and a tangible proof of concept rather than abstract considerations, which we view as a valuable facilitator of dialogue.

The question of validity also raises concerns regarding the absence of three key informants during the final evaluation. Their presence may have altered the findings and challenged key assumptions related to either flexibility or standardization. Furthermore, it raises questions about the trustworthiness of the informants' statements. Certain perspectives may have been emphasized – or omitted – due to the formalized setting of the evaluation. In addition, the qualitative and interpretive nature of the study means that the findings are heavily dependent on how we, as researchers, interpreted the empirical data. This introduces challenges related to researcher subjectivity. While we have sought to remain as faithful to the empirical material as possible, we cannot entirely rule out the possibility that our own biases and interests influenced which findings we chose to emphasize.

Finally, the fact that the final evaluation was conducted only once, rather than across multiple iterations, represents a limitation. As a result, it is impossible for us to determine whether the discussions continued beyond the evaluation session, whether organizational decisions subsequently

changed, or whether more a durable common language and bilateral understanding emerged over time. These reflections point to several important limitations of the study that should be considered when interpreting the findings.

7. Design Knowledge – Deriving Design Considerations & Principles

As described in the methodology section, the knowledge created from DSR-studies does not lie in the specific artifacts, but in the understandings derived from designing and evaluating artifacts (Vom Brocke et al., 2024). Similarly, DSR-studies should concern prescriptive *how to*-knowledge rather than theoretical and descriptive *what is*-knowledge. The overall aim of this section is to synthesize empirical findings from the interviews and the DSR-activities with relevant perspectives from the literature review.

As the literature suggest, the development of a broad IT strategy aligned with goals, missions and resources is not yet a prioritized topic in NPOs (Vogelsang et al., 2021; Cipriano & Za, 2024). In accordance, our case organization lacked a broad IT strategy favouring participatory involvement that extends beyond just IT-department. Throughout our engagement we encountered structural challenges visible in the silo structure of the organization and fragmented processes – hindering the redesign and rethinking of these processes through process automation. In accordance, we encountered a lack of “system thinking” and “process thinking” as well as a data standardisation problem and low data quality. As Radu et al. (2024) argue these can all be conceptualized as being embedded in the “organizational core – structure, culture, knowledge and communication” (Radu et al., 2024, p. 10). Importantly, the development of a *common language* and *bilateral understanding* can support *clear strategies and goals* and *process alignment* and thereby improve synergy and understandings across departments – including between IT and the rest of the organization (Radu et al., 2024). Further, this was thus seen as a beneficial entry point for addressing many of the challenges present in the organization. This is not to say that facilitating a *common language* will magically fix all the challenges. Yet, it was a favourable entry point for our DSR-study to make these misalignments clear and foster a *common* understanding for the future – to maintain mission sensitivity, organizational alignment and avoid drifting in unsuitable directions.

Consistent with the DSR approach, our contribution is intended to move beyond the primarily *descriptive* and theoretical perspectives presented by Vial (2019), Radu et al. (2024) and Cipriano & Za (2024) by offering prescriptive insights. In the following, the challenges we encountered will be organized into three overarching themes that highlight particularly important considerations for both

what to consider as an NPO (the tensions observed in our study – phrased as *design considerations*) as well as *how* to address these (*design principles*) in order to develop a *common language* and *bilateral understanding* to support digital transformation. The design- *considerations* and *principles* are visualised in figure 33 and figure 34. In the following we will present three overall tensions observed in the empirical data.

7.1. Tension 1: Standardization vs. Flexibility

At the centre of much of the friction we encountered during the design and evaluation phases was the unstructured data and the implications of standardizing the survey forms. Throughout all four months of our collaboration with the organization, these challenges were continuously designed, discussed, and evaluated internally between the Clinical Department and the IRL Department: “But when it comes to like an overall picture for the organization, so that is not there in Kobo we are still trying that if we can design this like a master forms, which can be rolled out in all the countries.” (Appendix A, p. 6). The IRL Department hoped to standardize the forms into a “Master Form” to make the process of preparing, cleaning, and analyzing the data less time-consuming. This would also enable them to conduct deeper analyses across countries, regions, and training sessions. Furthermore, it would support the Business Department’s, Communication and Programme Teams’ interest in the “bigger numbers,” such as overall increases in knowledge, skills, and confidence, as well as the number of completed trainings (Appendix D).

On the other hand, the Clinical Department expressed a particular interest in the “smaller details” and module-specific elements of the surveys in order to understand how the trainings had been conducted and explore “what went good and what didn’t go so well and [...] how can we improve” (Eke, Appendix D, p. 5), while also asking questions such as: “why didn’t they score good here? What happened? Like, what could we do differently?” (Appendix D, p. 9). This is further expanded upon in the final group-interview where Ileen – on behalf of Clinical Department – describes that “I think that is something that they want to maintain, that if we are doing a country-specific training, we want to tailor it to the country” (Appendix G, p. 7). This strongly indicates a resistance towards complete standardization from Clinical Department as they need the flexibility to *tailor* the surveys to specific countries. This tension is thus inherently connected to the concept of impact reporting and accountability as the data collected must – according to Clinical Department – reflect the specificity and contextualization of each country’s legislation and the specific goals of the training. This

complicates the standardization as flexibility must be present in the design of the surveys. This will be further elaborated on in *tension 2 – Impact Reporting vs. Social Impact*.

Conceptualized through the DT-process framework (Vial, 2019) in dialogue with the specific value-creation structure of NPOs (Cipriano & Za, 2024), the tension can be interpreted to express different understandings of how data should create value for the organization. On the one hand, Business Development and IRL are interested in scalable and comparable data to enable *deeper analysis* and *upward accountability* – and hereby ensure funding for the organization. On the other hand, Clinical Development are interested in using the data to improve and understand the actual social impact made through the training. The tension inherent in the discussion of standardization vs flexibility can thus be interpreted to revolve around foundational different understandings of what the data should support.

The internal discussions between the Clinical Team and the IRL Team regarding how the forms should be standardized can therefore be understood as discussions concerning how standardization may reshape the organizational value creation. By conceptualizing this as an organizational barrier to digital transformation, the findings highlight the importance of including and aligning stakeholders from different departments to gain a *common language* of how data and IT-systems can be implemented in the organization. Even a seemingly “simple” integration of a digital automation tool – and the related standardization of survey forms – may introduce unforeseen changes in both the operational setup and the organization’s broader value proposition. Had we not interviewed employees from the Clinical Department, there is no guarantee that these tensions would have been identified. This further emphasizes the importance of involving and aligning multiple departments and employees throughout the digital transformation process to avoid undesirable outcomes.

This is not to suggest that standardization and scalability are inherently problematic or without value. Rather, the findings highlight the risks associated with adopting a purely efficiency- and optimization-oriented logic within the distinct value-creation structures of NPOs. This tension ultimately informs the design consideration 1: *Balanced standardization*.

7.2. Tension 2: Accountability – Impact Reporting vs. Social Impact

As the literature suggests, digital transformation offers NPOs the potential to engage with stakeholders, funders, and beneficiaries in new ways through digital services and tools (Cordery et al., 2023; Rana & Cordery, 2024; Kaba, 2021). At the same time, this growing ability to develop new and creative ways of showcasing organizational impact may risk shifting attention away from the actual social impact itself. A key argument within the literature is that while digital technologies may improve transparency and impact reporting, they may also reinforce funder-centric accountability rather than beneficiary-centred social impact (Cordery et al., 2023). Furthermore, digitalized performance systems may transform social missions into “calculative performance objects”, thereby reducing the value creation of NPOs to measurable indicators alone (Rana & Cordery, 2024).

In our case study, all stakeholders – including the Clinical Team – expressed interest in developing a dashboard displaying the overall metrics from trainings. This empirical finding that all interviewed stakeholders – even the ones that raised concerns towards standardization – saw beneficial value in an overall dashboard showcasing impact measurement, was alluring. Through the different DSR-phases, we discovered that they were resistant towards the standardization as a prerequisite for this dashboard. This indicates the importance of visualizing and communicating the trade-offs connected to an automated driven dashboard. Consequently, stakeholder enthusiasm for a dashboard should not necessarily be taken at face value. Rather, the findings highlight the importance of making the operational and data-related implications and trade-offs associated with this ambition as visible as possible before implementation decisions are made.

We found that the concept of *common language* has the potential to be an-entry point to address this tension by establishing a middle-ground where both *upward* and *downward accountability* is considered. By rooting the establishment and supporting a common language in a context specific artifact, we facilitated a group-discussion informing how and to what extent standardization and flexibility can be incorporated into their current process of impact reporting through a process automation perspective. This fostered beneficial discussion regarding what should count as “impact” and how to collect and potentially automate this in a meaningful way. As expressed by Atana, specific importance should concern the quality and trust in the data: “what I’m interested in is a level of standardization that facilitates automation to free up time, but also that we do it in a way that we can,

we have trust in the data... that it's... it's actually representing what we mean it to represent.” (Atana, Appendix G, p. 10). Atana’s focus can be seen as reflecting an awareness to not succumb to the *performative calculative practices*. Through critical dialogue with the literature, we suggest that this should be considered when NPOs reflect upon how to implement process automation in relation to impact reporting. As such, this tension informs design consideration 2: *Reflexive impact reporting*.

Facilitating discussions concerning the relationship between impact reporting and social impact may also help address the challenges visible in the empirical findings, where the Clinical Department simultaneously expressed enthusiasm for a KoboToolbox dashboard while also resisting the standardization of survey forms. The attractive idea of digital visualization and automation (a dashboard) is not without consequences or trade-offs.

7.3. Tension 3: Process Clarity Vs. Organizational Flexibility

Another central empirical finding in our study lies in the complexity of fully understanding the process. As seen in the three iterations, we had to do multiple interviews with different stakeholders to understand and map the process in a meaningful way. When we asked who ‘owns’ the process at the group interview Ulrika replied:

“Yeah, and it's actually a good question. I'm pretty sure that originally it was clinical team who owned it, because you're the ones who do the training. Yeah, I'm not sure if that's still something we need to decide, because I think it's a mixture together, as you're saying now, project and clinical, like it's programme and clinical who work together, they've all merged together now and then“
(Ulrika, Appendix D, p. 6)

Besides no specific clarity regarding who owns the process and is responsible for it, our main collaboration department, IRL, is situated as a supportive ‘middle man’ that do not own the data nor the process but supports other departments in cleaning, organizing and understanding the data. The seemingly ‘simple’ process thus turned out to be cross-departmental, fragmented and quite unorganized. As argued earlier, an automatization of an unstructured and non-repetitive process is not favourable (Nunes et al., 2026; Hofmann et al., 2020).

In our study, we found that clarification regarding what processes serve what goal was highly important in understanding how and if process automation was possible. In the final group-interview the new interim manager of IRL highlighted that “things drift over time, right? We do more questions and develop our training and so on... So it's always good to just reassess.” (Atana, Appendix G, p. 12). In this regard, thinking along the possibilities of process-automatization can be seen as an entry point for an organization to assess and agree on if it is meaningful to streamline specific impact reporting processes. Process automation is thus a beneficial entry-point for discussing if there are actual advantages in having a clarified processes or a more flexible organizational set-up in specific contexts such as impact reporting.

On one hand, this flexibility might be meaningful and useful in specific cultures and organizations. On the other hand, unstructured processes can also foster confusion and misalignment within an organization (Dumas et al., 2018). Our DSR-process suggest the importance of establishing an overview of processes within the organization. Without understanding the core processes and the social mission and how these connect to other processes, there is a risk of focusing on operational efficiency and obscuring the organizations social mission (Haddad et al., 2016). Thus, process improvement – such as our process landscape, AS-IS and TO-BE BPMNS – serve as an important analytical step to ensure process clarification before assessing the possibilities of process automation. In relation to this we derived design consideration 3: *Process Clarification*.

7.4. Design Considerations & Design Principles

Inspired by the empirical tensions described above, we derived three design considerations and three design principles. These are intended to inform the question regarding how to support *common language* and *bilateral communication* in NPOs. The ambition is – through design considerations and principles – to inform NPOs and practitioners on how to: “improve understanding and synergy across different levels.” (Radu et al., 2024, p. 7). Thus, instead of seeing communication only as a barrier to digital transformation, we find that it can resolve the tensions mentioned above. Concretely, the following design principles are specific *communication tools* (Radu et al., 2024) that an organization can utilize to address the NPO-specific tensions – described as *design considerations*.

Design Consideration	Intended outcome
Balanced standardization	Digital initiatives should balance standardization with the need for contextual adaptation and local practice requirements.
Reflexive impact governance	Organizations should continuously reflect on and negotiate what constitutes meaningful impact and how it is measured and communicated.
Process Clarification	Organizations should establish clarity regarding process ownership, workflows, and responsibilities prior to implementing digital or automated solutions.

Figure 33: Design Considerations

Design principle	Intended outcome
Multi-stakeholder representation	Digital solutions should be designed through the inclusion of multiple stakeholder groups to support common language and organizational alignment.
Proof-of-concept embedded in real-world	Proof-of-concept artifacts should be embedded within real-world work practices to enable stakeholders to identify practical opportunities, constraints, and unintended consequences.
Visualize trade-offs	Organizations should explicitly visualize trade-offs between competing priorities (e.g. BPMN-models)

Figure 34: Design Principles

The design considerations and principles should be interpreted with consideration of the study's contextual limitations. While the case study approach enabled an in-depth exploration of the phenomenon, the results are primarily transferable to organisations operating in similar contexts. However, the intention of the study is not statistical generalisation, but rather analytical generalisation, where insights from the case are linked to broader theoretical understandings. Consequently, the findings may contribute to future qualitative and quantitative research within NPOs and related organisational settings. Furthermore, the study provides practical insights into factors that may hinder digital transformation processes in the context of impact reporting, which can support practitioners in planning and implementing transformation initiatives.

8. Discussion

In this section, we reflect upon and contextualize our data-collection, methodological approach, analytical findings, as well as the strengths and limitations associated with these. To do this, we will first briefly highlight our main findings.

By adopting the DSR-approach we aimed at interacting with a “real-world challenge” (Vom Brocke et., al, 2024) present when an NPO aims at implementing a seemingly ‘simple’ automation tool to automate impact reporting. Through the different DSR-iterations and through critical dialogue with the relevant literature, we found that the challenges we met could be conceptualized as a lack of *common language* across departments. In accordance, our artifact became a proof-of-concept *communicative* tool aimed at creating common language and synergy across different departments in the organization (Radu et al., 2024). Furthermore, it helped illuminate three specific tensions present in the organization in relation to automation-initiatives aimed at improving impact reporting. These tensions are further conceptualized as three *design considerations* important for practitioners to consider when implementing digital technologies to improve impact reporting. Subsequently, we presented three prescriptive *design principles* that conceptualize how to address these specific considerations and hereby support process alignment and a clear IT-strategy for the future.

8.1. Revisiting the Design Consideration & Design Principles

Rather than being primarily theoretical and descriptive, as is the framework proposed by Cipriano and Za (2024), or focusing exclusively on identifying the barriers that NPOs face when implementing digital technologies (Radu et al., 2024), the specific contribution of our study lies in its prescriptive orientation. Our findings suggest that three key tensions emerge when NPOs seek to leverage process automation to improve impact reporting. We conceptualize these as *design considerations*. In addition, we propose three *design principles* that our study indicates may be useful for illuminating and potentially addressing these tensions. Thus, whereas Radu et al. (2024) identify barriers to digital transformation in NPOs, our contribution lies in providing prescriptive guidance that can inform practitioners on how to address *communication*-related barriers and navigate the tensions associated with process automation and impact reporting.

The Technology: Not the real barrier

Across all three tensions, we argue that they fundamentally reflect value-creation trade-offs rather than purely technological trade-offs (Cipriano & Za, 2024). While these discussions are often framed in terms of digital tools, data infrastructures, or process automation technologies, the underlying tensions, we argue, concern competing organizational priorities related to impact reporting, as well as the absence of a common language and a shared understanding of the implications associated with these priorities. The challenge is therefore not simply a matter of identifying the most advanced technological solution, but rather of determining what kind of value the organization seeks to create and preserve through its digital transformation efforts. In this sense, the tensions revolve around balancing efficiency with NPO-specific contextual sensitivity (Haddad et al., 2016), comparability with flexibility, and automation with meaningful representations of social impact.

Consequently, the design principles emphasize the importance of critically reflecting on what may be gained and lost through increased standardization and automation. Radu et al., (2024) highlight how a key technical barrier for NPOs is a lack of data quality which is directly linked to a standardization problem. Thus, we encourage critical reflection when dealing with standardization and automation in the context of impact reporting. For example, while standardized data structures may enable process automation, scalability, and cross-training comparisons, it risks simultaneously reducing the organization's ability to capture contextual nuances and locally relevant insights. Similarly, while automation can reduce manual work and improve operational efficiency, it risks shaping what counts as a social impact or a successful mission (Rana & Cordery, 2024). Standardized data do not simply 'represent' impact. It actively shapes it. Therefore, NPOs must critically consider what counts as impact before standardizing the data towards this.

Proof-Of-Concept: An Entry-Point for Facilitating Common Language

Our findings further suggest that these value-creation trade-offs are often difficult for stakeholders to fully comprehend when discussed solely at an abstract or conceptual level. During the evaluation, several of the most productive discussions emerged when stakeholders were presented with concrete visualizations of how process automation could potentially be implemented and what organizational consequences such implementation might entail. This indicates that stakeholders may find it easier to engage with complex trade-offs when these are materialized through tangible artifacts rather than discussed as hypothetical possibilities. Such artifacts make abstract trade-offs visible and discussable

across organizational groups with different responsibilities, perspectives, and interests. By translating strategic considerations into concrete technological possibilities, proof-of-concept artifacts can help stakeholders develop a common language for discussing the implications of process automation and digital transformation.

Furthermore, proof-of-concept artifacts can play a critical role in supporting future digital initiatives. They allow stakeholders to move beyond assumptions and speculative discussions by illustrating how different design choices may affect organizational processes, data practices, and impact reporting activities. In doing so, they serve as an entry-point for common language that can contribute to process alignment and the development of clear strategies (Radu et al., 2024).

Finally, we argue that organizations should establish clarity regarding process ownership, workflows, and responsibilities prior to implementing digital technologies to improve impact reporting (Nunes et al., 2026). Throughout our study, several challenges stemmed not from technological limitations but from ambiguities surrounding who was responsible for different activities, how data moved across departments, and what objectives the reporting process was intended to achieve. Without a common language of existing processes and responsibilities, digital technologies risk automating inefficiencies, reinforcing organizational silos, or generating additional complexity. Process clarification should therefore be viewed as a prerequisite for successful digital transformation (Dumas et al., 2018), as it provides the foundation upon which technological solutions can be meaningfully designed, implemented, and evaluated.

Social Mission vs. Common Language

An inherent challenge for NPOs lies in balancing the pursuit of their social mission with investments in organizational development and strategic reflection. Unlike commercial organizations, NPOs often operate under significant resource constraints, making it difficult to justify activities that do not directly contribute to mission delivery (Nahrkhalaji et al., 2019). Creating common knowledge across departments, stakeholders, and organizational levels requires time, attention, and resources that could otherwise be allocated to beneficiaries and program activities.

This raises an important question: can NPOs justify investing resources in developing common language of organizational challenges and opportunities? From a short-term perspective, such efforts

may appear to divert scarce resources away from the social mission. However, from a longer-term perspective, a lack of common language may result in fragmented decision-making, misaligned priorities, and inefficient digital initiatives that ultimately undermine the organization's ability to create social value.

Consequently, the tension is not simply between social mission and organizational reflection. Rather, it concerns whether investments in facilitating common language should be viewed as a cost that competes with mission delivery or as an enabling capability that supports the organization's ability to fulfil its mission more effectively over time. Determining the appropriate balance remains a central challenge for NPOs seeking to navigate digital transformation while remaining faithful to their social purpose. This highlights an important avenue for future research: the development of communication tools that are both financially viable and less resource intensive.

8.2. Theoretical Implications – Contribution to Existing Literature

Having discussed our study's contribution to design knowledge within the field in the section above, we now move on to consider how our findings relate to the frameworks and theories that informed our artifact. As stated earlier, the literature on the topic is quite scarce. As such, the present study contributes to an emerging research stream. However, the relative scarcity of prior literature necessitates caution in interpreting and generalizing the theoretical implications, as further empirical investigation is needed to substantiate and refine the insights generated.

Digital Transformation & NPO-specific Value-Creation Logics

As Cipriano & Za argues, NPOs differ from other organizations based on “the primacy of nonfinancial outcomes and the absence of incentives to increase productivity” (Cipriano & Za, 2024, p. 736). Accordingly, they argue that DT processes of NPOs are poorly understood in relation to their specific preconditions and unique value-creation structures (Cipriano & Za, 2024). Without understanding the peculiarities and specific elements that characterises NPOs and their engagement with digital transformation, initiatives “may fail to deliver the expected value” (Cipriano & Za, 2024, p. 736). Our study supports this general understanding, that the specific value-creation structure of NPOs must be considered when engaging with digital transformation initiatives. Without considering this, digital transformation processes might mutate and evolve in challenging directions. Informed by Rana &

Cordery (2024) our specific case regarding utilizing digital technologies to improve impact reporting emphasized how a seemingly simple digital tool could potentially transform social missions into *calculative performance objects* (Rana & Cordery, 2024) and not only measure impact, but actively shape and organize what counts as “impact”. The tensions within the organization seemingly reflect more than just technological challenges. They might just reflect conflicting value-creation perspectives.

Similarly, our findings support the question raised by Cipriano & Za (2024) regarding if NPOs can adopt digital transformation strategies grounded in different organizational value-creation setups. It could be argued that the tension between standardization and flexibility reflects the coexistence of multiple value-creation logics within the organization, some of which prioritize operational efficiency while others emphasize contextual adaptation and mission fulfilment. As such, our findings support their call for adopting NPO-specific strategies for engaging with digital transformation processes (Cipriano & Za, 2024). These findings further imply that caution is warranted when applying Vial's (2019) framework to the NPO context, as certain assumptions underlying the framework may not fully align with the characteristics of nonprofit organizations.

Process Automation: A Socio-Cultural Challenge

Furthermore, our findings support existing literature on RPA implementation. Prior studies have emphasized that successful implementation depends on organizational readiness, stakeholder alignment, and a clear understanding of existing processes (Hofmann et al., 2020; Nunes et al., 2026). Our findings are consistent with these observations. In particular, the tensions identified between standardization and flexibility, as well as the differing interpretations of what constitutes meaningful impact, illustrate how challenges associated with process automation extend beyond technical implementation. Rather, the findings suggest that establishing a common language regarding organizational objectives, reporting practices, and process ownership may constitute an important prerequisite for process automation and RPA initiatives in nonprofit settings. In this regard, our study supports existing arguments that process automation projects are socio-technical endeavours requiring organizational alignment in addition to technological capabilities (Hofmann et al., 2020), while extending these insights by demonstrating how such alignment is closely connected to negotiations surrounding accountability and impact reporting in NPO contexts.

Towards a ‘Common Language’

Our findings further support the overall finding by Radu et al. (2024) emphasizing that *communication* is an important barrier to address in relation to digital transformation initiatives in NPOs. By synthesizing this argument with the importance of considering the *peculiar* value-creation structure of NPOs (Cipriano & Za), our findings suggest that the *communication* barrier can be conceptualized as a beneficial entry-point for addressing and negotiating the specific tensions and challenges that exist when NPOs take on digital transformation initiatives. As such, our study suggest that communication is more than just a barrier. Rather, the category of communication appears to constitute a valuable analytical entry point for illuminating, assessing and understanding the broader organizational dynamics that shape digital transformation processes.

Taken together, these findings suggest that communication may play a more foundational role in NPOs engagement with digital transformation. Rather than being understood primarily as a barrier, we find that communication could constitute a beneficial mechanism through which NPO actors can negotiate their particular understandings of impact, accountability, and value creation. In this regard – we argue – the concept of common language may represent a critical prerequisite for successful digital transformation initiatives by aligning organizational objectives, social missions, reporting practices and desired outcomes. Thus, facilitating a common language can mitigate how social missions are interpreted, operationalized, and ultimately translated into measurable representations of impact. By highlighting these interdependencies, the present study contributes to the literature on the digital transformation initiatives in NPOs by conceptually linking the concepts of common language, nonprofit value-creation logics, and the construction of calculative performance practices.

8.3. Methodological Limitations

Complexity of Engaging with the Real-World

One of the advantages of the DSR approach is that it requires researchers to engage directly with real-world problems and design artifacts aimed at addressing specific organizational challenges. The strength of adopting the DSR approach in this assignment lies in its ability to reveal the complexity of engaging with the real world in all its nuances. The challenges highlighted throughout our assignment may not have become visible had it not been for the inherent emphasis within DSR on designing and evaluating an artifact intended to solve a real-world problem. This problem-solving orientation ultimately fostered practice-near and actionable prescriptive knowledge that can be applied by both practitioners and researchers.

However, adopting the DSR approach also introduced several challenges. For starters, the real-world focus of DSR – although often highly beneficial and valuable – also carries certain implications. The real world is not always predictable or cooperative. In our study, this created significant challenges, as we continuously had to adapt and redefine both the problem and the scope of the assignment. A central challenge during the initiation phase revolved around developing a comprehensive understanding of the organization, its processes, and its strategic ambitions. Numerous projects and ambitions, ranging from the "5-year digital strategy" and "cascading effects" to various future-state visions and ideal solutions, were repeatedly discussed, and all appeared relevant to the assignment.

Consequently, we spent considerably more time than anticipated moving back and forth between Activity 1 (Problem Identification and Motivation) and Activity 2 (Define Objectives of a Solution). As a result, less time was available for designing, demonstrating, and evaluating the artifact than we had initially hoped. While we acknowledge the value of this iterative characteristic of DSR, it also has time-consuming implications. In an ideal scenario, we would have liked to spend more time exploring how a master form could be designed to balance the tension between flexibility and standardization, while also conducting additional evaluations of the artifact.

Furthermore, the final evaluation highlighted how rapidly circumstances can change within an NPO. For example, the former Head of IRL had been replaced by an interim manager without our knowledge. In addition, several key informants who had been expected to participate in the final evaluation were ultimately absent. This significantly affected our original evaluation plan. As such,

important perspectives and discussion points may have been lost due to their absence from the final interview.

Balancing Relevance & Rigor

A further challenge that we continuously reflected upon was balancing rigor (being academically, theoretically, and methodologically grounded) with relevance (addressing a real-world problem), while ensuring that our project was neither overly practical and susceptible to the “consultancy trap” nor excessively theoretical with limited practical impact. We sought to balance these two aspects by iteratively allowing the design and evaluation phases to inform, and be informed by, relevant theory.

One might argue that our designed artifact does not directly solve a tangible problem, as it cannot be instantiated in its current form. On the other hand, we argue that the design and evaluation of this proof of concept facilitated a much-needed cross-departmental dialogue, highlighting the inherent complexity and often uncertain outcomes associated with implementing a process automation solution in an NPO context to improve impact reporting.

Researcher Bias

Another challenge worth mentioning is the inevitable researcher subjectivity. DSR researchers are deeply involved in all aspects of understanding the problem, designing a meaningful artifact, and evaluating the proposed solution. In other words, we inevitably shape the solution throughout all phases of the DSR process. As such, there is no guarantee that the same artifact and the same prescriptive knowledge would have emerged had different researchers engaged with the organization.

A further source of bias relates to our technological competencies. Had we possessed deeper practical experience with the specific IT tools used by the organization, we might have been able to design a different artifact, which in turn could have resulted in different – or potentially fewer – challenges. Consequently, DSR is a highly contextual approach, with all the implications that entails. An ambition to implement a process automation solution may work smoothly in other countries or organizational contexts. However, through our critical engagement with the relatively scarce literature on the topic, it became evident that many of the challenges we encountered are common within the NPO sector (Radu et al., 2024; Cipriano & Za, 2024; Vogelsang, 2021).

Discussing our Evaluation Strategy

Following Venable et al.'s (2016) extended DSR evaluation framework, our evaluation strategy can be characterized as a predominantly formative and naturalistic ex post evaluation conducted under significant contextual constraints. Rather than evaluating a fully implemented artifact through quantitative performance indicators, we evaluated a proof-of-concept artifact through stakeholder interaction, discussion, and reflection. This choice was influenced by several contextual factors highlighted by Venable et al. (2016), including the exploratory nature of the artifact, the limited timeframe of the project, restricted organizational resources, and the fact that the artifact during different iterations progressively changed its orientation towards a proof-of-concept *communicative tool* rather than immediate operational deployment.

More specifically, the evaluation combined scenario-based evaluation and informed argumentation (Hevner, 2004). The artifact functioned as a plausible future scenario that enabled stakeholders to discuss the implications of process automation technologies in context of impact reporting, while assessments of its utility and relevance were informed by existing literature on digital transformation, process automation, and nonprofit accountability. In Venable et al.'s terms (2016), the evaluation therefore prioritized contextual relevance and organizational learning over experimental rigor or predictive testing through quantitative measurements. As such, our evaluation strategy was inspired by what Venable et al. conceptualize as a *human-risk and organizational-risk* evaluation rather than a *technical-risk* evaluation (Venable et al., 2016, p. 83). As such, we adapted the evaluation strategy to explore whether the artifact could support organizational understanding and alignment, not whether the automation architecture actually works.

This evaluation strategy aligns closely with the purpose of the artifact. The artifact was not designed to increase efficiency, but to facilitate organizational discussions concerning standardization, flexibility, impact conceptualization, and future digital strategy. Consequently, criteria such as "common language," "discussion facilitation," and "strategic reflection" were considered more appropriate indicators of success than traditional measures of efficiency and performance. From this perspective, the evaluation findings suggest that the artifact achieved a degree of utility and efficacy by making previously implicit tensions visible and by stimulating discussions around these through concrete and tangible technical possibilities and requirements.

However, criticisms can be raised regarding this evaluation strategy. First, the evaluation relies heavily on stakeholder descriptions and researcher interpretations. While we argue that the artifact facilitated valuable discussions, such statements do not necessarily demonstrate that the artifact solved the identified organizational problem. Our evaluation may overstate organizational impact by treating the emergence of discussion as an indicator of improvement. Venable et al. (2016) note that evaluations emphasizing perceived usefulness may provide weaker evidence of artifact effectiveness than evaluations demonstrating observable outcomes.

Second, the evaluation exhibits limitations regarding internal validity. It remains difficult to establish whether the discussions observed were caused by the artifact itself or whether they would have emerged regardless due to ongoing organizational developments. The arrival of a new interim department head, who already appeared highly motivated to discuss data integration and standardization, illustrates this challenge. Consequently, the artifact may have acted as a catalyst for discussions that were already likely to occur rather than as their primary cause.

Third, the evaluation's participant composition introduces potential bias. Several key stakeholders whose perspectives informed earlier design stages were absent from the final evaluation. In particular, the absence of representatives from the IT, Clinical, and Programme departments may have limited the diversity of viewpoints represented in the findings. It is therefore possible that alternative concerns, implementation barriers, or disagreements remained underrepresented.

Fourth, the ex-post evaluation was conducted as a single event. While we did have repeated cycles of design, development and refinement of our artifact and its objectives, Venable et al. (2016) emphasize that DSR-studies often benefit from repeated cycles of evaluation. Because the ex-post evaluation occurred only once, it cannot determine whether the identified discussions translated into organizational learning, strategic decisions, behavioural change, or longer-term adoption of the proposed ideas. The findings therefore only provide indications of immediate stakeholder reactions rather than sustained organizational impact.

Nevertheless, Venable et al. (2016) argue that evaluation strategies should be judged relative to the context in which they are conducted. Given the exploratory nature of this project, the organizational constraints, and the artifact's purpose oriented towards facilitating discussion and reflection, a

naturalistic formative evaluation was considered most appropriate. The contribution of the evaluation therefore lies less in demonstrating that the artifact definitively solves the identified problem and more in showing how it helped surface organizational tensions, reveal competing assumptions, and create a platform for dialogue concerning future digital transformation efforts.

8.4. Future Research

As the study is based on a single case organization and a proof-of-concept artifact, future research should investigate the applicability of these design principles across other NPO contexts. Examining organizations with different missions, sizes and governance structures could help refine and further develop the proposed principles. This aligns with the general call for examining which aspects link to digital transformation outcomes in NPOs (Trischler & Li-Ying, 2022).

Furthermore, future research could adopt a longitudinal perspective. This could entail examining how organizations apply and sustain the proposed design principles over time. While the artifact developed in this study facilitated discussion and reflection regarding process automation trade-offs regarding impact reporting, the long-term effects on organizational alignment and digital strategy development remain unknown. Future studies could therefore investigate how common language and bilateral understanding evolve throughout implementation processes and whether investing in supporting a common language contributes to more successful digital transformation outcomes.

Finally, research could critically examine the sustainability and resource implications of the proposed design principles presented in this study. While stakeholder involvement, workshops, iterative evaluations, and cross-departmental alignment proved highly valuable throughout the DSR approach, such approaches may simultaneously require substantial organizational resources that smaller or resource-constrained NPOs may struggle to allocate. Consequently, further research is needed to investigate how organizations can balance the benefits of the design considerations and principles with the inherent cost of facilitating continued communication and bilateral understanding.

9. Conclusion

This thesis set out to explore how an artifact can be designed to support impact reporting in NPOs and explore what design considerations and principles can be derived from this. Through a Design Science Research approach, the study engaged with an NPO interested in automating an impact reporting process.

Throughout the DSR-process, impact reporting was conceptualized through the interconnected perspectives consisting of (1) digital transformation in NPOs, (2) accountability and (3) process automation. These perspectives offered complementary conceptualizations for how to design an artifact capable of capturing the inherent complexity embedded in the ambition of improving impact reporting through process automation technologies. Taken together, these perspectives enabled a richer understanding of the broader organizational implications associated with redesigning impact reporting processes.

Although the project initially focused on implementing process automation technologies, the iterative design process revealed that impact reporting is deeply intertwined with organizational structures, accountability requirements, data practices, and differing understandings of value creation. Consequently, the final artifact evolved into a proof-of-concept that combined process automatization, data visualizations, future-state process models, and hereby clarifying trade-offs to support organizational reflection and discussion.

Across the three DSR iterations, the different DSR-activities progressively contributed to a deeper understanding of how an artifact could support the organization and what the artifact should accomplish. The first iteration revealed challenges related to fragmented processes, unclear responsibilities, and inconsistent data structures. The second iteration revealed internal organizational discrepancy regarding what impact reporting should accomplish as well as differing perspectives on how to accomplish this. The third iteration demonstrated how the artifact could facilitate stakeholder discussions regarding the implications of automation, standardization, accountability, and future digital initiatives. Rather than producing a fully implementable technical solution, the DSR-process highlighted the importance of creating an artifact that could support stakeholders collectively to assess the potential and challenges of different design choices before implementation.

Across these iterations, three tensions emerged. Based on these tensions, three design considerations were derived. First, *balanced standardization* emphasizes the need to balance comparable and scalable data structures with local flexibility and contextual relevance. Second, *reflexive impact governance* highlights the importance of continuously reflecting upon what constitutes meaningful impact and how digital systems may not only measure and represent these but actively shape these understandings. Third, *process clarification* emphasizes the need to establish clear responsibilities, workflows, and process ownership prior to implementing digital technologies to improve impact reporting.

To address these considerations, the study proposes three design principles. (1) *Multi-stakeholder representation* emphasizes the inclusion of diverse organizational perspectives throughout the design process. (2) *Real-world embedded proof-of-concept artifacts* highlight the value of grounding discussions in tangible and context-specific representations of future possibilities. Finally, (3) *visualization of trade-offs* stresses the importance of making competing priorities and potential consequences visible to stakeholders before implementation decisions are made.

Overall, the study suggests that supporting impact reporting in NPOs is not primarily a technological challenge but an organizational one. The findings indicate that establishing a *common language* may constitute a critical prerequisite for successful digital transformation in NPOs, as it enables stakeholders to negotiate competing priorities and develop shared understandings of how digital technologies can support operational efficiency while also balancing both accountability requirements and the broader social mission of the organization.

While digital technologies may provide new opportunities for efficiency, analysis, and accountability, their successful implementation depend heavily on a contextual understanding of the complex nature of NPOs. The contribution of this thesis therefore lies not only in the design of an artifact, but also in the design knowledge generated through the iterative process, resulting in three design considerations and three design principles that may guide future impact reporting and digital transformation initiatives in nonprofit organizations.

10. References

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